

The Global Capitalism Index:

A Multidimensional Index for Measuring Market Systems

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Abstract

This paper introduces the Global Capitalism Index (GCI), a multidimensional measure of cross-national variation in market systems. The GCI conceptualizes capitalism as a multidimensional institutional construct composed of eight subindices: Strength of Property Rights and Private Ownership, Market-Supporting Policy, Labor Market Openness, Market Competition, Capital Market Sophistication, Depth and Stability of Banking System, New Business Formation and Growth, and Free Flow of Goods and Capital. The composite GCI covers 170 countries across 2009–2025, with annual coverage ranging from 158 to 169 countries depending on data availability. Subindex coverage ranges from 153 countries (Labor Market Openness) to 175 (Property Rights, Market Competition). Using harmonized and internationally comparable data, each subindex is constructed via adaptive principal component analysis. The composite index is constructed by a second PCA using the eight subindices. 95 percent bootstrap confidence intervals are reported for every country-year score. We assess internal coherence, robustness to alternative aggregation choices and benchmark the GCI against existing measures. We then discuss how the GCI correlates with external variables of interest and illustrate four varieties of capitalism the eight-dimensional space reveals.

JEL Classifications: C1, C43, P1, P5, P16

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Introduction

For decades if not centuries, scholars, civil servants, and the public have struggled for consensus on exactly what capitalism is and how it operates. Even the scholarly scions of capitalism have produced widely varied definitions. Milton Friedman emphasized private property and profit making. However, he regularly fell back on amorphous concepts such as “free enterprise” or “a system of co-operation without coercion, where individuals make their own choices, and the market aggregates those choices.”¹ Joseph Schumpeter described capitalism as a system defined by innovation and entrepreneurship. In so doing, Schumpeter focused on systemic outputs rather than the structures that produce them.² Nobel prize-winner Friedrich Hayek found the term ‘capitalism’ so difficult that he largely avoided it, preferring instead to emphasize “the market-based order.”³ Widely considered the father of modern capitalism, Adam Smith emphasized private property and the objective rule of law. Yet unlike the others, Smith explicitly focused on specialization, competition, and markets free from interference from both public and private actors.⁴

The lack of consensus on capitalism among even its strongest supporters presents an important question: is trying to define such a conflicted and amorphous concept worth the effort to do so? We believe it is. Despite the deficient understanding of capitalism’s shape and meaning, it is a concept that holds tremendous cultural, political, and historical authority. A rising number of academic conferences seek to understand the economic power of capitalism and its role in societies across geography and time. Social and political movements often unite around shared views of capitalism, both positive and negative. Capitalism is a controversial subject and one that spurs debate across disciplines. The lack of a lingua franca through which to have discussions and debates means that evidence-based conversations are impossible. Without a common model of what capitalism is and how it works, we cannot debate its benefits and weaknesses, much less study how its components interact to promote or hamper societal outcomes.

Recent scholarly debates on capitalism have emphasized its relationship (or lack thereof) with democracy. There is a substantial body of literature defining democracy, but scholarly effort to define and measure capitalism has been difficult and fractured.⁵ For all his writings on capitalism and its relationship with human progress, Karl Marx rarely if ever used the word “capitalism,” but typically referred to “bourgeois society” or “the reign of capital.”⁶ Several prominent scholars, including Nobel Prize-winning economist Daron Acemoglu, even propose retiring the term altogether.⁷ Depending on their definition of these terms, observers can view democracy and capitalism can be viewed as compatible or incompatible with each other.

Understanding the material role of capitalism and its role in society requires a clear definition and framework for understanding the subject itself. The existing literature focuses on the extent of state involvement in the economic system. In their defining scholarship on the varieties of capitalism,

¹Milton Friedman, *Capitalism and Freedom* (Chicago: University of Chicago Press, 1962).

²Joseph A. Schumpeter, *Capitalism, Socialism and Democracy* (New York: Harper & Brothers, 1942).

³Friedrich A. Hayek, *The Road to Serfdom* (Chicago: University of Chicago Press, 1944).

⁴Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

⁵Robert F. Bruner, “What Is Capitalism? Why Definitions Matter,” Darden Ideas to Action (University of Virginia Darden School of Business, July 21, 2022), <https://ideas.darden.virginia.edu/what-is-capitalism>; Joyce Appleby, *The Relentless Revolution: A History of Capitalism* (New York: W. W. Norton, 2010); Sven Beckert, “Interchange: The History of Capitalism,” *Journal of American History* 101, no. 2 (September 2014): 503–536; Jonathan Levy, *Ages of American Capitalism: A History of the United States* (New York: Random House, 2021).

⁶Karl Marx, *Capital: A Critique of Political Economy*, Volume I, trans. Fowkes (Penguin Classics, 1976 [1867]).

⁷Daron Acemoglu, “Capitalism,” in *Economic Ideas You Should Forget*, ed. Bruno S. Frey and David Iselin (Cham: Springer International Publishing, 2017), 35–38.

Peter Hall and David Soskice argue that nations develop institutions that vary in the extent of their government intervention and planning.⁸ Hall and Soskice’s varieties of capitalism framework clusters countries into liberal market economies (LMEs) and coordinated market economies (CMEs), each characterized by distinct institutional and regulatory configurations. Although this descriptive framework provides value, it is still largely based on qualitative comparisons. The literature lacks a systematic and quantitative measurement tool capable of capturing institutional variation and conducting analyses of modern capitalism.

The Global Capitalism Index (GCI) provides a new theory and quantitative model of capitalism. The GCI is comprised of 213 datasets across 33 domains, and 8 independent subindices. We compile and weight all datasets from 2009 to 2025 via an adaptive principal components analysis (PCA) methodology covering 170 countries. Annual coverage at the composite index level ranges from 158 to 169 countries depending on data availability. Subindex coverage ranges from 153 countries (Labor Market Openness) to 175 (Property Rights, Market Competition).⁹ The GCI is structured in multiple layers: individual datasets aggregate to form domains and those collections of datasets form their respective subindex’s score. Finally, the eight subindices generate a composite Global Capitalism Index score. We will continually update the GCI with each new year of data available.

We view capitalism as a system that privileges the ability to deploy capital in open markets as freely as possible, while minimizing constraints from both public and private sources. Capitalism can and should be thought about as the combination of many different factors that may seem contradictory on the surface. Our model of capitalism unites theoretical and empirical foundations, based on a comprehensive examination of the existing literature (see below). This model delivers a shared language for capitalism, but it also provides the tools to compare and investigate similarities and patterns within the components of capitalism itself. The eight subindices are listed in Table 1 (below).

Subindex	Domains	Datasets
Strength of Property Rights & Private Ownership	7	31
Market-Supporting Policy	2	20
Labor Market Openness	3	13
Market Competition	3	31
Capital Market Sophistication	7	29
Depth & Stability of Banking System	2	22
New Business Formation & Growth	6	56
Free Flow of Goods & Capital	3	11
Total	33	213

Table 1: Structure of the Global Capitalism Index

Scholars and practitioners disagree on whether capitalism is best characterized by the activities that constitute its operation (voluntary exchange, open market pricing, private accumulation, wage labor) or by the institutional infrastructure that makes those activities possible.¹⁰ We share the

⁸Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001).

⁹five countries appear in individual subindices but do not meet the threshold for a composite score.

¹⁰Robert F. Bruner, “What is Capitalism? Why Definitions Matter,” Bruner Blog, Darden School of Business, October 2024.

view that voluntary exchange and market pricing are vital observable features of any functioning capitalist economy. However, we argue that these features do not emerge or sustain themselves spontaneously but rather depend on institutional preconditions that secure and enable the free deployment of capital.¹¹

The GCI therefore defines and measures capitalism by assessing a system’s institutional preconditions rather than its results. This reflects the GCI’s methodological commitment to, whenever possible, measure *inputs* to capitalist economic activity rather than the *outputs* that activity generates. Outputs can arise from combinations of non-systemic factors while structural forces constitute the system itself. A country in which voluntary exchange and open market prices occur on an isolated or transient basis is not *necessarily* a capitalist economy. What determines the type of economic system is the structure that produces, sustains, and renders predictable the outputs we associate with it. A country that scores highly across the GCI’s eight subindices is, by construction, one in which voluntary exchange and market pricing are highly likely to occur on a consistent and durable basis.

Our institutional approach also opens analytical space for a more nuanced characterization of market systems. Academics and practitioners deploy conventional designations of capitalist, non-capitalist, and “mixed economy” with abandon.¹² These designations are analytically limited and do not provide insight on which combinations of institutional features distinguish market systems from each other. We find existing designations of “state-capitalist” or “liberal market capitalist” to be unhelpful—“the state” does many things, and measuring capitalism by “state involvement” without understanding what the state is doing risks conflating political designation with market structure.

In a forthcoming publication, we develop a series of market modalities that uses our subindex structure to characterize the institutional configurations that move beyond those proposed in the existing varieties of capitalism literature. We aim to create a new analytical framework that groups capitalist systems not by state involvement in the aggregate, but rather a sophisticated understanding of what each state does in each context and how effective it is at doing so.¹³

To create the GCI, we have collected publicly accessible datasets from international organizations, academic publications, and private sector sources, and harmonized them for analysis.¹⁴ We subject all data to our adaptive principal components analysis process to reduce data dimensionality¹⁵ and combine structurally different datasets into unified measures of each element of the GCI.¹⁶

¹¹Douglass C. North, *Institutions, Institutional Change and Economic Performance* (Cambridge: Cambridge University Press, 1990).

¹²We argue that, with a few outliers like North Korea or Venezuela, most economies are “mixed,” thus making outright designations of “capitalist” and “non-capitalist” irrelevant.

¹³Initial analysis has yielded four quantitative models of capitalism: 1) Entrepreneurial Capitalism, 2) Corporate Capitalism, 3) Institutional Capitalism, and 4) Commercial Capitalism. Forthcoming research on this subject can be found at www.capitalismindex.com.

¹⁴This process is detailed in the Data Collection and Harmonization section.

¹⁵Data dimensionality reduction constitutes representing the original dataset(s) using fewer features or dimensions while still capturing as much of the original properties as possible.

¹⁶This approach is designed to accommodate the reality that institutional dimensions of capitalism are typically related but not identical. Often, domains within a subindex (for example, Rule of Law, Lack of Corruption, and Contract Enforcement within the Strength of Property Rights and Private Ownership subindex) are conceptually distinct but systematically correlated. Our index construction process does not assume orthogonality among input variables and allows us to capture the common variance shared across correlated inputs and produces components that reflect that shared variation while downweighting redundancy. If Rule of Law, Lack of Corruption, and Contract Enforcement were perfectly collinear, our principal components analysis process would effectively treat them as a

Literature Review

Empirical cross-national indices have become foundational infrastructure for quantitative social science. These datasets provide operational means for testing hypotheses across institutions, regimes, and markets. Composite indices have gained popularity because they can contribute to a variety of research designs at the country-year level including panel regressions, instrumental variable identification, and difference-in-differences models. The widespread use of these measures reflects a practical research need: scholars require cross-national and longitudinal proxies for complex institutional environments. Composite indices allow researchers to compare otherwise heterogeneous countries, states, sectors, and time periods using standardized empirical measures. In so doing, the index itself becomes an implicit theory of institutions. The choice of components, weights, aggregation rules, missing-data procedures, and conceptual boundaries can and often do affect results. For that reason, improvements in index construction are not merely technical; they can alter substantive conclusions across a wide body of research.

Across the areas covered by the GCI, the landscape of existing indices can be organized into two broad families: 1) specific measures of economic freedom and/or capitalism and its components, and 2) institutional, governance, or political system indices.

Indices of economic institutions measure the rules, regulations, and policy environment within which markets operate. The Fraser Institute’s *Economic Freedom of the World* (EFW) Index¹⁷ and the Heritage Foundation’s *Index of Economic Freedom* (IEF)¹⁸ are the most prominent indices providing broad measures of market-oriented or “capitalist” systems. The World Bank’s Worldwide Governance Indicators (WGI),¹⁹ the World Bank’s *Doing Business* indicators (discontinued in 2021 following methodological controversies but historically central to the literature),²⁰ the *PRS International Country Risk Guide* (ICRG),²¹ and the Harvard Growth Lab’s *Economic Complexity Index*²² all explore specific elements of market systems, such as business structures, commercial output, and state-business interaction.

The second group of indices focus on political institutions and evaluate the rules and practices of political participation, contestation, accountability, and rights. The Varieties of Democracy (V-Dem) datasets represent the gold standard for composite indices, both in terms of systematic coverage and methodological rigor.²³ Freedom House’s *Freedom in the World* survey,²⁴ the Bertelsmann

single signal. To the extent that they are correlated but not identical, the methodology extracts both their shared dimensions and any independent variation each contributes. This process is detailed thoroughly in the Compositional Methodology section.

¹⁷James Gwartney, Robert Lawson, Joshua Hall, Ryan Murphy, et al., *Economic Freedom of the World: 2025 Annual Report* (Vancouver: Fraser Institute, 2025).

¹⁸Heritage Foundation, *2026 Index of Economic Freedom* (Washington, DC: The Heritage Foundation, 2026).

¹⁹Daniel Kaufmann, Aart Kraay, and Massimo Mastruzzi, “The Worldwide Governance Indicators: Methodology and Analytical Issues,” *Hague Journal on the Rule of Law* 3, no. 2 (2011): 220–246; World Bank Policy Research Working Paper No. 5430 (September 2010).

²⁰World Bank, *Doing Business* reports, 2003–2020.

²¹The PRS Group, *International Country Risk Guide*, (East Syracuse, NY: The PRS Group, 1980–present)

²²Ricardo Hausmann, César A. Hidalgo, et al., *The Atlas of Economic Complexity: Mapping Paths to Prosperity* (Cambridge, MA: MIT Press, 2014).

²³V-Dem Institute, University of Gothenburg, <https://www.v-dem.net>; (V-Dem) dataset draws on more than 4,200 country experts, covers 202 countries from 1789 through the present, contains 531 indicators and 251 indices in version 16.

²⁴Freedom House, *Freedom in the World 2025: The Uphill Battle to Safeguard Rights* (Washington, DC: Freedom House, 2025) <https://freedomhouse.org/report/freedom-world>.

Stiftung’s Transformation Index,²⁵ and the Economist Intelligence Unit’s Democracy Index²⁶ also provide reliable metrics of political right and social institutions.

The use of composite indices in empirical research is vast and growing. The Fraser Institute’s official citation tracker records 6,033 cataloged academic citations between 2011 and 2025, with 1,571 citations in the four years between 2022–2025.²⁷ In 2024 alone, the tracker logs 501 citations.²⁸ Fraser’s accounting found that of 402 articles citing the index, 198 used it as an independent variable, and over two-thirds reported “good” outcomes from higher scores.²⁹ While V-Dem does not report direct citations, they note that the primary dataset has seen over 400,000 downloads since 2016.³⁰ “The Worldwide Governance Indicators: Methodology and Analytical Issues” in the *Hague Journal on the Rule of Law*, the foundational methodology paper anchoring the WGI dataset, has become a foundational citation in growth, governance, and finance regression analyses. Put together, these composite indices have received hundreds of thousands of citations in a wide range of peer-reviewed publications.

Composite indices provide, at least in theory, a sophisticated independent variable against which scholars can measure a range of output variables to better understand their function and operation. While not comprehensive, the regularly studied outputs can be organized into the following categories:

1. Economic Growth and Development
2. Entrepreneurship, Innovation, and Business Dynamism
3. Comparative Political Economy and Varieties of Capitalism
4. Democracy, Political Institutions, and Governance
5. Capital Markets and Financial Development
6. Trade, Foreign Investment, and Globalization
7. Methodology, Measurement, and Index Construction

The empirical literature on the institutional determinants of long-run growth and development feature widespread use of cross-national indices. Acemoglu, Johnson, and Robinson’s landmark paper “The Colonial Origins of Comparative Development: An Empirical Investigation” (2001) used components of the PRS Group’s Political Risk Index (PRI) to measure institutional development as a determinant of cross-country variation in income per capita. They also ran tests against elements of both Fraser’s EFW and Heritage’s IEF indices.³¹ The research that grew out of this work

²⁵Bertelsmann Stiftung, ed., *Transformation Index BTI 2024: Governance in International Comparison* (Gütersloh: Verlag Bertelsmann Stiftung, 2024).

²⁶Economist Intelligence Unit, *Democracy Index 2024: What’s Wrong with Representative Democracy?* (London: The Economist Intelligence Unit, 2025), <https://www.eiu.com/n/campaigns/democracy-index-2024/>.

²⁷Fraser Institute, “Citations in Professional Literature of the Fraser Institute’s Economic Freedom Research,” <https://efotw.org/economic-freedom/citations>.

²⁸Ibid.

²⁹Joshua C. Hall and Robert A. Lawson, “Economic Freedom of the World: An Accounting of the Literature,” *Contemporary Economic Policy* 32, no. 1 (2014): 1–19.

³⁰V-Dem Institute, *Democracy Report 2025* (Gothenburg: University of Gothenburg, 2025); V-Dem Institute, “V-Dem in Numbers 2024,” February 25, 2025, <https://www.v-dem.net/news/v-dem-in-numbers-2024/>.

³¹Daron Acemoglu, Simon Johnson, and James A. Robinson, “The Colonial Origins of Comparative Development: An Empirical Investigation,” *American Economic Review* 91, no. 5 (December 2001): 1369–1401.

culminated in *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*, which placed institutional measurement at the center of development economics.³² Similarly, Hall and Jones’s QJE paper “Why Do Some Countries Produce So Much More Output per Worker than Others?” (1999) featured a bespoke composite metric of social infrastructure.³³ While they did not use the EFW or IEF in their models, Hall and Jones demonstrated that institution-based indices could help scholars understand cross-country variation in a wide array of output metrics. Rodrik, Subramanian, and Trebbi’s “Institutions Rule: The Primacy of Institutions over Geography and Integration in Economic Development” (2004) likewise demonstrated the importance of comprehensive and rigorous cross-national indices to operationalize the examination and analysis of institutional quality.³⁴

In fact, many studies in the growth and development literature specifically used Fraser’s EFW and Heritage’s IEF as proxies for market-supporting institutions. Easton and Walker (1997) demonstrate that EFW scores correlate with income levels.³⁵ De Haan and Sturm’s dual studies (2000 and 2001) qualified the claim, finding that change in economic freedom, rather than levels, hold a robust relationship with growth.³⁶ Meta-analyses by Doucouliagos and Ulubasoglu (2006 and 2008) further demonstrated the relationship between economic freedom, economic growth, and human capital.³⁷ Likewise, De Haan, Lundström, and Sturm’s “Market-Oriented Institutions and Policies and Economic Growth: A Critical Survey” (2006) surveyed 33 empirical studies including the EFW and IEF, concluding that market-oriented institutions exhibit a strong relationship to economic growth.³⁸

A parallel literature uses cross-national institutional indices to study entrepreneurship and new business formation. Djankov, La Porta, Lopez-de-Silanes, and Shleifer’s “The Regulation of Entry” (2002) served as the canonical precursor to the World Bank’s Doing Business indicator. This research used measurable regulatory burdens as institutional variables which overlap with economic freedom subcomponents concerning business regulation. They assembled new cross-country data on the procedures, time, and cost of starting a business in 85 countries and found that countries with heavier entry regulation have higher corruption and larger unofficial economies, but not higher quality public or private goods. Klapper, Laeven, and Rajan’s “Entry Regulation as a Barrier to Entrepreneurship” (2006) performed a similar feat, demonstrating that costly regulation hampers the creation of new firms.³⁹ Together these papers demonstrated the power and usefulness of cross-

³²Daron Acemoglu and James A. Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (New York: Crown Business, 2012).

³³Robert E. Hall and Charles I. Jones, “Why Do Some Countries Produce So Much More Output per Worker than Others?” *Quarterly Journal of Economics* 114, no. 1 (February 1999): 83–116.

³⁴Dani Rodrik, Arvind Subramanian, and Francesco Trebbi, “Institutions Rule: The Primacy of Institutions over Geography and Integration in Economic Development,” *Journal of Economic Growth* 9, no. 2 (June 2004): 131–165.

³⁵Stephen T. Easton and Michael A. Walker, “Income, Growth, and Economic Freedom,” *American Economic Review* 87, no. 2 (May 1997): 328–332.

³⁶Jakob de Haan and Jan-Egbert Sturm, “On the Relationship between Economic Freedom and Economic Growth,” *European Journal of Political Economy* 16, no. 2 (June 2000): 215–241; Jan-Egbert Sturm and Jakob de Haan, “How Robust Is the Relationship between Economic Freedom and Economic Growth?” *Applied Economics* 33, no. 7 (2001): 839–844.

³⁷Hristos Doucouliagos and Mehmet Ali Ulubasoglu, “Economic Freedom and Economic Growth: Does Specification Make a Difference?” *European Journal of Political Economy* 22, no. 1 (March 2006): 60–81.; Hristos Doucouliagos and Mehmet Ali Ulubasoglu, “Democracy and Economic Growth: A Meta-Analysis,” *American Journal of Political Science* 52, no. 1 (January 2008): 61–83.

³⁸Jakob de Haan, Susanna Lundström, and Jan-Egbert Sturm, “Market-Oriented Institutions and Policies and Economic Growth: A Critical Survey,” *Journal of Economic Surveys* 20, no. 2 (April 2006): 157–191.

³⁹Simeon Djankov, Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer, “The Regulation of Entry,” *Quarterly Journal of Economics* 117, no. 1 (February 2002): 1–37.; Leora Klapper, Luc Laeven, and Raghuram Rajan, “Entry Regulation as a Barrier to Entrepreneurship,” *Journal of Financial Economics* 82, no. 3 (December

nation composite indices in metricizing developmental forces. In so doing, they spawned not only the World Bank’s Doing Business indicators but also a cross-national comparative literature on institutional factors and new business formation and growth.

The new literature on firm creation has made extensive use of composite, cross-national indices. McMullen, Bagby, and Palich’s “Economic Freedom and the Motivation to Engage in Entrepreneurial Action” (2008) connects Fraser’s EFW to differential rates of opportunity-motivated versus necessity-motivated entrepreneurship across countries.⁴⁰ Bjørnskov and Foss’s “Economic Freedom and Entrepreneurial Activity: Some Cross-Country Evidence” (2008) and subsequent papers use EFW data to distinguish the role of small government from the role of legal infrastructure in explaining cross-national variation in startup activity.⁴¹ Aidis, Estrin, and Mickiewicz (2008, 2012) and Estrin and Mickiewicz (2011) use Heritage’s IEF and Polity IV alongside Global Entrepreneurship Monitor (GEM) data to examine institutional entrepreneurship across geographies. In the following years, Estrin, Korosteleva, and Mickiewicz (2013) and Estrin, Mickiewicz, and Stephan (2013) established further relationships between institutional measures and new business growth.⁴² Acs, Estrin, Mickiewicz, and Szerb (2018) use multiple composite indices including GEM to further integrate the entrepreneurship and growth literatures.⁴³

Where the growth and entrepreneurship literatures treat institutional measures as continuous regressors, the comparative political economy tradition typically uses them to classify types of capitalism. The foundational text is Hall and Soskice’s *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (2001). Hall and Soskice distinguished “liberal market economies” from “coordinated market economies” along dimensions of corporate governance, industrial relations, education and training, inter-firm relations, and employer–employee relations. Hall and Soskice argued that complementarities among these spheres produce distinctive patterns of comparative advantage.⁴⁴ Amable (2003) broadened their typology using factor analysis on institutional indicators across labor markets, finance, social protection, education, and product markets. Hall and Gingerich (2009) operationalized an empirical model of the Hall and Soskice typology, demonstrating the association of institutional complementarities with superior economic performance.⁴⁵ Streeck and Thelen (2005) and Schneider (2009) added new geographic and theoretical complexity to the scholarly understanding of institutional diversity and change.⁴⁶ While

2006): 591–629.

⁴⁰Jeffery S. McMullen, D. Ray Bagby, and Leslie E. Palich, “Economic Freedom and the Motivation to Engage in Entrepreneurial Action,” *Entrepreneurship Theory and Practice* 32, no. 5 (2008): 875–895.

⁴¹Christian Bjørnskov and Nicolai J. Foss, “Economic Freedom and Entrepreneurial Activity: Some Cross-Country Evidence,” *Public Choice* 134, no. 3–4 (March 2008): 307–328.

⁴²Ruta Aidis, Saul Estrin, and Tomasz Mickiewicz, “Institutions and Entrepreneurship Development in Russia: A Comparative Perspective,” *Journal of Business Venturing* 23, no. 6 (November 2008): 656–672; Saul Estrin and Tomasz Mickiewicz, “Institutions and Female Entrepreneurship,” *Small Business Economics* 37, no. 4 (2011): 397–415; Ruta Aidis, Saul Estrin, and Tomasz Mickiewicz, “Size Matters: Entrepreneurial Entry and Government,” *Small Business Economics* 39, no. 1 (2012): 119–139; Saul Estrin, Julia Korosteleva, and Tomasz Mickiewicz, “Which Institutions Encourage Entrepreneurial Growth Aspirations?” *Journal of Business Venturing* 28, no. 4 (2013): 564–580; Saul Estrin, Tomasz Mickiewicz, and Ute Stephan, “Entrepreneurship, Social Capital, and Institutions: Social and Commercial Entrepreneurship across Nations,” *Entrepreneurship Theory and Practice* 37, no. 3 (2013): 479–504.

⁴³Zoltán J. Acs, Saul Estrin, Tomasz Mickiewicz, and László Szerb, “Entrepreneurship, Institutional Economics, and Economic Growth: An Ecosystem Perspective,” *Small Business Economics* 51, no. 2 (2018): 501–514.

⁴⁴Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001).

⁴⁵Peter A. Hall and Daniel W. Gingerich, “Varieties of Capitalism and Institutional Complementarities in the Political Economy,” *British Journal of Political Science* 39, no. 3 (2009): 449–482; Bruno Amable, *The Diversity of Modern Capitalism* (Oxford: Oxford University Press, 2003).

⁴⁶Wolfgang Streeck and Kathleen Thelen, eds., *Beyond Continuity: Institutional Change in Advanced Political*

this literature tends to rely less on the Fraser EFW and Heritage IEF indices, it demonstrates the importance of, and need for, new, rigorous, and wide-ranging institutional indices.

The democracy and governance literature features strident empirical critiques of existing indices and, as a result, has produced significant improvements to index creation methods. Munck and Verkuilen (2002) provided the canonical critique, showing that even the most carefully constructed democracy indices suffer significant weaknesses during conceptualization, measurement, and aggregation. As such, they called for composite indices that combine methodological precision, rigor, and scope, a call we believe we have answered with the GCI.⁴⁷ Treier and Jackman (2008) underscore the value and difficulty in producing accurate composite indices, demonstrating that even the most respected indices miss the target when accounting for—in the rare cases that they calculate it—measurement error.⁴⁸ Pemstein, Meserve, and Melton (2010) attempted to answer that call by producing a Bayesian latent-variable approach to synthesize 10 extant scales into the Unified Democracy Scores (UDS), with the explicit goal of producing measurement that is reliable while supplying quantitative estimates of uncertainty.⁴⁹ The UDS has faced mixed reviews.

A new literature dedicated to correcting the failings of its forebearers has yielded significant advances in the measurement methodologies necessary to capture complex institutional systems in a single metric. Cheibub, Gandhi, and Vreeland’s “Democracy and Dictatorship Revisited” (2010) made the case that existing measures of democracy are not interchangeable and that the choice of measure should be guided by theoretical and empirical underpinnings. Soon thereafter, Coppedge et al. (2011) laid the conceptual foundation for V-Dem by proposing a historical, multidimensional, disaggregated, and transparent approach.⁵⁰ The subsequent V-Dem methodological publications, including Pemstein, Marquardt, Tzelgov, Wang, and Krusell’s (2015) “The V-Dem Measurement Model” working paper, have now become the field’s methodological paragon.⁵¹ Kaufmann, Kraay, and Mastruzzi (2010, 2011) became the methodological reference for the WGI by documenting the unobserved components model that aggregates several hundred underlying variables from public, private, and NGO data sources into six dimensions of governance. This model covered more than 200 countries from 1996 to the present while reporting margins of error for each country-year.

The cross-national institutional turn in the finance literature began with La Porta, Lopez-de-Silanes, Shleifer, and Vishny’s landmark papers “Legal Determinants of External Finance” (1997) and “Law and Finance” (1998). These papers examined legal rules covering shareholder and creditor protection, the origin of those rules, and the quality of enforcement in 49 countries. They show that common law countries provide the strongest protections and civil law countries the weakest,

Economies (Oxford: Oxford University Press, 2005); Ben Ross Schneider, “Hierarchical Market Economies and Varieties of Capitalism in Latin America,” *Journal of Latin American Studies* 41, no. 3 (2009): 553–575.

⁴⁷Gerardo L. Munck and Jay Verkuilen, “Conceptualizing and Measuring Democracy: Evaluating Alternative Indices,” *Comparative Political Studies* 35, no. 1 (February 2002): 5–34.

⁴⁸Shawn Treier and Simon Jackman, “Democracy as a Latent Variable,” *American Journal of Political Science* 52, no. 1 (2008): 201–217.

⁴⁹Daniel Pemstein, Stephen A. Meserve, and James Melton, “Democratic Compromise: A Latent Variable Analysis of Ten Measures of Regime Type,” *Political Analysis* 18, no. 4 (2010): 426–449.

⁵⁰José Antonio Cheibub, Jennifer Gandhi, and James Raymond Vreeland, “Democracy and Dictatorship Revisited,” *Public Choice* 143, no. 1 (April 2010): 67–101; Michael Coppedge, John Gerring, David Altman, Michael Bernhard, Steven Fish, Allen Hicken, Matthew Kroenig, Staffan I. Lindberg, Kelly McMann, Pamela Paxton, Holli A. Semetko, Svend-Erik Skaaning, Jeffrey Staton, and Jan Teorell, “Conceptualizing and Measuring Democracy: A New Approach,” *Perspectives on Politics* 9, no. 2 (June 2011): 247–267.

⁵¹Daniel Pemstein, Kyle L. Marquardt, Eitan Tzelgov, Yi-ting Wang, and Joshua Krusell, “The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data,” V-Dem Working Paper No. 21 (University of Gothenburg, 2015 and subsequent updates).

with concentration of ownership negatively related to investor protections.⁵² La Porta, Lopez-de-Silanes, Shleifer, and Vishny (1999) and Djankov, La Porta, Lopez-de-Silanes, and Shleifer (2003, 2004, and 2008) extended the analytical program across regulatory areas with similar results.⁵³ Beck, Demirgüç-Kunt, and Levine (2003) and their companion contributions on financial structure, development, and access to finance anchored the World Bank’s Financial Development and Structure dataset.⁵⁴ Put simply, this literature demonstrated a deep and systematic relationship between political and legal institutions and financial performance, thus underscoring the importance of systematizing metrics that capture and explain the interaction of these interdisciplinary variables.

The empirical trade literature also relies on a small set of cross-national composite indices. Sachs and Warner (1995) introduced a binary openness measure based on tariff levels, non-tariff barriers, black-market exchange-rate premia, and state monopoly of exports.⁵⁵ Wacziarg and Welch (2008) updated the Sachs-Warner classification to 2001 and reported that trade liberalization is associated with average annual growth rates 1.5 percentage points higher in the post-liberalization period.⁵⁶ Globerman and Shapiro (2002) used the WGI to show that governance quality is a robust determinant of foreign direct investment inflows and outflows.⁵⁷ In study after study, scholars have used composite cross national indices to provide multidimensional analysis that models real world interactions across space and time.

Lastly, a methodological literature featuring the EFW, IEF, and other similar measures has arisen specifically to examine the viability of using economic freedom as a proxy for capitalism. Cohen’s “Is ‘Economic Freedom’ the Same as Free-Market Capitalism? A Decompositional Analysis of the Economic Freedom of the World Index” (2020) rightly argued that Fraser’s EFW is not strictly a measure of market capitalism but rather “an amalgam of at least two distinct though related concepts: economic liberalism and good governance.” This conceptual conflation, Cohen argues, creates serious interpretive problems for empirical analysts using the EFW to evaluate laissez-faire policies.⁵⁸ Likewise, Berggren (2003) and De Haan, Lundström, and Sturm (2006) both note that the EFW conflates the size of government with the quality of legal institutions.⁵⁹ The OECD and Joint Research Centre’s Handbook on Constructing Composite Indicators: Methodology and User Guide (2008) catalogs the tradeoffs in normalization, weighting, and aggregation choices.

⁵²Rafael La Porta, Florencio Lopez-de-Silanes, Andrei Shleifer, and Robert W. Vishny, “Legal Determinants of External Finance,” *Journal of Finance* 52, no. 3 (July 1997): 1131–1150; La Porta et al., “Law and Finance,” *Journal of Political Economy* 106, no. 6 (1998): 1113–1155.

⁵³Rafael La Porta, Florencio Lopez-de-Silanes, Andrei Shleifer, and Robert W. Vishny, “The Quality of Government,” *Journal of Law, Economics, and Organization* 15, no. 1 (1999): 222–279; Simeon Djankov, Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer, “Courts,” *Quarterly Journal of Economics* 118, no. 2 (May 2003): 453–517; “The Regulation of Labor,” *Quarterly Journal of Economics* 119, no. 4 (November 2004): 1339–1382; Rafael La Porta, Florencio Lopez-de-Silanes, and Andrei Shleifer, “The Economic Consequences of Legal Origins,” *Journal of Economic Literature* 46, no. 2 (June 2008): 285–332.

⁵⁴Thorsten Beck, Asli Demirgüç-Kunt, and Ross Levine, “Law and Finance: Why Does Legal Origin Matter?” *Journal of Comparative Economics* 31, no. 4 (December 2003): 653–675.

⁵⁵Jeffrey D. Sachs and Andrew Warner, “Economic Reform and the Process of Global Integration,” *Brookings Papers on Economic Activity* 1995, no. 1: 1–118.

⁵⁶Romain Wacziarg and Karen Horn Welch, “Trade Liberalization and Growth: New Evidence,” *World Bank Economic Review* 22, no. 2 (2008): 187–231.

⁵⁷Steven Globerman and Daniel Shapiro, “Global Foreign Direct Investment Flows: The Role of Governance Infrastructure,” *World Development* 30, no. 11 (November 2002): 1899–1919.

⁵⁸Joseph N. Cohen, “Is ‘Economic Freedom’ the Same as Free-Market Capitalism? A Decompositional Analysis of the *Economic Freedom of the World Index*,” *Journal of Institutional Economics* 16, no. 6 (2020): 713–730.

⁵⁹Niclas Berggren, “The Benefits of Economic Freedom: A Survey,” *Independent Review* 8, no. 2 (Fall 2003): 193–211.

Furthermore, the Handbook warns that simple average composite indices with equal weights are vulnerable to substitutability across dimensions and to weak validity when underlying components are highly correlated.⁶⁰

The democracy-centric Pemstein–Meserve–Melton (2010) and Treier–Jackman (2008) models featuring Bayesian latent variable approaches, alongside the gold standard V-Dem measurement model (Pemstein et al. 2015), pose a stark contrast to existing economic freedom-capitalism indices. Until now, no measure of market economic systems, including the EFW and IEF, have taken a similar, methodologically-rigorous approach to index creation.

Improvement on Existing Indices

The Global Capitalism Index (GCI) extends the widely respected methodological approaches cataloged above to the measure of capitalist economies. The GCI provides a theoretically and empirically sound metric across 8 individual subindices of capitalism—plus a ninth composite capitalism score—and in so doing addresses the methodological limitations of the Fraser EFW and Heritage IEF indices. We contend that a gold standard capitalism index must establish a comprehensive theoretical model of capitalism, measure it with geographic and temporal completeness, and do so with absolute methodological transparency and statistical stability. The GCI provides a comprehensive global dataset defined by transparency and empirical rigor that we hope will anchor analyses of capitalist economies for generations.

The EFW and IEF both operationalize *economic freedom* in a way that Cohen (2020) demonstrates conflates a libertarian commitment to limited government with separable institutional features such as the rule of law, contract enforcement, and the quality of regulation. This conflation is not a minor measurement nuisance. It integrates the theoretical claim that smaller government equals more capitalism directly into the measurement instrument. As a result, these indices do not measure a comprehensive model of capitalism, but rather adherence to a specific political libertarianism.

It is vital to note that both the EFW and IEF indices self-identify as metrics of “economic freedom,” not capitalism. The issue lies with the broader conflation of economic freedom indices with measurements of capitalism in both the scholarly literature and public media. This conflation is not the fault of the creators of EFW or IEF—they do not claim that their indices or anything other than measures of “economic freedom.” The problem lies in the lack of a rigorous and comprehensive metric of capitalism itself. Anecdotal evidence collected by the authors of this paper suggest that scholars understand the differences between metrics of economic freedom and capitalism, but they use the EFW or IEF as a proxy because no alternative exists.⁶¹

We do not claim that the EFW or IEF are illegitimate products, but rather that they simply measure a different concept and should be treated as such. Following Douglass North’s (1990) institutional economics and Karl Polanyi’s (1944) demonstration that markets are institutional constructs requiring energetic, but limited, state support, we theorize and measure capitalism as an institutional system, not the simple absence of the state.⁶² Including outputs as inputs makes the

⁶⁰OECD and Joint Research Centre, *Handbook on Constructing Composite Indicators: Methodology and User Guide* (Paris: OECD Publishing, 2008).

⁶¹In just one example, in the volume edited by one of the authors of this paper entitled, *Can Democracy and Capitalism Be Reconciled?* (Oxford University Press, 2025), no fewer than three published and 2 other early draft chapters used the Fraser EFW index as a proxy for capitalism despite its clear labeling as a metric of “economic freedom.”

⁶²Karl Polanyi, *The Great Transformation: The Political and Economic Origins of Our Time* (Boston: Beacon Press, 1944).

indices definitionally correlated with macroeconomic outcomes, biasing growth regressions toward a positive coefficient on “economic freedom” regardless of any underlying causal relationship.

The GCI separates institutional inputs (rules, regulations, market structures) from derivative outputs (GDP, inequality, innovation, etc.), enabling empirical analyses to test for the latter with less worry of circularity or endogeneity. The GCI’s subindex-centric architecture also separates institutional inputs from systemic outputs against which the subindex or broader composite index can be externally validated. In doing so, the GCI is the first major cross-national index to seriously measure the Hall and Soskice insight that varieties of capitalism rest in their institutional configurations and are not reducible to a single *laissez-faire* dimension.

While we contest the theoretical justification for using economic freedom indices as proxies for capitalism, the EFW and IEF also feature a series of methodological problems that justify a reconsideration of their broader usage. The EFW and IEF construct their scores using simple averages across collections of datasets and normative judgments about relative weightings of subindices. The EFW takes the simple average of 45 datasets to derive 25 components, simple averages of those components within each of five areas, and then another simple average to aggregate the five areas into a composite topline score. The EFW creators explicitly state that “theory provides us with some direction about elements that should be included in the five areas and the summary index, but it does not indicate what weights should be attached to the components within the areas or among the areas in the construction of the summary index.”⁶³ They also note that they “investigated several methods of weighting the various components, including principal component analysis and a survey of economists,” but conclude that “the summary index is not very sensitive to alternative weighting methods” and therefore “make no attempt to weight the components in any special way.”⁶⁴ This insensitivity finding is itself a substantive empirical claim. If PCA produces nearly identical results to their equal weighting methodology, it is because the EFW’s components are heavily intercorrelated, which is precisely the justification for using dimensionality reduction methods that extract the common institutional signal while reporting the level of variation contributed by each input.

The creators of the IEF take a similar approach, but instead of conducting three layers of simple averaging they execute one simple average of 12 components to derive the final composite score. IEF creators score each of the 12 components 0–100, “weighted equally so that the overall score will not be biased in favor of any one component or policy direction.”⁶⁵ Equal weighting of components is not neutral, but rather represents a normative choice to equally distribute importance across domains. The IEF methodology acknowledges that “12 economic freedoms obviously interact, but the exact mechanisms of this interaction are not clearly definable.” Rather than conducting empirical tests to determine minimum thresholds, levels of contribution, or dependent, complementary, or supplementary relationships between the elements of their model, the IEF team concludes that “these questions, while valid, are beyond the scope of our fundamental mission” and asks readers to “study, weight, and integrate [the data] as they see fit.”⁶⁶ In contrast, we believe that these questions are imperative. We specifically design our weighting scheme to determine and understand these relationships.

⁶³James Gwartney, Robert Lawson, Joshua Hall, Ryan Murphy, et al., *Economic Freedom of the World: 2023 Annual Report* (Vancouver: Fraser Institute, 2023), 7. The methodological text is substantively unchanged across recent editions.

⁶⁴Gwartney et al., *Economic Freedom of the World: 2023 Annual Report*, 7.

⁶⁵Heritage Foundation, “Methodology,” in *2026 Index of Economic Freedom* (Washington, DC: The Heritage Foundation, 2026), 404.

⁶⁶Heritage Foundation, “Methodology,” 405.

Data normalization and harmonization is just as important as component integration, but both the EFW and IEF deploy erratic and theoretically unjustified modes of doing so. The IEF uses a wide variety of normalization formulas across its twelve components. However, it documents no theoretical justification for why these functional forms differ across components, why the α coefficients take the values they do (the IEF describes $\alpha = 0.03$ simply as “a coefficient to control for variation among scores”), or why Government Spending’s quadratic functional form treats “zero government spending as the benchmark,” despite their acknowledgment that this function can create “artificially high scores for underdeveloped countries.”⁶⁷

The EFW’s normalization is more uniform across components with most using a min-max transformation to a 0-to-10 scale. However, the relevant $V_{\max}$ and $V_{\min}$ values likely vary meaningfully across the index. Many EFW components anchor their bounds to 1990 reference data, meaning the EFW normalizes country scores in 2024 against a fundamentally different global distribution. Several components use bespoke categorical scoring (the top marginal tax rate, for example, is scored via a 12×4 matrix mapping tax-rate brackets against income threshold brackets) or hard cutoffs (money growth above 50% receives a score of zero regardless of other conditions).⁶⁸ This inconsistent method has the potential to produce significant measurement error, which also cannot be tracked due to a lack of documentation.

Furthermore, neither the EFW nor the IEF teams publish tests of endogeneity, multicollinearity, or other statistical maladies in their flagship indices. They do not provide documentation on the collection of the raw data that comprise these indices, nor do they publish replicable data on their normalization, harmonization, or scaling processes. So far as can be seen, their methods have not been validated or tested by external scholars or reviewers. This methodological opacity precludes evaluation of theoretical precision and compromises validity testing and inference assessment. Also, neither the EFW or IEF include uncertainty estimates at the country-year level. This combination of methodological flaws result in the problems outlined in Table 2 (below).

It is vital that we make clear the real consequences of arbitrary weighting in composite indices. The use of arbitrary weights, including simple averages, in index construction means that index composition relies on prior assumptions, expert opinion, or convenience as opposed to empirical examination of the available evidence.⁶⁹ In contrast, the GCI combines hundreds of observable variables to capture the relationships between the latent structures and the relationships between all items included in our model, without relying on arbitrary weightings.⁷⁰ Assigning arbitrary weights to variables in latent factor construction introduces significant bias and can undermine the validity

⁶⁷The IEF deploys a standard min-max transformation for Property Rights, Judicial Effectiveness, Government Integrity, and Business Freedom; a quadratic penalty function $\text{Score} = 100 - \alpha(x)^2$ for Tax Burden and Government Spending (with $\alpha = 0.03$), and again for Fiscal Health but with different α values of 2 for deficit and 0.01 for debt; a relative-to-world-average formula $\text{Score} = 50 \times (\bar{x}/x_i)$ for seven of the nine Labor Freedom sub-parts, with binary 0/100 coding for the remaining two; a square-root inflation penalty plus a subjective 0–20-point price-control deduction for Monetary Freedom; a min-max plus qualitative NTB deduction for Trade Freedom; and purely categorical deduction-from-100 rubrics for Investment Freedom (with pre-defined penalties of 5, 10, 15, 20, or 25 points per restriction type) and Financial Freedom (an eleven-tier qualitative scale from 0 to 100 in increments of 10, with each tier corresponding to a verbal description of state interference); Heritage Foundation, “Methodology,” 395–404, and 397.

⁶⁸Gwartney et al., *Economic Freedom of the World: 2023 Annual Report*, 249–262.

⁶⁹See Becker, William, Michaela Saisana, Paolo Paruolo, and Ine Vandecasteele. 2017. “Weights and Importance in Composite Indicators: Closing the Gap.” *Ecological Indicators* 80: 12–22; Saisana, M., A. Saltelli, and S. Tarantola. 2005. “Uncertainty and Sensitivity Analysis Techniques as Tools for the Quality Assessment of Composite Indicators.” *Journal of the Royal Statistical Society: Series A* 168 (2): 307–323; Saisana, M., B. d’Hombres, and A. Saltelli. 2011. “Ricky Numbers: Volatility of University Rankings and Policy Implications.” *Research Policy* 40 (1): 165–177.

⁷⁰For a summary of our index construction and PCA process, see the Compositional Methodology section.

Issue		Description
Reduced Validity		The index may fail to accurately capture the underlying construct, leading to invalid inferences.
Distorted Rankings		Countries or units may be misclassified or ranked incorrectly, potentially affecting policy recommendations or resource allocation.
Lack of Transparency		Subjective weighting choices are difficult to justify and replicate, reducing the credibility of the index.
Sensitivity to Subjectivity		Results may be highly sensitive to researcher preferences rather than grounded in empirical relationships.
Amplified Error	Estimation	Subjective weights can increase estimation variance, making the index less reliable and more sensitive to noise in the data.

Table 2: Risks of Subjective Weighting in Composite Indices

of the resulting index. Consider an index constructed from n observed variables $x_1, x_2, \dots, x_n$ where each is assigned an arbitrary weight $w_1, w_2, \dots, w_n$:

$$I = \sum_{i=1}^n w_i x_i.$$

If any w_i does not accurately represent the true relationships between the observed variables and the underlying construct η , the resulting index I will diverge from η and cause bias. Instead, the latent construct should be represented using weights that reflect true factor loadings λ_i :

$$\eta = \sum_{i=1}^n \lambda_i x_i.$$

The bias introduced by using arbitrary weights can be measured as:

$$\text{Bias} = \mathbb{E}[I - \eta] = \sum_{i=1}^n (w_i - \lambda_i) \mathbb{E}[x_i].$$

Unless the assigned weights exactly match the true factor loadings ($w_i = \lambda_i$ for all i), the constructed index will fail to represent the latent construct correctly, leading to distortions in interpretation and analysis. Beyond bias, arbitrary weighting can also amplify estimation errors. A well-constructed index should minimize the variance of the estimation error, defined as $\epsilon = I - \eta$. The variance of this error is given by:

$$\text{Var}(\epsilon) = \text{Var}(I) + \text{Var}(\eta) - 2 \text{Cov}(I, \eta).$$

By substituting $I = \sum_{i=1}^n w_i x_i$ and $\eta = \sum_{i=1}^n \lambda_i x_i$, we obtain:

$$\text{Var}(\epsilon) = \sum_{i=1}^n (w_i - \lambda_i)^2 \text{Var}(x_i) + 2 \sum_{i < j} (w_i - \lambda_i)(w_j - \lambda_j) \text{Cov}(x_i, x_j).$$

The weights should match closely with the real factor loadings ($w_i = \lambda_i$) to keep the error variance manageable. The differences reveal themselves as additional variance and weaken the reliability

of the model. By extension, arbitrary weighting guesses how the weights may fit the loadings and necessarily distorts the data frame’s covariance matrix Σ . Assume that σ_{ij} is the covariance between x_i and x_j . The covariance between the index I and item x_k is given by:

$$\text{Cov}(I, x_k) = \sum_{i=1}^n w_i \sigma_{ik}.$$

The estimated covariance $\text{Cov}(I, x_k)$ may not reflect the true relationships between x_k and η if the weights are used arbitrarily, resulting in inaccurate estimations of the relationships between variables.

The consequences of arbitrary weighting manifest in multiple ways. The index may not accurately reflect the construct it is intended to measure, leading to invalid inferences and reduced validity. Countries or units may be misclassified or ranked incorrectly, affecting policy recommendations or resource allocation. Arbitrary choices are difficult to justify and replicate, reducing the credibility of the index and creating a lack of transparency. Results may be highly sensitive to the subjective preferences of the researcher rather than grounded in empirical relationships. Finally, arbitrary weights can increase the variance of the estimation error, making the index less reliable and more sensitive to noise in the data. Put simply, arbitrary weighting in index construction can introduce bias, inflate error, and undermine the interpretability and usefulness of the resulting measure. Fortunately, measurement methods such as PCA offer an alternative by deriving weights from empirical observations of relationships among variables.

The theoretical and methodological problems described above result have significant consequences for many veins of scholarly inquiry. The widespread use of indices with measurement bias and/or methodological imprecision undermines the accuracy and interpretability of resulting scholarly work. However, the lack of rigorous alternatives force graduate students and Nobel laureates alike to rely on the EFW and IEF or abandon their inquiries. While one could argue that flawed studies are better than nothing, the use of these indices embed a distinct measurement bias and ideologically fraught understanding of capitalism into empirical work. Furthermore, the dominance of the EFW and IEF perpetuate a false sense of measurement consensus. Since disparate analyses use the same indices as central elements of their models, consistent findings may reflect shared measurement error rather than convergence within and across disciplines.

The Global Capitalism Index provides a significant improvement over EFW and IEF in measuring the performance of, and adherence to, capitalist principles. The GCI’s model of capitalism, as opposed to “economic freedom,” provides a more accurate conception of the GCI’s component elements and theoretical scaffolding. Furthermore, the GCI’s quantitative approach provides a more accurate capture of the underlying variation in the data, and thus more accurate scoring across the 8 subindices and the topline capitalism metric. We would expect some natural overlap with EFW and IEF due to shared concepts and datasets. When we plot the GCI against them, we find a correlation of the GCI to EFW of 0.807 and GCI to IEF of 0.826. The correlation between the two economic freedom indices is higher at 0.873. These correlations illustrate that the Global Capitalism Index captures a significant level of differentiation from these existing indices while maintaining an expected degree of similarity.

The GCI’s correlations to EFW (0.807) and IEF (0.826) demonstrate an integral point: economic freedom and capitalism are related but not synonymous.⁷¹ The qualitative and quantitative conflation of economic freedom with capitalism can and has biased academic and policy research towards

⁷¹These correlations are detailed further in the Results section.

laissez-faire conceptions of capitalism. It has also disguised the broader institutional frameworks needed to sustain free markets. We agree that state involvement in a capitalist system, past a certain point, is fundamentally deleterious to capitalism and the immense benefits that come with it. Nevertheless, we reject a model of capitalism in which the relationship between capitalism and government activity is linear and downward sloping (Figure 1, below). Rather, we contend that the state has a role in preserving the institutions that capitalism requires. Likewise, we contend that free and competitive markets are vital for capitalist economies to operate and that, in many cases throughout history, the suppression of free and competitive markets has come from private forces just as much as public ones.⁷² The GCI’s model of the relationship between government and capitalism therefore resembles an inverted-parabolic curve (Figure 2): economies can become more capitalist as they move away from minimal government activity—up to a moderate point of what Alexander Hamilton described as “limited but energetic government”—past which further state involvement can erode market functioning.⁷³

We use this non-linear relationship between government and capitalism as a theoretical construct throughout the index. Under a strictly linear understanding, certain domains would appear to work against one another. Instead, we group them together within the same subindex to capture the balance. For example, labor market flexibility and labor protections are often treated as competing objectives. However, we see them as essential complements to a functioning labor market. For a small number of datasets, we operationalize the non-linear framing quantitatively during scaling.⁷⁴

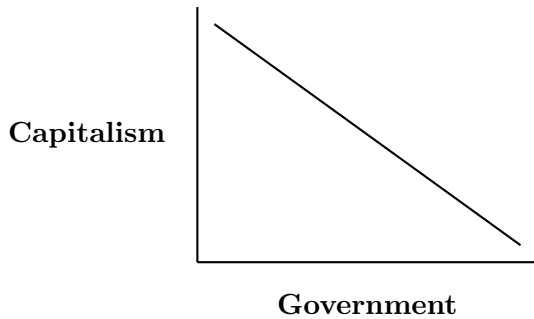


Figure 1: “Economic Freedom” Index Model

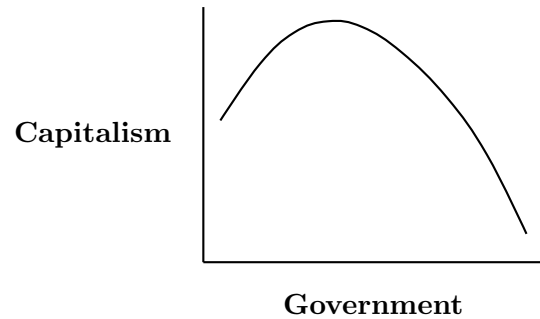


Figure 2: Global Capitalism Index Model

Theory

Capitalism is a system that privileges the ability to deploy capital in open markets as freely as possible, while minimizing constraints arising from both public and private sources. An emphasis on the systemic nature of capitalism is vital—capitalism does not arise naturally, but is rather the

⁷²Martin Wolf, *The Crisis of Democratic Capitalism* (New York: Penguin Press, 2023).

⁷³Alexander Hamilton, Report on Manufactures (Philadelphia: U.S. Government Printing Office, 1791), reprinted in *The Papers of Alexander Hamilton*, ed. Harold C. Syrett, vol. 10 (New York: Columbia University Press, 1966), 230–371.

⁷⁴For two datasets in the Strength of Property Rights and Private Ownership subindex, we employ a parabolic maximization approach where a more central value yields a score of 10 and deviations away from that score (to the positive or negative direction) receive lower scores. To properly designate the central value for a given dataset, we conducted an extensive literature review of the varieties of capitalist models and chose a collection of countries that represent the breadth of possible systems. We then take that list of countries, gather their average scores for each year of a given data series, and compute the simple average of all those values so that each model receives equal weight. That aggregate average per year is then treated as the central value for the dataset during 0–10 scaling. Please see Appendix C for more details of this process.

intentional compilation of institutions that interlock and reinforce each other. Well-functioning institutions guard markets from uncertainty, opportunism, and rent extraction, while providing the stability and infrastructure that markets require. Adam Smith rightly argued that the “invisible hand” of individuals pursuing their own interests can promote profound economic and societal gains. However, Smith also understood that capitalism could not function without effective institutions to protect against monopolies, price fixing, gatekeeping, and cronyism, all of which could come from powerful governments *and* firms alike.⁷⁵ Smith railed against both the English crown and the private East India Company, warning that both public and private actors could and would suppress competition, constrain economic initiative, and undermine capitalist dynamism. As such, our theory of capitalism balances ‘freedom from’ undue interference and ‘freedom to’ undertake activity at both the public and private level.⁷⁶

Smithian institutions create and sustain an ecosystem in which market participants have equal opportunity to deploy their financial, physical, or human capital with minimal constraint. While traditional definitions of capitalism tend to favor financial or physical capital, we maintain that all three forms of capital hold equal *ex ante* value. Open markets serve as the channels for this deployment of capital and the receipt of information via relatively unconstrained prices. The “relatively unconstrained” qualifier (or “as freely as possible” in our broader definition) is vital. A system can be capitalist even if “the ability to deploy capital in open markets” is not absolute. The question is about degrees—do prospective obstacles *prevent* (not just make harder in some circumstances) the ability to “deploy capital in open markets.”

Each subindex (described below) can stand alone as the best measures of their independent subjects currently available. When synthesized, the GCI’s eight subindices create as complete, complex, and nuanced a formulation of capitalist systems as currently exist.

Strength of Property Rights and Private Ownership

Property rights and the private ownership of the means of production in all its forms—physical, financial, human, and intellectual—are fundamental to the free deployment of capital. A lack of legal assurances of ownership and freedom from arbitrary impediments to capital deployment or accumulation introduce uncertainty as to whether owners will profit from the successful deployment of their capital. If owners doubt that they will benefit from their risks, they will less likely to take them. This does not mean that firms or individuals do not have inherited advantages that allow them to seize market share from competitors. It does, however, mean that owners of capital will not be forced to part with their physical or intellectual property without their consent.⁷⁷

Guarantee freedom from arbitrary or forced dispossession creates an environment in which those with capital will benefit from, and thus make, investments. Classical theory has long held property rights as a precondition for voluntary exchange. Adam Smith argued that property rights and security of ownership creates the necessary incentives for commercial venturing and capital deployment.⁷⁸ He asserted that “commerce and manufactures can seldom flourish long in any state which does not enjoy a regular administration of justice... and people do not feel secure in the possession

⁷⁵ Adam Smith, *The Wealth of Nations* (New York: Random House, 1965); Glory M. Liu, *Adam Smith’s America: How a Scottish Philosopher Became an Icon of American Capitalism* (Princeton, NJ: Princeton University Press, 2022).

⁷⁶ Joseph E. Stiglitz, *The Road to Freedom: Economics and the Good Society*. New York: W. W. Norton, 2024.

⁷⁷ Armen A. Alchian, “Some Economics of Property Rights,” *Il Politico* 30, no. 4 (1965): 816–829.

⁷⁸ Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

of their property.”⁷⁹

Smith’s logic presages the Schumpeterian concept of creative destruction. Schumpeter argued that economic dynamism demands the right of firms to “keep their production processes secret, have their trademarks protected...and obtain patents.”⁸⁰ Without guarantees of ownership, inventors will not experience the incentives that encourage venturing and risk-taking. Even more important, investors will not provide the capital to convert an innovation into an accessible product. John Locke famously wrote that people form governments to protect “the mutual preservation of their lives, liberties, and estates,” placing property rights as a core driver for political, social, and economic organization.⁸¹ Put simply, if new entrants do not believe that their intellectual and/or physical property is safe from seizure or theft, they will have neither the means nor desire to challenge and displace old, ossified companies.

Building on the classical fundamentals of property rights, modern scholars emphasize the importance of codifying these rights within institutions. Douglass North defined institutions as “humanly devised constraints” that “reduce uncertainty by establishing a stable structure to human interaction.”⁸² By lowering uncertainty and creating incentives, institutional property rights establish the “rules of the game” necessary for transforming possession into ownership, ownership into capital, and capital into market systems. Besley and Persson (2009, 2011) connect property rights to broader state capacity by demonstrating that the protection of property rights is inseparable from the fiscal and legal capacity of states to enforce contracts, collect revenue, and constrain rent extraction.⁸³ Acemoglu and Johnson (2005) distinguish property rights institutions, ones which protect individual rights from expropriation, from “contracting institutions,” which facilitate private transactions and agreements. They find that property rights institutions have a direct effect on long-run economic growth, investment, and financial development.⁸⁴ In further work, Acemoglu, Johnson and Robinson (2005) show that differences in property rights regimes can drive differences in per capita income. Inclusive institutions which broadly protect property rights and enforce the rule of law are essential for sustainable economic growth.⁸⁵

Empirical scholarship further extends the classical framework by identifying the specific mechanisms through which property rights affect economic activity. Field (2007) shows that titled households substantially increase work hours outside the home as the security of title freed labor previously committed to property defense for commercial activity.⁸⁶ Goldstein and Udry (2008) found that Ghanaian farmers facing insecure property rights forewent investments in soil fertility, particularly within groups without political connections.⁸⁷ Delineated and protected rights to property do not

⁷⁹Ibid, Book V, Chapter III, paragraph 7.

⁸⁰Joseph A. Schumpeter, *Capitalism, Socialism and Democracy* (New York: Harper & Brothers, 1942).

⁸¹John Locke, *Two Treatises of Government*, ed. Peter Laslett (Cambridge University Press, 1988 [1690]).

⁸²Douglass C. North, *Institutions, Institutional Change, and Economic Performance* (Cambridge: Cambridge University Press, 1990), 6.

⁸³Timothy Besley and Torsten Persson, “The Origins of State Capacity: Property Rights, Taxation, and Politics,” *American Economic Review* 99, no. 4 (September 2009): 1218–1244; Timothy Besley and Torsten Persson, *Pillars of Prosperity: The Political Economics of Development Clusters* (Princeton: Princeton University Press, 2011).

⁸⁴Daron Acemoglu & Simon Johnson, 2005. “Unbundling Institutions,” *Journal of Political Economy*, University of Chicago Press, vol. 113(5), pages 949–995, October.

⁸⁵Daron Acemoglu, Simon Johnson, and James A. Robinson. 2005. “Institutions as the Fundamental Cause of Long-Run Growth.” *Handbook of Economic Growth*, vol. 1A, 385–472.

⁸⁶Erica Field, “Entitled to Work: Urban Property Rights and Labor Supply in Peru,” *Quarterly Journal of Economics* 122, no. 4 (November 2007): 1561–1602.

⁸⁷Markus Goldstein and Christopher Udry, “The Profits of Power: Land Rights and Agricultural Investment in Ghana,” *Journal of Political Economy* 116, no. 6 (December 2008): 981–1022.

simply lead individuals to enhance their own holdings, but also incentivizes them to engage in economic activity that benefits the broader growth-facilitating ecosystem.

Property rights also serve as a vital barrier between the state and the market. Locke and other theorists maintained that property rights provide the legal infrastructure through which citizens and commercial actors combat state expropriation without cause.⁸⁸ Property rights have both formal and practical dimensions, and both are important considerations for accurate systemic measurement. Formal de jure institutions, laws, and courts, must exist for the execution of rights while also having the capacity for de facto enforcement and freedom from practical corruption.⁸⁹ Strong property rights also give individuals optionality when larger firms impinge on their ability to engage in market activity without constraint. While property rights regimes do not guarantee that smaller parties will regain their access to markets, empirical work shows that systems with strong property rights see greater competition and less arbitrary action by large private firms against their smaller peers.

The operative core of capitalist property rights is the capacity for, and practice of, private ownership of the means of production. Marxist theory distinguishes capitalism not just by the existence of property rights, but also by the vast majority of capital being owned by non-government entities. This private ownership thus disperses political, social, and economic interests, driving holders of capital toward competition as opposed to unified, coordinated action.⁹⁰ Public ownership, regardless of its nature, serves a distinctly different purpose than the capitalist aims of efficiency, productivity, and profit-making. State ownership often prioritizes components of distributional equity, social welfare, or national security as those in power see it. This difference in objectives does not mean that all public ownership is inherently anti-capitalist. Smith acknowledges the crucial role of the state in the provision of public goods that facilitate markets but are not profitable for private investors. Our Market-Supporting Policy subindex captures the effectiveness of states' provision of public goods that are "advantageous to a great society...of such a nature that the profit could never repay the expense."⁹¹ However, the capitalist impulse towards profit and competition dissipates when state control becomes an end in and of itself and transforms the economic logic away from allocative efficiency toward centralized command.⁹² For capitalist incentives to remain tangible and secure, market forces must remain the dominant economic force and publicly owned capital should support market expansion, integration, and activity.

The administrative commitment to recognize and enforce property rights requires a functioning rule of law and legal order upon which individuals, firms, and the state can claim or dispute legal rights of ownership. This subindex captures both the de jure legal strength of property rights and the de facto levels of private ownership and legal enforcement through a combination of 31 datasets. The domains of the Strength of Property Rights and Private Ownership subindex are:

1. Intellectual Property Protection
2. Lack of Corruption
3. State Stability

⁸⁸ John Locke, *Two Treatises of Government*, ed. Peter Laslett (Cambridge University Press, 1988 [1690]).

⁸⁹ Douglass C. North, *Institutions, Institutional Change, and Economic Performance* (Cambridge: Cambridge University Press, 1990).

⁹⁰ Karl Marx, *Capital: A Critique of Political Economy*, Volume I, trans. Fowkes (Penguin Classics, 1976 [1867]).

⁹¹ Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

⁹² Douglass C. North, *Institutions, Institutional Change, and Economic Performance* (Cambridge University Press, 1990).

4. Political Expression
5. Rule of Law
6. Contract Enforcement
7. Private Ownership

For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Market-Supporting Policy

A fundamental premise of the Global Capitalism Index is that well-functioning markets are not naturally occurring and cannot be perpetual without foundations laid by governing institutions. As a result, capitalism does not benefit from the absolute absence of government, but rather the alignment of state policy toward market creation, integration, operation, and expansion. While markets *can* function in the absence of market-supporting policy, the empirical literature demonstrates that those markets are less efficient, more volatile, and grow slower.⁹³

In classical articulations of the state of nature, market systems are not possible. Thomas Hobbes argued that “there is no place for Industry” in the state of nature “because the fruit thereof is uncertain; and consequently no Culture of the Earth; no Navigation, nor use of the commodities that may be imported by Sea...” As a result, Hobbes wrote, “there be no propriety, no dominion, no mine and thine distinct; but only that to be every man’s that he can get, and for so long as he can keep it.”⁹⁴

Markets—or property rights for that matter—cannot exist in this environment. Thus, governments must tame the state of nature to provide that environment in which property rights exist and goods and services can be exchanged. Since markets do not occur naturally, public and private actors must provide a broad architecture for markets to emerge and function. We designed this subindex to measure that architecture.

This subindex, and the GCI in general, distinguishes between government support for market structures and direct government action within them. Policies that provide connective infrastructure, common rules, stable and safe modes of exchange, and security for market participants all empower capitalist activity. However, states can use their profound extra-market advantages to become a direct player in, and fundamentally bias the market itself. This type of direct state involvement discourages private investment and competition, and often devolves towards price-setting or the allocation of capital toward non-productive, politically-driven ends. As a result, market logic begins to shift away from allocative efficiency towards political appeasement and rent-seeking behavior.

The difference between market-supporting policy and direct market involvement is a fine line, but a very important one. As Kathleen Thelen (2014) shows, the stability of capitalist systems depends on institutional arrangements that support both the autonomy of markets and the legitimacy of the state.⁹⁵ The ideal market-supporting policy exists in an equilibrium where the state is active

⁹³Dave Donaldson, “Railroads of the Raj: Estimating the Impact of Transportation Infrastructure,” *American Economic Review*, 108, no. 4–5 (April 2018): 899–934; Daron Acemoglu, Simon Johnson, James A. Robinson, and Yungyong Thaicharoen, “Institutional Causes, Macroeconomic Symptoms: Volatility, Crises and Growth,” *Journal of Monetary Economics* 50, no. 1 (January 2003): 49–123.

⁹⁴Thomas Hobbes, *Leviathan*, ed. Richard Tuck (Cambridge: Cambridge University Press, 1991).

⁹⁵Kathleen Thelen, *Varieties of Liberalization and the New Politics of Social Solidarity*. Cambridge University Press, 2014.

enough to encourage market functioning, but disciplined enough to avoid direct involvement in it.

As stated above, this aligns with the Smithian notion that markets require the state provision of public goods, infrastructure, and guardrails. Smith’s insistence that market systems cannot exist in an environment of asymmetric competition, arising from both public and private sources, requires a robust institutional framework that protects against monopolistic and rent-seeking behavior.⁹⁶ Smith’s argument implies not only the provision of public goods but also a regulatory capacity to prevent predatory market conditions.

The historical literature demonstrates that economic development in Europe and North America relied heavily on the state’s willingness to support market systems while not becoming a player in them. In the post-Revolutionary United States, deficient interstate infrastructure, combined with weak state capacity at the Federal level, resulted in a non-existent domestic market. At the same time, conflicting monetary regimes, incohesive trade policies, and a lack of coordination on interstate commerce produced a Thunderdome of commercial conflict between the American states and commercial actors within them. As such, an American domestic market did not begin to emerge until the federal Constitution of 1787 reformed rulemaking, created a monetary union, and established a national policymaking framework that empowered the creation of commercial infrastructure via direct investment and the elimination of local, county, and state impediments.⁹⁷ Similar policy action supercharged British market integration through regulatory reform and government incentives for transportation network creation in the 1830s and 1840s.⁹⁸

Modern scholarship continues to extend this understanding of the constructed nature of markets and the need for targeted government provisions to ensure continuity. In their *Varieties of Capitalism*, Hall and Soskice (2001) find that both liberal market economies and coordinated market economies rely on strong institutional involvement.⁹⁹ Liberal market economies utilize anti-monopoly regulations and public research contracts. In contrast, coordinated market economies rely on business associations, sectoral coordination, and vocational training systems to coordinate economic activity among firms. Despite their structural differences, both systems require governments to establish and enforce the “rules of the game” for institutional configurations to coordinate behavior, protect economic rights, and manage systemic failures.

Governments that balance technocratic involvement and private sector encouragement often build market-supporting policies that are insulated from rent-seeking while remaining engaged with industry.¹⁰⁰ Dani Rodrik (2011) argues that states with deficient policy frameworks experience increased political and economic fragility amidst globalization.¹⁰¹ Ha-Joon Chang (2002) demonstrates that most high-income nations implemented highly active policy during their developmental phases to build and sustain markets.¹⁰² Capitalism requires and flourishes with effective and disciplined state involvement to support productivity and limit rent extraction.

Beyond protection from market failures and provision of public goods, governments can create the

⁹⁶Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

⁹⁷See forthcoming, Scott C. Miller, *Minerva Rising!: The Revolutionary Origins of American Economic Power* (Chicago: University of Chicago Press, 2027).

⁹⁸“Investor Reactions to Legislative Liberalization and the Run-up in British Share Prices, 1844-1845,” with Vaska Atta-Darkua and Robert F. Bruner, *Financial History Review*, Vol. 31, No. 1 (2025), 1-35.

⁹⁹Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001).

¹⁰⁰Peter Evans, 1995. *Embedded Autonomy: States and Industrial Transformation*. Princeton University Press.

¹⁰¹Dani Rodrik, 2011. *The Globalization Paradox: Democracy and the Future of the World Economy*. W. W. Norton.

¹⁰²Ha-Joon Chang, 2002. *Kicking Away the Ladder: Development Strategy in Historical Perspective*. Anthem.

preconditions for new markets to develop that would not otherwise exist. Mariana Mazzucato (2013) emphasizes the role of the entrepreneurial state in creating initial markets in technology, energy, health, and other industries where public utility might outweigh immediate private profit incentives.¹⁰³ Rodrik (2007) reinforces this perspective, arguing that effective states engage in strategic coordination to identify and grow comparative advantages while maintaining broader market competition.¹⁰⁴ This does not contradict market logic, but rather, by process of state-enabled discovery, expands it into new frontiers.¹⁰⁵

We include multiple infrastructure metrics such as road network density, liner shipping connectivity, and air transport freight as measures of market construction and integration.¹⁰⁶ We also include our new “Hamilton Indicator,” which proxies a government’s ability to use commercial and trade policy to create markets for essential goods while maintaining a competitive environment.¹⁰⁷

The domains of the Market-Supporting Policy subindex are:

1. Market Initiatives
2. Operational and Institutional Quality

For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Labor Market Openness

The ability to freely deploy one’s naturally endowed form of capital—human capital, or labor—is fundamental to the logic and efficacy of capitalist systems. Classical scholarship recognizes human labor as capital, integral to a capitalist system. Long before the emergence of industrialized capitalism, John Locke argued that when an individual mixes their labor with natural resources, they remove those resources from common ownership and make them their own.¹⁰⁸ This labor theory of value justifies human labor as a form of capital. Henry George extended this logic even further, arguing that labor alone creates all economic value, and the free exercise of labor is an essential precondition for well-functioning markets.¹⁰⁹ In his criticism of capitalist systems, Marx understood labor to be a “commodity which its possessor, the wage-worker, sells to the capitalist.”¹¹⁰ Though Marx critiques the system, he identifies possessing and selling one’s labor as an inherently

¹⁰³Mariana Mazzucato, *The Entrepreneurial State: Debunking Public vs. Private Sector Myths* (London: Anthem Press, 2013).

¹⁰⁴Dani Rodrik, *One Economics, Many Recipes: Globalization, Institutions, and Economic Growth*. Princeton University Press, 2007.

¹⁰⁵Robert Wade, 1990. *Governing the Market: Economic Theory and the Role of Government in East Asian Industrialization*. Princeton University Press; Albert Hirschman. 1958. *The Strategy of Economic Development*. Yale University Press.

¹⁰⁶Utilities like electricity, railways, and communications are important for the transmission of capital but pose methodological challenges. Those industries are often private-led and difficult to parse out what constitutes direct public sector support. We also find health spending and education to be more peripheral. While they are obviously essential for human prospering, we do not view them as in and of themselves vital for the development and sustenance of market systems. As such, we argue that the development of human capital is less about the architecture of market systems than the operations and activities within those systems.

¹⁰⁷The theoretical basis for the Hamilton Indicator can be found in the Free Flow of Goods and Capital section, and the methodology is detailed in the accompanying GCI Codebook.

¹⁰⁸John Locke, *Two Treatises of Government*, ed. Peter Laslett (Cambridge: Cambridge University Press, 1988 [1690]).

¹⁰⁹Henry George, *Progress and Poverty* (New York: D. Appleton and Co., 1879).

¹¹⁰Karl Marx, *Wage Labor and Capital*, trans. Frederick Engels (New York: International Publishers, 1933 [1847]).

capitalist undertaking, wherein the worker can “leave the capitalist as often as he chooses” and the firm can “discharge him as often as he sees fit.”¹¹¹ Polanyi similarly placed commodified human labor as key to capitalism. In his analysis of industrial-era England, he asserts that “not until 1834 was a competitive labor market established... hence industrial capitalism as a social system cannot be said to have existed before that date.”¹¹² For Polanyi, the commodification of labor as a tradeable input was an institutional innovation achieved through deliberate legal and political restructuring. It requires formalization of property in one’s own person and mobility across industries and borders. In the Schumpeterian and neoclassical traditions, human labor is conceptualized as a form of capital, wherein individuals bear their embedded knowledge, skills, and training as capital analogous to physical or financial capital to sell in the marketplace to firms.¹¹³ Similarly, the ability of employers to allocate human capital as they see fit is crucial to the successful operation of market systems. In order to achieve allocative efficiency, suppliers and demanders of human capital must both be able to freely deploy it where most valued. This process is only possible when social, economic, and political institutions guarantee capital access, mobility, and non-discrimination.

To achieve allocative efficiency, both suppliers and demanders of human capital must be able to freely deploy it where it produces the most value. This process is only possible when social, economic, and political institutions guarantee human capital access, mobility, and non-discrimination. Well-functioning capitalist systems cannot include forced labor and unequal work arrangements as they inherently constrain the free exercise and deployment of human capital. Capitalism implicitly requires an equality in the terms of labor, if not necessarily in the labor itself, to sustain the market in which human capital is converted into tangible value. Arbitrary constraints such as slavery, gender or race-based discrimination, or unequal legal treatment, artificially undervalue human capital by preventing equal bidding in an open market.¹¹⁴ Becker’s (1957) labor theory

¹¹¹Karl Marx, *Wage Labor and Capital*, trans. Frederick Engels (New York: International Publishers, 1933 [1847]).

¹¹²Karl Polanyi, *The Great Transformation: The Political and Economic Origins of Our Time* (Boston: Beacon Press, 1944).

¹¹³Gary S. Becker, *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*, 3rd ed. (Chicago: University of Chicago Press, 1993 [1964]).

¹¹⁴There has been much scholarly debate in recent years about the complicated relationship between slavery and capitalist institutions, especially in the historical literature. Works including Edward E. Baptist, *The Half Has Never Been Told: Slavery and the Making of American Capitalism* (New York: Basic Books, 2014); Sven Beckert and Seth Rockman, eds., *Slavery’s Capitalism: A New History of American Economic Development* (Philadelphia: University of Pennsylvania Press, 2016); Seth Rockman, *Scraping By: Wage Labor, Slavery, and Survival in Early Baltimore* (Baltimore: Johns Hopkins University Press, 2009); Caitlin Rosenthal, *Accounting for Slavery: Masters and Management* (Cambridge: Harvard University Press, 2018); and Calvin Schermerhorn, *The Business of Slavery and the Rise of American Capitalism, 1815-1860* (New Haven, CT: Yale University Press, 2015) have sought to interweave the institution of slavery with the development of capitalist economies. As Beckert said in his 2014 article in the *Chronicle of Higher Education* entitled “Slavery and Capitalism,” “Historians long depicted slavery as an aberration from, or a precursor to, capitalism. That’s changing.” While we see immense value in this literature and agree that slavery or labor coercion can take place within broadly capitalist economies, we reject the premise that slavery is intrinsically capitalist. We believe that capitalist institutions or operational methods can be deployed to enact or perpetuate slavery, while rejecting the conception that coerced or unfree labor makes an economic system “more capitalist.” Since economic systems are not monolithic, all include institutional elements that are at odds with their broader structural makeup. (For example, robust and competitive black markets occurred in the Soviet Union while monopolistic and state-owned enterprises exist in the United States.) Put another way, we would argue that if you have two identical economies, with the single difference being that one incorporated slavery and the other rejected it, the one without slavery would be the more capitalist. We believe this for several reasons. First, slavery and coerced labor has been a feature of a wide variety of economic systems throughout history. Economies from ancient Gaul to the Mongol and Incan Empires, to the early modern West all incorporated unfree labor, despite their economic systems having little else in common. Our aim with the GCI is to isolate input variables that are inherent and unique to capitalist systems so that this model can be measured against exogenous variables (including the presence of slavery). Second, we reject the theoretical premise that a system of labor that rejects property rights and open

views wage differentials due to gender or race as “taste-based discrimination”¹¹⁵ that explicitly distorts labor markets. The structure of capitalism necessitates egalitarian work arrangements with unimpeded deployment and allocation of labor. For example, contemporary research shows that gender inequalities not only constrain labor mobility but also attenuates aggregate productivity and economic growth.¹¹⁶ It is worth noting that we considered measures of labor action activity, including ILO’s days not worked due to strikes per 1000 workers, however, we ultimately did not include this measure because of a fundamental interpretive ambiguity: a high number of strike days could indicate that workers possess and are able to exercise meaningful labor rights, which would suggest a well-functioning institutional environment, or it could indicate that labor relations are dysfunctional and bargaining mechanisms have broken down, which would suggest institutional failure. Conversely, a low number of strike days could reflect either effective dispute resolution and cooperative labor relations or the suppression of labor rights by the state or employers. Days lost to strikes is an outcome of the labor relations environment not a feature of the institutional infrastructure itself. It is important to note that this domain measures labor market availability outcomes rather than migration policy openness directly. Globally comparable, annually updated measures of migration policy do not currently exist at the scale required by this index. Therefore, we proxy the conditions under which labor supply can expand and circulate using the share of foreign-born workers in the labor force and perceived ease of filling vacancies. We acknowledge that these outcome variables are reflections of both policy choices and structural economic conditions. This subindex measures the ability of individuals to deploy their human capital without constraint, whether those constraints arise from structural discrimination or organized barriers to free allocation.

Labor market competition and human capital turnover are necessary for capitalist functioning. Like firms competing in a consumer market, labor markets require an open supply of new entrants into the market with minimal barriers to entry to experience increased competition. The Schumpeterian notion of creative destruction requires an availability of new market participants to create competition, incentivize development, and adjust to shifting demand.¹¹⁷ It is worth noting that the Schumpeterian logic applied here focuses on the capacity of workers to move when beneficial, not the absolute level of labor market turnover. Forced and excessive turnover can erode human capital investment and reduce productivity, which in turn erode the function of capitalist labor markets. This domain is intended to capture whether workers possess the option to allocate their human capital freely, not whether they are compelled to do so.

Keynesian and neoliberal economic models both recognize the short-run macroeconomic dependence on the labor market’s ability to adjust wages and employment to business cycles.¹¹⁸ Contemporary scholarship shows that reducing impediments to internal and cross-border labor migration helps provide optimal labor supply and productivity. Autor, Dorn, and Hanson (2013) show that heavily segmented labor markets increase regional inequalities and social dislocation, preventing allocative

labor markets, impinges on robust competition, suppresses the development of sophisticated financial institutions and capital markets, and results in diminished entrepreneurial activity, is structurally capitalist. Finally, we contend that human capital—i.e human potential value creation manifested through the execution of one’s own labor—is capital and that any private or public barriers to the free execution of it is contrary to capitalist principles. Therefore, we model any slavery or forced labor as a linear negative value in our calculation of capitalist adherence.

¹¹⁵Gary S. Becker, *The Economics of Discrimination*, 2nd ed. (Chicago: University of Chicago Press, 1971 [1957]).

¹¹⁶Claudia Goldin, *Career and Family: Women’s Century-Long Journey toward Equity*. Princeton University Press, 2021.

¹¹⁷Joseph A. Schumpeter, *Capitalism, Socialism and Democracy* (New York: Harper & Brothers, 1942).

¹¹⁸John Maynard Keynes, *The General Theory of Employment, Interest, and Money* (London: Macmillan, 1936); Friedrich A. Hayek, *The Road to Serfdom* (Chicago: University of Chicago Press, 1944).

efficiency.¹¹⁹ Empirical work on global labor mobility demonstrates that both internal and cross-border labor mobility enables improved human capital allocation, productivity, and greater levels of innovation.¹²⁰

It is worth noting that there is a theoretical tension between labor flexibility and labor protections, where each implies a different level of labor regulation. Our intent by including them in one subindex is to more accurately capture the necessary balance between free deployment of human capital and its sustainability in the marketplace. Without protections from forced labor, coercion, or inequitable arrangements, providers of human capital are unable to deploy it as they see fit. Therefore, this subindex does not conceptualize market flexibility and protections as opposing poles, but rather as crucial institutional complements. Our adaptive PCA process is designed to accommodate the potential tension between these concepts by deriving principal components based on variation in the subindex’s entire collection of data. Our approach allows us to distinguish countries that fall into different cells of a 2×2 matrix: high flexibility/high protections (flexicurity models), high flexibility/low protections (minimal regulation models), low flexibility/high protections (rigid labor markets with strong unions), and low flexibility/low protections (dysfunctional markets). By deriving weights empirically through PCA rather than assuming these dimensions oppose each other, we can reward countries that achieve both dimensions well. The component structure reveals whether these dimensions actually trade off in practice or can coexist.

If human labor is understood as a capital input, labor markets must be open, inclusive, and adaptable to support a capitalist economic system in which workers and firms respond dynamically to incentives. A fully realized capitalist system empowers the owner of that capital to deploy it as freely possible, with minimal subjection to constraints from both state authorities and private actors. This subindex measures the practical extent to which labor markets empower human capital owners to direct the use of their assets.

Based on the theoretical justifications outlined above, the domains of the Labor Market Openness subindex are: 1) Labor Market Flexibility, 2) Labor Protections, and 3) Labor Availability. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Market Competition

The Global Capitalism Index places significant value on the presence of vibrant competitive dynamics within the market for goods and services. The prospect of earning market “rewards” through the development of superior technologies, production processes, organizational techniques, or investment strategies, balanced against the inherent risks of failure and/or loss in markets, constitutes a primary incentive that drives individual- and firm-level growth. In many ways, competition is as unique to capitalism as any quality in the GCI. All other forms of economic organization—socialism, communism, mercantilism, and feudalism to name a few—revolve around the consolidation of economic operations in the hands of a small group of coordinated producers. Only in capitalism is the economic system designed to incentivize and protect modes of competition instead of coordination.

¹¹⁹ Autor, David H., David Dorn, and Gordon H. Hanson. “The China Syndrome: Local Labor Market Effects of Import Competition in the United States.” *American Economic Review* 103, no. 6 (2013)

¹²⁰ Michael Clemens. “Economics and Emigration: Trillion-Dollar Bills on the Sidewalk?” *Journal of Economic Perspectives* 25, no. 3 (Summer 2011): 83–106; Francesc Ortega and Giovanni Peri. “Openness and Income: The Roles of Trade and Migration.” *Journal of International Economics* 92, no. 2 (March 2014): 231–251; Giovanni Peri. “Immigrants, Productivity, and Labor Markets.” *Journal of Economic Perspectives* 30, no. 4 (Fall 2016): 3–30.

The GCI operates on the premise that capitalist economies require a high level of competition, or a freedom to compete, even though “excess competition” can reduce profit maximization. The equilibrium between competition and profit maximization is an important point of analysis, but we believe that the level of competition in an economic system is a greater representation of the capacity for capitalist enterprise than the output variables of profits or growth.¹²¹ Even when coming at the expense of short-term profits, competition produces numerous positive systemic externalities in the medium- and long-term. Assuming that robust competition policy is coupled with a policy regime in which uncompetitive firms can fail in an orderly manner, high levels of market competition will produce optimal aggregate effects over time.

As a result, we focus on the systemic perspective, measuring the institutional arrangements by which a society fosters broad-based competition rather than measuring the outcomes of that competition. We emphasize that constraints on competitive activity can come from forces both public and private, and thus this subindex measures the composite capacity of an economic system to foster competition free from constraints from both government and monopolists. For this reason, we include measures of regulatory capacity aimed at promoting competition within the system, with the understanding that while those regulations could have deleterious effects in other areas, they will be captured in other areas of the index.

In a macroeconomy with many firms and individuals competing for market gain and its financial benefits, processes of growth and innovation replacing the obsolete with the innovative are examples of “creative destruction.” Joseph Schumpeter described this “creative destruction” as a process “that incessantly revolutionizes the economic structure from within, incessantly destroying the old one, incessantly creating a new one.”¹²² In this sense, creative destruction constitutes a foundational mechanism of capitalism and, by extension, a central component of both the Global Capitalism Index and its Market Competition subindex. It is worth acknowledging that the relationship between market competition and innovation incentives is theoretically contested. The Schumpeterian view holds that the prospect of temporary monopoly rents motivates firms to invest in innovative technologies, while the competing Arrowian perspective argues that competitive pressure is itself the primary driver of innovation.¹²³ This subindex is designed to measure purely the institutional capacity for competitive dynamism and whether entrants and incumbents can compete and innovate rather than prescribing an optimal level of market concentration.

Building on Mazzucato and Tancioni (2005), the Market Competition subindex recognizes that while firm entry and exit are key traditional indicators of creative destruction, changes in firms’ relative market positions (the “turbulence”) also provide critical evidence of dynamic competition.¹²⁴ Mazzucato (2005) emphasizes the analytical importance of a market share instability index (e.g., Hymer and Pashigian 1962); although the Hymer and Pashigian framework is organized by industry rather than geography, its logic can still inform our cross-national approach.¹²⁵ As Mazzucato (2005) concludes, “when creative destruction occurs through competence destroying

¹²¹To conflate growth with capitalism is an analytic error. In most cases, there is a high correlation between capitalist, economic systems, and economic growth, but they are not the same thing and should not be assessed as such.

¹²²Joseph A. Schumpeter, *Capitalism, Socialism and Democracy* (New York: Harper & Brothers, 1942).

¹²³Kenneth J. Arrow, “Economic Welfare and the Allocation of Resources for Invention,” in *The Rate and Direction of Inventive Activity: Economic and Social Factors*, ed. Richard R. Nelson (Princeton, NJ: Princeton University Press, 1962), 609–626.

¹²⁴Mariana Mazzucato, Massimiliano Tancioni. *Indices that Capture Creative Destruction: Questions and Implications*. In: *Revue d’économie industrielle*, vol. 110, 2e trimestre 2005. pp. 199–220.

¹²⁵Stephen Hymer and Peter Pashigian (1962). *Turnover of Firms as a Measure of Market Behavior*. *Review of Economics and Statistics* 44: pp. 82–87.

innovations, we should expect to see a correlation between economy wide growth and market share instability... due to the uneven nature of technological change, what matters is relative change (and the associated instability) not absolute change.”¹²⁶ Drawing from this precedent, we have estimated economy-wide growth (and thus, market instability) as year-over-year GDP growth. Accordingly, while perfect cross-country measures of market turbulence remain limited, the inclusion of proximate indicators provides a methodologically robust alternative for capturing the essence of creative destruction within the Market Competition subindex. It is also worth noting that many economies, particularly advanced ones, employ institutional mechanisms to cushion the most disruptive effects of creative destruction, like bankruptcy protections, social safety nets, and active labor market policies. These buffers can attenuate observable signals of firm turnover without necessarily reducing the underlying competitive dynamism of the system. The subindex therefore focuses on the capacity for creative destruction rather than its most visible manifestations.

Creative destruction, as a process of continual economic renewal, depends fundamentally on the broader forces of innovation and research that generate technological and organizational advancements within markets. Without sustained investment in research and development (R&D), the mechanisms of market turnover and productivity growth central to capitalism would stagnate. Aghion et al. (2009) find that “technologically advanced entry threats spurs innovation incentives in sectors close to the technological frontier—successful innovation allows incumbents to prevent entry. In laggard sectors it discourages innovation—increased entry threat reduces incumbents’ expected rents from innovating.”¹²⁷ This dynamic highlights that innovation both drives and is driven by competitive pressure, making R&D investment a key indicator of creative destruction and, by extension, macroeconomic growth. Fieldhouse and Mertens (2024) reinforce this relationship, identifying a strong causal link between federal nondefense R&D appropriations and business-sector productivity growth.¹²⁸

Gumus and Celikay (2015) examine data from 52 countries over a 14 year period and find that “R&D expenditure has a positive and significant effect on economic growth for all countries in the long run, which is consistent with the relevant literature.”¹²⁹ Although short-run effects vary, especially among developing nations, their results confirm that R&D consistently enhances long-term growth. Similarly, Henriquez et al. (2023) show that “R&D has a positive effect on process and product innovation in Latin American firms”, emphasizing its policy and strategic implications.¹³⁰ Finally, Mohnen et al. (2021) demonstrate a “relatively strong complementarity between R&D and organizational innovation,” suggesting that institutional capacity and knowledge management amplify the productivity of research investments.¹³¹ Collectively, these studies provide strong theoretical and empirical support for incorporating R&D as a core explanatory variable in discussions of innovation-driven growth and the dynamics of creative destruction, across both advanced and

¹²⁶Mariana Mazzucato, Massimiliano Tancioni. Indices that Capture Creative Destruction: Questions and Implications. In: *Revue d’économie industrielle*, vol. 110, 2e trimestre 2005. pp. 199-220.

¹²⁷Philippe Aghion, Richard Blundell, Rachel Griffith, Peter Howitt, and Susanne Prantl. The Effects of Entry on Incumbent Innovation and Productivity. 2009. *Review of Economics and Statistics* 91, no. 1. pp. 20–32.

¹²⁸Andrew J. Fieldhouse and Karel Mertens, “The Returns to Government R&D: Evidence from U.S. Appropriations Shocks,” Federal Reserve Bank of Dallas Working Paper 2305 (May 2023, revised November 2024).

¹²⁹Erdal Gumus & Ferdi Celikay. (2015). R&D Expenditure and Economic Growth: New Empirical Evidence. *Margin The Journal of Applied Economic Research*. 9. 205-217. 10.1177/0973801015579753.

¹³⁰Ortiz Henriquez, Rodrigo Esteban, Fernando A. Crespo, Cristian Geldes, Tiago Alves Ferreira, and Mauricio Castillo-Vergara. 2023. Impact of R&D on the Innovation of Products and Processes in Latin Countries. *Axioms* 12, no. 2: 149.

¹³¹Pierre Mohnen, Michael Polder, and George van Leeuwen. Information and Communications Technology, R&D, and Organizational Innovation: Exploring Complementarities in Investment and Production. *National Bureau of Economic Research*. April 2021. pp 299-322.

developing economies. R&D expenditure is measured here as aggregate investment relative to GDP rather than disaggregated by funder or target industry. The distinctions between broad public funding, targeted industrial policy investments, and private expenditure are theoretically meaningful but no globally comparable and disaggregated data exists at the scale required for this index.

Complementing the logic of creative destruction, capitalism presupposes a baseline regulatory infrastructure that preserves incentives for success while ensuring the enforcement of consequences for market failure. Effective regulatory frameworks thus play an essential role in sustaining competitive market dynamics. In the context of expanding digital and knowledge-based economies, Nicoletti et al. (2023) observe that “there is considerable consensus, among academic and policy-makers, that ex-ante regulatory policies are needed to avoid competition being stifled in these markets, with a negative impact on productivity and innovation.”¹³² This position underscores the notion that regulation, when appropriately designed, can enhance rather than hinder competition by setting transparent rules that encourage innovation and deter monopolistic behavior.

Similarly, Litina et al. (2021) introduce an important distinction between types of regulatory intervention. They find that “protective regulation”, i.e., regulation that creates predictability and scope for markets to work, confers a positive effect on innovation, while “restrictive regulation”, i.e., regulation that endeavors to spell out negative behaviors, confers a negative effect on innovation.”¹³³ This distinction reinforces that “regulation” is not a monolithic construct, but rather a spectrum of policy instruments whose impact depends on design, intent, and implementation. Taken together, these insights justify the inclusion of a Regulatory Analysis domain within the Market Competition subindex. This domain evaluates not the quantity of regulation, but the quality and orientation of regulatory frameworks, further distinguishing between those that support market dynamism and those that constrain it.

All economies rest fundamentally on the participation and productivity of their labor force. Without active and capable workers, meaningful market activity, and by extension competition, cannot exist. Yet it is the quality of that labor force (skills, education, training, collaboration, and other intangible attributes) that differentiates national economies and shapes the efficiency of market outcomes. Although comprehensive cross-country measurement of labor-market characteristics remains challenging, this domain seeks to capture as holistic an understanding as possible of human capital competitiveness through legal, cultural, and social dimensions. This domain places particular emphasis on survey-based indicators, which provide valuable insight into the more subjective aspects of human capital (such as managerial competence, collaboration norms, willingness to take risks, and adaptability) that are not easily quantifiable.

Our premise that human capital competitiveness underpins market dynamism draws on an extensive body of research. The Global Talent Competitiveness Index (GTCI), published by INSEAD, the Descartes Institute for the Future, and the Human Capital Leadership Institute, served as a foundational reference for conceptualizing human capital as an input to Market Competition. In particular, the GTCI’s consideration of both input variables (e.g., education, vocational training) and output variables (e.g., productivity, innovation capacity) offers a comprehensive framework for evaluating workforce performance.¹³⁴ Building on this foundation, the Human Capital Com-

¹³²Giuseppe Nicoletti, Cristiana Vitale, and Carolina Abate. *Competition, Regulation and Growth in a Digitized World – Dealing with Emerging Competition Issues in Digital Markets*. OECD. 2023.

¹³³Anastasia Litina, Christos A. Makridis, Georgios Tsiachtsiras. Do product market reforms raise innovation? Evidence from Micro-data across 12 countries, *Technological Forecasting and Social Change*, Volume 169, 2021, 120841, ISSN 0040-1625. <https://doi.org/10.1016/j.techfore.2021.120841>.

¹³⁴Bruno Lanvin, Felipe Monteiro, Michael Bratt, and Anna Henry. *The Global Talent Competitiveness Index 2023*.

petitiveness domain of the Market Competition subindex extends prior work by integrating recent survey-based data from business executives worldwide.

Each domain of the subindex integrates a mixture of measurable “hard” data variables, and perception-based “soft” survey data. The proportional blend of these data types is determined by their conceptual relevance to the respective domain and the broader availability of reliable sources. Based on the theoretical justifications outlined above, the domains of the Market Competition subindex are: 1) Creative Destruction, 2) Regulatory Analysis, and 3) Human Capital Competitiveness. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Capital Market Sophistication

A foundational pillar of the Global Capitalism Index is the existence, strength, breadth, and resilience of capital markets. We capture these four elements under the top-line construct of “capital market sophistication.” Measuring capital market sophistication allows us to not only assess current and historical market activity, but also to evaluate the potential for future capital formation and/or baseline financial accessibility. In constructing this subindex, we make a deliberate effort to recognize alternative models of capital market development (market-based, state-led, etc.) and to fully acknowledge diverse institutional perspectives. Consistent with other components of the Global Capitalism Index, our methodology integrates both “hard” quantitative indicators and “soft” survey-based measures. This dual approach allows us to capture a wide spectrum of market characteristics, ranging from highly observable metrics (such as the density of ATM and bank branches) to more qualitative dimensions, including the perceived extent to which financial services meet business needs, or the assessed availability of venture capital.

While this methodology enables a multidimensional view of capital market activity, important limitations remain. Most notably, cross-country data on mergers and acquisitions activity and venture capital investment are sparse, uneven, and inconsistently defined across sources.

The Access and Participation domain of capital market sophistication focuses on the extent to which individuals and firms can access formal financial infrastructure and actively participate in financial transactions. This domain is operationalized through two complementary dimensions: (a) the presence of financial infrastructure, and (b) the prevalence and accessibility of financial transactions. Together, these dimensions capture both the supply-side availability of financial services and the demand-side utilization of those services within an economy.

The joint consideration of infrastructure and transactional activity allows for meaningful distinctions across stages of financial development. Economies with limited physical and digital financial infrastructure typically exhibit low levels of transactional engagement, reflecting early stages of financial system development. Conversely, economies with extensive infrastructure but relatively modest transaction volumes may signal structural bottlenecks, regulatory frictions, or underutilization of available financial services. At the upper end of the spectrum, economies characterized by both dense infrastructure and high transactional intensity are indicative of mature, highly intermediated financial systems in which capital markets can drive economic growth, risk allocation, and enterprise financing. By explicitly juxtaposing infrastructure and transactional activity, this framework captures not only the depth of financial access, but also the effectiveness with which that access translates into real economic participation.

By examining enabling inputs alongside observable market outputs, this framework establishes a

baseline assessment of the capacity for capital markets to exist and attain a level of “sophistication” defined above. This distinction is particularly critical in a global context, where financial systems frequently diverge from traditional Western, bank-centered or market-based models.¹³⁵ Capturing and normalizing cultural divergences requires evaluating access and participation through multiple, complementary indicators. Accordingly, we have constructed this subindex to capture structural differences through the inclusion of both infrastructure- and transaction-based variables, allowing for a more nuanced and internationally comparable assessment of capital market development.

Our examination of total capital market depth and maturity focuses on equity, bond, venture capital, and mergers and acquisitions (M&A) markets. Within public bond and equity markets, current data availability permits a detailed examination of market scale, trading activity, regulatory context, and asset-quality characteristics. Indicators including stock market turnover and mutual fund assets as a percentage of GDP provide a holistic portrait of financial market scope, capturing both the corporate financing dimension and the breadth of investor participation.

We include mutual fund assets as a percentage of GDP to capture a complementary dimension of market participation and the development of nonbank financial intermediation. Following Svirydzenka (2016), we treat mutual fund assets-to-GDP as a core indicator of financial institution depth. Higher mutual fund penetration reflects the capacity of nonbank intermediaries to mobilize household savings into listed securities, broadening the investor base and enhancing risk-sharing. Consistent with IMF and World Bank frameworks, deeper nonbank intermediation is associated with greater equity-market breadth and liquidity.¹³⁶ While other financial assets like ETFs, CEFs, and REITs play a similar and growing role, mutual funds’ ubiquity across time and geography allow for a more consistent and expansive cross-nation evaluation of capital market depth.

In addition to market scale, participation, and institutional depth, we assess market functioning through measures of risk and regulation. To do so, we analyze data comprising stock market volatility and survey measurements on regulation of security exchanges. Volatility provides insight into prevailing uncertainty and into the transmission of monetary policy across financial systems.¹³⁷

Finally, we capture the governance of financial markets through survey-based indicators designed to reflect institutional guardrails. Given the heterogeneity in legal traditions, supervisory capacity, and state involvement across financial systems, survey-based measures provide a pragmatic means of capturing cross-country variation in regulatory effectiveness and market oversight. We rely on survey data for two reasons: first, to reflect the diversity of regulatory models and the perspectives of market participants operating within them; and second, due to the limited availability of consistent cross-national, quantitative measures of regulatory enforcement. While imperfect, these indicators represent the most consistent proxies available globally for evaluating the regulatory foundations of equity market depth and maturity.

We assess bond market depth and scope using data on corporate bond markets, supplemented by broader assessments of debt maturity and investment horizons and general lending conditions

¹³⁵The case of China illustrates this point clearly. Despite exhibiting extremely high levels of digital transaction activity, with 82% of adults making a digital payment in 2021, China simultaneously maintains one of the world’s largest unbanked populations. As of 2021, estimates suggest that there were over 130 million unbanked individuals in the nation. Demirgüç-Kunt, Asli, Leora Klapper, Dorothe Singer, and Saniya Ansar. 2022. *Global Findex Database 2021: Financial Inclusion, Digital Payments, and Resilience in the Age of COVID-19*. Washington, DC: World Bank.

¹³⁶Katsiaryna Svirydzenka. *Introducing a New Broad-based Index of Financial Development*. IMF Working Paper. January 2016.

¹³⁷Roni Bhowmik and Shouyang Wang. Stock Market Volatility and Return Analysis: A Systematic Literature Review. *Entropy* (Basel). 2020 May 4;22(5):522.

within the financial system. We examine the structural composition of national bond markets through (a) corporate bond issuance volume to GDP, and (b) corporate bond average maturity.

Corporate bond issuance volume relative to total economic output (GDP) allows for meaningful cross-country comparison of the role played by bond markets in financing real economic activity. Higher levels of corporate bond issuance relative to GDP are generally associated with highly sophisticated capital markets in which firms can substitute market-based finance for bank credit.

Likewise, longer average maturities in corporate bond markets signal greater bond market sophistication. In economies with well-developed capital markets, longer average maturities represent higher investor confidence, credible macroeconomic policy frameworks, and a substantial presence of long-horizon institutional investors.¹³⁸ Longer maturities also reduce rollover and liquidity risk, thereby supporting financial stability and predictable return profiles. Time series variation in average corporate bond maturity reflects the interaction between investment horizons, asset lifecycles, and leverage dynamics.¹³⁹ At the national level, average bond maturity provides indirect insight into private investment structure, asset lifecycles, and the temporal profile of credit demand. In many bank-centered financial systems, commercial banks serve as primary intermediaries in domestic bond markets by underwriting, holding, and distributing corporate and government debt in ways that substitute for independent capital markets. The subindex captures bond market activity through market-based indicators but we acknowledge that the balance between bank-intermediated and market-intermediated debt financing varies meaningfully across national financial systems.

Beyond bond-specific indicators, we incorporate the interest rate spread as a measure of broader lending conditions.

The interest rate spread, defined as the difference between lending and deposit rates, represents a consumer-facing indicator of capital market efficiency and financial intermediation. In highly sophisticated financial systems, competitive pressures and stable macroeconomic conditions compress this spread, aligning lending and deposit rates. By contrast, wider spreads may reflect elevated credit risk, weak competition, policy distortions, or fragile depositor confidence. In such settings, banks require greater compensation for extending credit while needing to offer higher incentives to mobilize household savings. Persistent or extreme spreads therefore signal frictions in financial intermediation consistent with lower levels of capital market sophistication.

It is important to note that in many nations around the world, the practice of lending for interest may not be in accordance with state-accepted parameters of financial activity. To compensate for differences in usury limitations across national cultural and legal systems, we have complemented interest rate-driven data with a survey-based assessment of venture capital availability that does not explicitly mention the charging of interest.¹⁴⁰ This survey question is more inclusive of alternative financial systems (such as bartering, sukuk, certificate, time contracting, or microfinance) that may or may not be based on interest rates, but still provide both short- and long-term credit.¹⁴¹

¹³⁸Under standard maturity-matching practices, extended maturities often reflect demand from pension funds, insurance companies, and sovereign wealth funds seeking duration alignment with long-term liabilities.

¹³⁹“Firms issue debt with a maturity that matches the useful life of new assets and an amortization schedule that reflects the need to free up debt capacity as capital ages. . . also [implying] that both leverage and debt maturity should be negatively related to capital age while both the duration of debt cycles and debt maturity should be positively related to the useful life of assets” in Thomas Geelan et al. *Asset life, leverage, and debt maturity matching*. Journal of Financial Economics. April 2024.

¹⁴⁰“In your country, how easy is it for start-up entrepreneurs with innovative but risky projects to obtain equity funding? [1 = Extremely difficult; 7 = Extremely easy].”

¹⁴¹Executive Opinion Survey. World Economic Forum.

The liquidity and resilience of a capital market is a vital element of its overall sophistication and effectiveness. Liquidity is a precondition for efficient price discovery, financial asset creation, and open market functioning. Markets that lack sufficient liquidity are prone to amplified volatility and diminished investor confidence. By “resilience”, we mean the system’s ability to adapt to short-term volatility and changes to the macro environment, as well as the long-term capacity to return to normal operations amidst crisis situations. Together, liquidity and resilience determine the degree to which capital markets can sustain their intermediation function under adverse conditions. The indicators selected for this domain are therefore designed to capture both the market-facing dimension of liquidity and the systemic dimension of financial resilience.

For cross-national comparisons, we include measures of (a) systemic reserve adequacy, (b) country risk premiums, and (c) the percentage of liquid liabilities to GDP. Using these three metrics, we have greater capacity to assess countries, regions, and trends on a globalized scale. We include two country risk premium metrics due to small, but meaningful differences in methodology. The first of the country risk premiums spans all available years of data and is defined as the Adjusted Default Spread or Rating-based Default Spread multiplied by bond market volatility. The second country risk premium metric begins in 2012 and is based on CDS spreads. Including both metrics provides a more detailed portrait of national risk profiles and includes data for more nations over the covered period.¹⁴²

It is worth noting that banks are, of course, a central institution within any capital market ecosystem. Empirically, economies that score highly on capital market sophistication tend to feature large, well-capitalized banking sectors. However, we find no compelling theoretical basis for disaggregating the banking sector as a standalone component within this subindex. Our measurement framework is designed to capture what is done within capital markets rather than the actors present. Whether credit is extended by a commercial bank, a non-bank lender, or a family member, what matters for the purposes of this index is the volume, accessibility, and sophistication of financial activity. The sole exception to this principle is our inclusion of mutual fund assets as a percentage of GDP, which we retain because it remains the best available proxy for the depth of financial intermediation across a wide cross-section of countries and time periods. However, we do believe that the depth and stability of the banking system is a crucial component of capitalist systems and measure it as a separate subindex in the GCI.

Based on the theoretical justifications outlined above, the domains of the Capital Market Sophistication subindex are: 1) Access and Participation – Infrastructure, 2) Access and Participation – Transactions, 3) Equity Market Depth & Scope, 4) Bond Market Depth and Scope, 5) Venture Capital Availability, 6) M&A Prevalence, and 7) Liquidity and Resilience. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Depth and Stability of Banking System

Capitalist systems depend on banking infrastructure to effectively circulate capital. The banking system stores value, creates credit, and maintains confidence in the financial system’s stability and robustness. The ability of a banking system to penetrate the real economy and maintain stability in the face of shocks is vital for well-functioning capitalist enterprise. Adam Smith argued that the banking system “renders a greater part of that capital active and productive which would

¹⁴²For more details on country risk premium methodology and data sourcing, please see the Capital Market Sophistication appendix and datasheet.

otherwise be dead and unproductive.”¹⁴³ A deep banking system mobilizes and circulates capital for productive use. In his early work, Schumpeter positions the banking system as the “ephor of the exchange economy”¹⁴⁴ to create credit and purchasing power for entrepreneurial activity. Schumpeter’s model of economic dynamism requires the structural ability to save and obtain credit. Similarly, Gerschenkron (1962) argues that banking depth supplies the “long-term credit necessary for industrialization” and financial intermediation in early capitalist development.¹⁴⁵

The institutionalization of banking creates an accessible technology through which risk can be pooled and voluntary exchange can be made more accessible. Banks transform credit maturity at a systemic level, borrowing within short timeframes and lending within longer ones, intermediating credit for firms and households and tying the stability of the financial system to the expectation of future profit.¹⁴⁶ The historical development of banking institutions parallels the rise of capitalist systems. In early modern Europe, proto-banking establishments provided credit facilities that allowed for international trade and industrial economic activity.¹⁴⁷ Banking infrastructure established mechanisms to measure creditworthiness and solvency, which created trust in impersonal credit markets and allowed for broader access beyond individual financial relationships. The English and Dutch financial innovations of the late 17th century institutionalized banking systems and increased capacity for monetary stability and state fiscal credibility.¹⁴⁸ Gerschenkron’s historical overview of industrialization underscores the role of the banking system as a substitute for missing market mechanisms. Where stock markets or venture capital were absent or underdeveloped, banks and state financial institutions provided investment coordination in industrial sectors and infrastructure.¹⁴⁹ Banking systems thus function not only as intermediaries but as key drivers of capitalist growth. We use the concentration of assets among the five largest banks in an economy as a proxy for banking system depth. Deeper, more developed banking systems tend to feature larger, more capitalized leading institutions that reflect broader financial sector expansion and capacity.

We assess not only the extent to which a banking system penetrates the broader economy, but also its institutional stability. Since bank operations contain inherent points of fragility (maturity mismatch, collateral risk, leverage, run dynamics, etc.), their robustness to shocks—and the broader economic and societal effects of those shocks—remain a vital point in analyzing the extent to which banking systems empower capitalist enterprise. When banks fail or credit freezes, uncertainty leads to contractions in the money supply, consumer demand, and the flow of capital.¹⁵⁰ Banks are “endogenously unstable” and prone to excessive leverage during economic upswings and rapid credit unwinding during downturns.¹⁵¹ Charles Kindleberger’s (1978) study of financial crises reinforces this logic, showing how credit expansion and contraction generate the manias and crashes arising from capitalist development.¹⁵² This tendency necessitates institutional stabilizers, like central banks, deposit insurance, and lender-of-last-resort facilities, to sustain confidence and prevent

¹⁴³Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

¹⁴⁴Schumpeter, Joseph A. *The Theory of Economic Development: An Inquiry into Profits, Capital, Credit, Interest, and the Business Cycle*. Translated by Redvers Opie. Cambridge, MA: Harvard University Press, 1934 [1911].

¹⁴⁵Alexander Gerschenkron, *Economic Backwardness in Historical Perspective: A Book of Essays* (Cambridge, MA: Belknap Press of Harvard University Press, 1962).

¹⁴⁶Charles Goodhart, *The Evolution of Central Banks* (Cambridge, MA: MIT Press, 1988).

¹⁴⁷Raymond de Roover, *The Rise and Decline of the Medici Bank: 1397–1494* (Harvard University Press, 1963).

¹⁴⁸Douglass C. North and Barry Weingast, “Constitutions and Commitment: The Evolution of Institutions Governing Public Choice in Seventeenth-Century England,” *Journal of Economic History* 49, no. 4 (1989): 803–832.

¹⁴⁹Alexander Gerschenkron, *Economic Backwardness in Historical Perspective: A Book of Essays* (Cambridge, MA: Belknap Press of Harvard University Press, 1962).

¹⁵⁰John Maynard Keynes, *The General Theory of Employment, Interest, and Money* (London: Macmillan, 1936).

¹⁵¹Hyman P. Minsky, *Stabilizing an Unstable Economy* (New Haven, CT: Yale University Press, 1986).

¹⁵²Charles P. Kindleberger, *Manias, Panics, and Crashes: A History of Financial Crises* (Basic Books, 1978).

systemic failure.¹⁵³ Banking systems must not simply be able to create and circulate credit, but also remain stable and operable in the midst of shocks. The stability of the banking system is a measure of the institutional capacity and ability to sustain credit markets over the long term. Importantly, systemic banking stability is driven primarily by the health of the largest, most systemically important institutions rather than the aggregate performance of all banks. A banking system may accommodate the failure of many smaller institutions while remaining fundamentally stable. System resilience is determined by the capitalization, liquidity, and operational continuity of the largest banks through which the majority of credit, deposits, and capital flows are intermediated. Bank profitability is a critical dimension of this stability. Consistently profitable banks are better positioned to absorb losses, build capital buffers, and sustain lending capacity during periods of financial stress. As a result, bank profitability is an important forward-looking indicator of systemic resilience. The Banking System Stability domain also captures capital adequacy, asset quality, and liquidity coverage ratios as measures of short-term shock absorption capacity.

The tension between banking depth and stability requires institutional oversight and, when appropriate, intervention to counteract cyclical trends, maintain credit, and avoid systemic crisis amid the volatility inherent in dynamic capitalist systems.¹⁵⁴ A shallow system cannot mobilize sufficient credit for innovation and growth, while an overextended one risks expanding beyond its capacity for stable management. As a result, effective banking systems rely on a delicate balance between autonomy and oversight. Independent regulatory bodies, like central banks, supervisory agencies, and macroprudential authorities, must maintain systemic discipline without stifling the economy.¹⁵⁵ Hall and Soskice’s (2001) *Varieties of Capitalism* supports this necessary balance between depth and stability. In coordinated market economies, relationships between banks, firms, and labor unions facilitate long-term credit. In liberal market economies, market-based banking systems provide credit and responsiveness to lending conditions. Despite their differences, both models require broad mobilization of financial capital and institutional confidence to enable capitalist production.¹⁵⁶ Banking depth and stability also depend heavily upon the state. Calomiris and Haber (2014) argue that banking systems are products of political bargains between governments, bankers, and borrowers. When that bargaining results in inclusive arrangements, banking systems tend to thrive. When those coalitions entrench elite privilege or instability, banking systems are more prone to crisis.¹⁵⁷ Strong state capacity also supports systemic stability by credibly backstopping banks, enforcing contracts, and maintaining monetary discipline.¹⁵⁸ However, state-directed lending, politically biased bailouts, or targeted financial repression can erode banking systems and distort credit allocation.¹⁵⁹

Scholarship after the Global Financial Crisis empirically supports the relationship between banking depth and stability. Schularick and Taylor’s (2012) analysis of fourteen advanced economies over the period 1870–2008 establishes that credit growth is a robust predictor of financial crises, particularly

¹⁵³Ben Bernanke, *Essays on the Great Depression* (Princeton, NJ: Princeton University Press, 2000).

¹⁵⁴Charles P. Kindleberger, *Manias, Panics, and Crashes: A History of Financial Crises* (Basic Books, 1978).

¹⁵⁵Paul Tucker, *Unelected Power: The Quest for Legitimacy in Central Banking and the Regulatory State* (Princeton University Press, 2018).

¹⁵⁶Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001).

¹⁵⁷Charles Calomiris and Stephen Haber, *Fragile by Design: The Political Origins of Banking Crises and Scarce Credit* (Princeton University Press, 2014).

¹⁵⁸Douglass C. North and Barry Weingast, “Constitutions and Commitment: The Evolution of Institutions Governing Public Choice in Seventeenth-Century England,” *Journal of Economic History* 49, no. 4 (1989): 803–832.

¹⁵⁹Dani Rodrik, *The Globalization Paradox: Democracy and the Future of the World Economy* (W.W. Norton, 2011).

during increased financialization post-WWII.¹⁶⁰ Mian, Sufi, and Verner (2017) show that household debt-to-GDP increases rather than corporate leverage predict declines in growth and employment across thirty countries.¹⁶¹ Reinhart and Rogoff’s (2009) survey of financial crises reinforces the logic that banking systems alternately enable capitalist growth and threaten its stability, and that the strength of institutions determine which effect is stronger.¹⁶² Admati and Hellwig (2013) argue that the Basel banking framework underweights equity capital requirements relative to their social benefits and that appropriately capitalized banks reduce systemic fragility without compressing credit provision.¹⁶³ Drechsler, Savov, and Schnabl (2017) detail how the structure of bank funding markets transmits monetary policy and shapes lending behavior through “deposit channels”.¹⁶⁴ These works emphasize the empirical justification for treating depth and stability as complementary institutional goals that together create a functioning banking system.

It is crucial to distinguish the Depth and Stability of the Banking System subindex from the Capital Market Sophistication subindex. Though interrelated, they each measure distinct domains of capitalism. Banking systems are the circulatory intermediaries of deposits and credit, transforming savings into productive investment through institutional trust and providing credit expansion. Capital markets are the mechanisms through which investors allocate capital and resources through price signals and sophisticated financial instruments. This subindex focuses on the institutional capacity for credit circulation, broader accessibility, and systemic resilience, not the efficiency or complexity of securities markets. It captures each of these crucial domains through datasets measuring the extent to which capital circulates through an economy as well as the regulatory and financial resilience of the banking system at large.

It is worth noting that this subindex includes measures of market depth but that these indicators capture domestic depth relative to a country’s own economy rather than a country’s share of global capital market activity. For example, the United States scores well on domestic depth measures, but the fact that a disproportionate share of international transactions are denominated in dollars and cleared through U.S. markets is not directly captured. Measures of global trading share were considered but excluded because the GCI is designed to measure the institutional environment for capitalism within a country rather than a country’s relative position in the global financial system. A country’s share of global trading activity is partly a function of institutional quality, but it also reflects reserve currency status and network effects that are better understood as outcomes of institutional strength rather than inputs to it. Including such a measure would risk conflating the institutional infrastructure that enables deep capital markets with the consequences of having them.

Based on the theoretical justifications outlined above, the domains of the Depth and Stability of Banking System subindex are: 1) Banking System Depth and 2) Banking System Stability. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

¹⁶⁰Moritz Schularick and Alan M. Taylor, “Credit Booms Gone Bust: Monetary Policy, Leverage Cycles, and Financial Crises, 1870–2008,” *American Economic Review* 102, no. 2 (April 2012): 1029–1061.

¹⁶¹Atif Mian, Amir Sufi, and Emil Verner, “Household Debt and Business Cycles Worldwide,” *Quarterly Journal of Economics* 132, no. 4 (November 2017): 1755–1817.

¹⁶²Carmen M. Reinhart and Kenneth S. Rogoff, *This Time Is Different: Eight Centuries of Financial Folly* (Princeton: Princeton University Press, 2009).

¹⁶³Anat Admati and Martin Hellwig, *The Bankers’ New Clothes: What’s Wrong with Banking and What to Do about It* (Princeton: Princeton University Press, 2013).

¹⁶⁴Itamar Drechsler, Alexi Savov, and Philipp Schnabl, “The Deposits Channel of Monetary Policy,” *Quarterly Journal of Economics* 132, no. 4 (November 2017): 1819–1876.

New Business Formation and Growth

Institutional support for firm formation, development, and growth is central to thriving capitalist systems. Entrepreneurship animates markets, creates innovation, incentivizes risk-taking, and promotes broader market competition. This subindex captures the institutional, human, and cultural conditions that support new business formation and enable entrepreneurial ventures to flourish. The entrepreneur occupies a central position in the culture of capitalism. Classical and modern theorists conceptualize the entrepreneur in different ways, yet they always converge on its structural importance.¹⁶⁵ Adam Smith believed that the entrepreneur catalyzed the division of labor, while Richard Cantillon presented the entrepreneur as a risk-bearing intermediary who enables efficiency gains and wealth creation.¹⁶⁶ Frank Knight built on this definition, emphasizing new business's role as a bearer of unquantifiable uncertainty and Schumpeter viewed the entrepreneur as not merely a market participant but an endogenous force for capitalist evolution.¹⁶⁷ Israel Kirzner (1973) framed entrepreneurs as agents of discovery, correcting market inefficiencies and equilibrating markets through their pursuit of perceived opportunities.¹⁶⁸ Together, these perspectives emphasize entrepreneurship as more than a byproduct of capitalist systems, but rather a precondition for their existence.

The institutional facilitation of new businesses is complex and multifaceted. Human capital is a key input to the entrepreneurial process. The embedded knowledge, skill, and education of a talent base supply productive capital and further entrepreneurial activity.¹⁶⁹ Endogenous growth theory underscores that the accumulation of human capital through education, experience, and research drives technological progress and new firm formation.¹⁷⁰ A concentration of human capital can exponentially increase business activity through entrepreneurial clusters¹⁷¹, in which strong talent ecosystems foster more resilient startups and institutional support.¹⁷² Audretsch and Feldman (1996) show that knowledge spillovers from geographic clusters of human capital foster "entrepreneurial cities" with high levels of firm creation.¹⁷³ The development of entrepreneurial education can similarly cultivate broader new business growth through talent development.¹⁷⁴ It is worth noting that we view education as a critical social endowment that strongly conditions economic performance, but not as a defining institutional component of capitalism itself. While education shapes how effectively capitalist systems operate, it does not constitute a market institution or policy mechanism in the same sense as our eight subindex topics. That said, we do incorporate measures of educational investment, attainment, and quality within this subindex, where human capital directly affects entrepreneurial capacity and firm creation as an enabling input to capitalist activity rather than as a core dimension of capitalism.

¹⁶⁵Mirjam Van Praag, *Some Classic Views on Entrepreneurship*, Tinbergen Institute Discussion Paper TI 2005-036/3 (Amsterdam: Tinbergen Institute, 2005).

¹⁶⁶Adam Smith, *The Wealth of Nations* (New York: Random House, 1965); Richard Cantillon, *Essai sur la Nature du Commerce en Général*, ed. and trans. Henry Higgs (London: Macmillan, 1931 [1755]).

¹⁶⁷Frank H. Knight, *Risk, Uncertainty, and Profit* (Boston: Houghton Mifflin, 1921).

¹⁶⁸Israel M. Kirzner, *Competition and Entrepreneurship* (Chicago: University of Chicago Press, 1973).

¹⁶⁹Gary S. Becker, *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*, 3rd ed. (Chicago: University of Chicago Press, 1993 [1964]).

¹⁷⁰Paul M. Romer, "Endogenous Technological Change," *Journal of Political Economy* 98, no. 5 (1990).

¹⁷¹Edward L. Glaeser, William R. Kerr, and Giacomo A. M. Ponzetto, "Clusters of Entrepreneurship," *Journal of Urban Economics* 67, no. 1 (2010).

¹⁷²Zoltan Acs, *Entrepreneurship, Innovation and Growth* (Oxford: Oxford University Press, 2006).

¹⁷³David B. Audretsch and Maryann Feldman, "R&D Spillovers and the Geography of Innovation," *American Economic Review* 86, no. 3 (1996).

¹⁷⁴Saras D. Sarasvathy, "Causation and Effectuation: Toward a Theoretical Shift from Economic Inevitability to Entrepreneurial Contingency," *Academy of Management Review* 26, no. 2 (2001): 243–263.

New business formation is a social phenomenon that requires levels of cultural acceptance and promotion, risk-taking, profit motivation, and innovation. The integration of entrepreneurship into broader society legitimizes the ideals of risk-taking, profit motivation, and innovation.¹⁷⁵ Max Weber’s conceptualization of the Protestant ethic connects Puritan cultural values of individuality and asceticism to the rise of capitalism.¹⁷⁶ Weber’s specific religious connotation aside, contemporary scholarship echoes this connection between cultural values and entrepreneurial activity. Societies that value risk-taking, have higher tolerance for failure, and emphasize individualism tend to have higher rates of business formation.¹⁷⁷ Audretsch and Moog (2022) explicitly tie democratic societies to entrepreneurship through their emphases on openness, mobility, and diversity.¹⁷⁸ Entrepreneurial culture functions as an informal institution that enables individuals to pursue new businesses and associated profits within established social boundaries.

Once entrepreneurs create new ventures, opportunities for firm competition and growth require access to a growing network of market systems. Firms must not only emerge but also be able to scale and expand into broader markets. Access to both domestic and international markets determines the potential for firm competition and growth. In modern capitalism, market access no longer progresses sequentially from domestic development to internationalization, but rather simultaneously through “born global” ventures.¹⁷⁹ These transnational firms benefit from open trade regimes, mobile labor, and broader market access.

Similarly, nations with deep access to risk capital foster greater entrepreneurial formation and scaling. The availability of startup capital provides both the early-stage financing to translate ideas into production and a signal of support for entrepreneurship by an economy.¹⁸⁰ Venture capital availability strongly predicts startup density, survival, and innovation output.¹⁸¹ Beyond purely early-stage venture capital, access to formal and informal credit systems allows entrepreneurs to avoid insurmountable liquidity constraints. In many developing and transitional economies, small and medium-sized enterprises (SMEs) rely on credit from banks, microfinance institutions, or informal lenders.¹⁸² The depth of these credit markets is a crucial determinant of total entrepreneurial activity.¹⁸³ These systems, including informal financial lending, can complement formal systems by expanding credit availability and risk diversification to a greater number of new businesses.¹⁸⁴ Access to capital, whether through venture capital, formal banking, or informal lending is an essential

¹⁷⁵Zoltan J. Acs and David B. Audretsch, *Handbook of Entrepreneurship Research* (Boston: Kluwer, 2005); Global Entrepreneurship Monitor (GEM), 2024/2025 Global Report (Global Entrepreneurship Research Association, 2025).

¹⁷⁶Max Weber, *The Protestant Ethic and the Spirit of Capitalism*, trans. Stephen Kalberg (Oxford: Oxford University Press, 2011 [1920]).

¹⁷⁷Global Entrepreneurship Monitor (GEM), 2024/2025 Global Report (Global Entrepreneurship Research Association, 2025); Geert Hofstede, *Culture’s Consequences: Comparing Values, Behaviors, Institutions, and Organizations Across Nations* (Thousand Oaks, CA: Sage, 2001).

¹⁷⁸David Audretsch and Petra Moog, “Democracy and Entrepreneurship,” *Entrepreneurship Theory and Practice* 46, no. 4 (2022).

¹⁷⁹Benjamin M. Oviatt and Patricia P. McDougall, “Toward a Theory of International New Ventures,” *Journal of International Business Studies* 25, no. 1 (1994).

¹⁸⁰Zoltan J. Acs and David Audretsch, *Handbook of Entrepreneurship Research* (Boston: Kluwer, 2005).

¹⁸¹Josh Lerner, and Antoinette Schoar, “Does Legal Enforcement Affect Financial Transactions? The Contractual Channel in Private Equity,” *Quarterly Journal of Economics* 120, no. 1 (2005): 223–246.

¹⁸²Thorsten Beck, Asli Demirgüç-Kunt, and Vojislav Maksimovic, “Financial and Legal Constraints to Firm Growth: Does Firm Size Matter?” *Journal of Finance* 60, no. 1 (2005).

¹⁸³King, Robert G., and Ross Levine, “Finance and Growth: Schumpeter Might Be Right,” *Quarterly Journal of Economics* 108, no. 3 (1993); Beck, Thorsten, and Asli Demirgüç-Kunt, “Access to Finance: An Unfinished Agenda,” *World Bank Economic Review* 22, no. 3 (2008).

¹⁸⁴Abhijit Banerjee, and Esther Duflo, *Poor Economics: A Radical Rethinking of the Way to Fight Global Poverty* (New York: Public Affairs, 2011).

foundation of capitalist dynamism.

The institutional complements to entrepreneurship play an important role in creating new economic activity.¹⁸⁵ Government policy can enable and support entrepreneurial activity, often through the diffusion of knowledge via research & development funding and reduced administrative burdens.¹⁸⁶ Cooperation by industry, research centers, and the state stimulates new ideas and thus provides avenues for the commercial realization of innovations.¹⁸⁷ The collaboration between these three entities stimulates private business growth by converting knowledge into entrepreneurial opportunities and providing sustainable markets for them.¹⁸⁸

New business formation and growth is not purely an output of capitalism, but rather the engine of capitalist dynamism. The vibrancy of a capitalist system depends on its ability to regenerate and sustain itself through entrepreneurship. Based on the theoretical justifications outlined above, the domains of the New Business Formation and Growth subindex are: 1) Human Capital, 2) Cultural Acceptance of Entrepreneurship, 3) Market Access, 4) Access to Startup Capital, 5) Government Supporting Policy, and 6) R&D Transfers. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Free Flow of Goods and Capital

Capitalist systems depend on the expansion and integration of markets within and across borders, enabled by the free movement of goods and capital. If, as Smith argued, the division of labor and specialization of capital is “limited by the extent of the market,” then the ability of human, physical, and financial capital to flow within and across borders is vital for effective capitalist systems.¹⁸⁹ The precondition of free movement must be applied to all forms of goods and capital to benefit from allocative efficiency. In order to benefit from comparative advantage, each country, as David Ricardo argues, “naturally devotes its capital and labor to such employments as are most beneficial to each”, including beyond its borders.¹⁹⁰ The free deployment of capital across borders reduces market distortions, increases the strength of private ownership, and encourages capital mobilization.¹⁹¹ The broader the market, the greater the specialization, and thus the greater potential for growth. This framework is inherently global, wherein widening the markets expands the potential for capital allocation. This logic applies not only to cross-border flows but to internal ones as well. Fragmented domestic markets can constrain the effective size of markets in otherwise open economies.

The open and free flow of goods and capital is the functional realization of this allocative efficiency. It signals the willingness of a given economy to integrate globally and the level of institutional capacity necessary to maintain access.¹⁹² Similarly, the ability of financial capital to flow across

¹⁸⁵William J. Baumol, “Entrepreneurship: Productive, Unproductive, and Destructive,” *Journal of Political Economy* 98, no. 5 (1990)

¹⁸⁶David Audretsch and Petra Moog, “Democracy and Entrepreneurship,” *Entrepreneurship Theory and Practice* 46, no. 4 (2022).

¹⁸⁷Zoltan J. Acs, “Patents, Knowledge Spillovers, and Entrepreneurship,” *Small Business Economics* 39, no. 4 (2012): 801–817; Mariana Mazzucato, *The Entrepreneurial State: Debunking Public vs. Private Sector Myths* (London: Anthem Press, 2013).

¹⁸⁸Henry Etzkowitz, *The Triple Helix: University-Industry-Government Innovation in Action* (New York: Routledge, 2008).

¹⁸⁹Adam Smith, *The Wealth of Nations* (New York: Random House, 1965).

¹⁹⁰David Ricardo, *On the Principles of Political Economy and Taxation* (London: John Murray, 1817).

¹⁹¹Milton Friedman, *Capitalism and Freedom* (Chicago: University of Chicago Press, 1962).

¹⁹²Alexander Gerschenkron, *Economic Backwardness in Historical Perspective: A Book of Essays* (Cambridge, MA:

borders with minimal restrictions facilitates risk diversification, broadens markets, and incentivizes active uses for stored savings or earnings. Additionally, the deployment of free human capital is a critical mechanism for capitalist systems to expand markets and incentivize market vibrancy. Hall and Soskice’s (2001) two capitalist models, liberal market economies and coordinated market economies, differ in how they achieve openness to global markets (the former largely through deregulation and the latter through coordinated export policy) but both models depend on broader participation in international trade.¹⁹³

Institutional commitment to broad, sustained trade participation requires a degree of state governance to mitigate potential shocks to the system and ensure national stability. In establishing the fundamentals of the early American economy, Alexander Hamilton believed in striking a balance between economic self-sufficiency and an open trade policy to sustain capitalism in the long-run. His *Report on Manufactures*¹⁹⁴ reflects this understanding of limited but energetic state interventions in trade policy to ensure domestic stability, and thus the ability of domestic firms to develop and produce goods and services for export, as well as helping “accelerate the transition to self-sustaining industrial capitalism.”¹⁹⁵ Contemporary scholarship on globalization reintroduces this tradeoff. Dani Rodrik’s (2011) globalization trilemma holds that national sovereignty, democracy, and broad economic integration cannot be maintained simultaneously.¹⁹⁶ The capacity to sustain trade openness is thus dependent on policies that reconcile external exposure with domestic stability. This logic is integrated into the subindex through the “Hamilton Indicator”, which yields high scores for low mean tariff rates with high levels of variation, weighted by a measurement of elite capture to account for the disequilibrium created by lobbying and policy objectives implemented without the public interest in mind. Deep trade integration extends the market logic of capitalism across borders, but risks state stability and domestic industry. Modern scholarship supports the distributional and political consequences of unfettered trade openness that motivates this logic. Goldberg and Pavcnik’s (2007) survey of globalization in developing countries demonstrates that trade liberalization has differential effects on skilled and unskilled workers and that distributional outcomes depend critically on domestic labor markets and educational institutions.¹⁹⁷ Autor, Dorn, and Hanson (2013) show that the geographic concentration of import competition can produce persistent local labor market dislocations even when aggregate national gains are positive, supporting Rodrik’s argument that sustained trade openness requires compensating domestic institutions.¹⁹⁸

The free flow of goods and capital is not solely determined by tariff policy or trade volume. IMF research on geoeconomic fragmentation (Aiyar et al. 2023) documents that the reversal of trade integration and globalization after 2018 through institutional decisions like tariffs, export controls, and foreign investment screening regimes had strong effects on capital flows and economic productivity. This reality underscores the importance of measuring institutional commitment to trade

Belknap Press of Harvard University Press, 1962).

¹⁹³Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001).

¹⁹⁴Alexander Hamilton, *Report on Manufactures* (Philadelphia: U.S. Government Printing Office, 1791), reprinted in *The Papers of Alexander Hamilton*, ed. Harold C. Syrett, vol. 10 (New York: Columbia University Press, 1966), 230–371.

¹⁹⁵Douglas A. Irwin, “The Aftermath of Hamilton’s ‘Report on Manufactures,’” *Journal of Economic History* 64, no. 3 (September 2004): 800–821.

¹⁹⁶Dani Rodrik, *The Globalization Paradox: Democracy and the Future of the World Economy* (W.W. Norton, 2011).

¹⁹⁷Pinelopi K. Goldberg and Nina Pavcnik, “Distributional Effects of Globalization in Developing Countries,” *Journal of Economic Literature* 45, no. 1 (March 2007): 39–82.

¹⁹⁸David H. Autor, David Dorn, and Gordon H. Hanson, “The China Syndrome: Local Labor Market Effects of Import Competition in the United States,” *American Economic Review* 103, no. 6 (October 2013): 2121–2168.

openness rather than assuming willingness or established levels of participation.¹⁹⁹ Economies may be formally open to international markets while remaining structurally constrained by narrow productive bases that limit their capacity to sustain diverse export participation. We use the Economic Complexity Index to capture the accumulated productive knowledge embedded across firms, workers, and institutions that enables an economy to generate a wide range of sophisticated goods. More complex economies can maintain participation across multiple export categories and global value chains, reducing dependence on narrow sectors and sustaining broad market integration even during periods of external disruption. This distinction between formal openness and structural capacity is central to measuring the genuine free flow of goods. The Free Flow of Goods and Capital subindex combines the capacity for sustained and integrated market flows through these three domains:

Based on the theoretical justifications outlined above, the domains of Free Flow of Goods and Capital subindex are: 1) Free Flow of Trade, 2) Free Flow of Capital, and 3) Free Flow of Human Capital. For full descriptions of each domain, see Appendix A. For a full list of the underlying datasets used in this subindex, see the accompanying GCI Codebook.

Compositional Methodology

The Global Capitalism Index organizes capitalism into eight areas, with each constructed as an independent subindex built on collections of domains. The GCI score is a composite measure empirically derived from those eight subindex scores via adaptive principle components analysis.²⁰⁰ This process uses the observed covariance structure rather than subjective judgment to measure each latent concept. By incorporating 17 years of data back to 2009, we capture the evolution of capitalist systems across time. Also, rather than focusing on a small subset of widely accepted capitalist (or non-capitalist) countries, we include 170 countries in our analysis. This process generates 0-100 scores for each country-year pair across the subindices and the composite index, along with bootstrapped confidence intervals. The subindex-to-composite design allows the components of capitalist economies to be studied separately as well as jointly. The sections below describe data harmonization, reliability scoring, the general adaptive PCA procedure, and the construction of the final composite index.

Data Collection and Harmonization

Every score in the GCI is the final product of an extensive data selection, collection, and normalization process. For each domain in a subindex (ex. Intellectual Property Protection within the Strength of Property Rights and Private Ownership subindex), we surveyed the empirical literature on that concept to identify the available collection of existing indices and measurements used by scholarly works. This ranges from international expert surveys (ex. V-Dem) to stock datasets (ex. trade to GDP ratio) to flow datasets (ex. net migration per year). Each domain within a subindex draws from up to 15 independent and publicly accessible data sources to ensure measure-

¹⁹⁹Shekhar Aiyar, Jiaqian Chen, Christian Ebeke, Roberto Garcia-Saltos, Tryggvi Gudmundsson, Anna Ilyina, Alvar Kangur, et al., “Gеоeconomic Fragmentation and the Future of Multilateralism,” IMF Staff Discussion Note SDN/2023/001 (January 2023).

²⁰⁰Principal component analysis is a dimensionality reduction method used to simplify the number of dimensions in a collection of potentially correlated variables datasets into a smaller set of variables, principal components (PC), that retain most of the original information. Each contributory element to a PC is given a year-adaptive factor load (ranging from -1 to 1). For our purposes, the factor load effectively functions as the final weight for that dataset or subindex. This is detailed further in the General Adaptive Principal Components Analysis Procedure section.

ment robustness and coverage. Each subindex draws from between 11 and 56 total datasets (see Table 1, above). By ensuring a breadth of data coverage for each domain of the GCI, we reduce overall sensitivity to unobserved errors even when elements of individual datasets are imperfect or inconsistent. We detail the domains that comprise each subindex in Appendix A, and all 213 underlying datasets in the accompanying GCI Codebook.

Differences in data coverage across the different domains are due to the complexity of each concept. A concept that is more easily measured in comprehensive and broadly available datasets (trade values, migration flows, etc.) can be covered with fewer, more directly related datasets. Where a concept has less direct measurements that are available (new business formation, market competition, rule of law, etc.), we assemble multiple datasets to better proxy the underlying the unobserved concept rather than search for a single source. The point is to measure the underlying institution through the joint pattern of many partial indicators, not through any one potentially imperfect one. We only use data from publicly accessible sources such as governmental and international organizations, academic publications, and private sector compilers.

Once datasets are compiled, the analysis begins by harmonizing them to ensure comparability. This entails transforming each data series into consistent and unified 0–10 scales so that higher values indicate stronger alignment with the thesis of the domain.²⁰¹ We used these consistent scale directions purely to investigate which datasets we wanted to include in a given subindex. Ultimately, datasets do retain their source scale direction because the resulting scores are unaffected. PCA recovers direction empirically rather than by assumption, so a reverse-coded series simply receives a negative weight on the component. The OECD FDI Regulatory Restrictiveness Index within Free Flow of Goods and Capital is the clearest instance, loading at -0.82 on that subindex’s first component against the Bertelsmann trade-openness measure at $+0.87$.

Data transformations use a combination of min-max normalization methods and z-score transformations.²⁰² For some datasets, we first took the natural log of values to ensure outliers did not dominate the scaled data. Data transformations (min-max normalization and Z-score scaling) utilize each dataset’s year-by-year minimum and maximum values. Year-specific data normalization allows us to compare data that would otherwise be incomparable, such as nominal values (GDP per capita, trade value of imports, etc.) and index or survey data (Executive Opinion Survey, World Bank expert assessments, etc.). This preserves year-on-year variation and avoids flattening legitimate changes in the cross-national distribution. Distorting or flattening that variation risks undermining the interpretability and credibility of our results. Many included datasets start and stop at different points and/or cover different sets of countries. We included datasets with coverage gaps in order to preserve as much information as possible. Missing values were handled by our index construction process as described with detail in the Period Identification and Clustering section. Each data transformation process per dataset is documented in the GCI Codebook.

For datasets where avoiding either extreme of the scale is preferable, we employ a parabolic maximization approach where a more central value yields a score of 10 and deviations away from that score (to the positive or negative direction) receive lower scores. This process is done for two datasets within the Private Ownership domain in the Strength of Property Rights and Private

²⁰¹For example, the nominal value of an intellectual property protection index (within the Protection of Property Rights subindex) that indicates the strongest possible level of IP protection is scaled to a 10. This is done across all datasets such that scores closer to 10 are aligned with higher levels of property rights protection.

²⁰²Min-Max normalizations transform data by rescaling them to a fixed range, in this case to 0-10, using the minimum and maximum values of the dataset per year. Z-Score transformations rescale datasets to a mean of 0 and a standard deviation of 1 and allows for easier comparison of data across different original scales and/or dispersion.

Ownership subindex.²⁰³

Reliability Scores

To account and adjust for varying levels of reliability and data quality across datasets, we scored each dataset from 1 (unreliable) to 5 (completely reliable) based on a weighted series of 6 factors (see Table 3). We then removed datasets scoring below 3/5 from the data pool and conducted a secondary review of its compilation method and source base. If a team of two researchers deemed individual components to hold potential value, we reassessed and graded decomposed data series for reliability. If any contributory source passed individual review by the same process, we reintegrated it back into the data pool.²⁰⁴ For example, we scored the Size of Government section of the Fraser EFW a 2/5, largely due to methodological opacity in the EFW’s data normalization, composition, and weighting methodologies. This score left it below our reliability threshold for inclusion in the Strength of Property Rights and Private Ownership subindex. However, after a detailed examination of the underlying elements of the EFW Size of Government section, we found it to be composed of datasets which met our threshold, including the V-Dem State Ownership indicator. We then integrated the reliable datasets into the subindex.

Reliability scores serve as both a filtering mechanism for prospective datasets and as an input to the PCA correlation matrix, as outlined in the Reliability Adjustment section.

Criterion	Weight
Data Accuracy & Reliability: This criterion focuses on the source’s trustworthiness, the accuracy of the data, and how consistent and reliable it is over time.	25%
Data Completeness & Granularity: This evaluates how complete each dataset is for each country and how detailed the information is.	25%
Data Transparency and Documentation: This evaluates how well the data is documented, including the availability of methodologies, sources, and standardization techniques.	25%
Timeliness and Frequency of Updates: The importance of timely data is critical, especially for fast-changing indicators like economic or political measures.	15%
Geographic Coverage: This core requirement is for complete geographic coverage of all countries. This criterion evaluates the breadth of coverage, including the inclusion of large and small countries, developing nations, and territories.	5%
Historical Depth: How far back does the data go, allowing for historical analysis and long-term trend identification?	5%

Table 3: Dataset Evaluation Criteria and Weights

²⁰³Please see Appendix C for more details on this process.

²⁰⁴We reviewed indices with good underlying data inputs but poor documentation on data transformation and index construction. In order to avoid any opacity, we gathered those datasets from their original sources and performed our own transformations.

General Adaptive Principal Components Analysis Procedure

A persistent challenge in index construction is the assignment of weights to component indicators. Researchers often assign arbitrary weights based on prior assumptions or normative judgments, but unless those weights coincide with the covariance structure among the variables, the resulting index does not recover the information in the data. Consider an index I constructed from n observed variables $x_1, x_2, \dots, x_n$ with assigned weights $w_1, w_2, \dots, w_n$:

$$I = \sum_{i=1}^n w_i x_i.$$

In a latent variable framework, the observed variables are assumed to be generated by an unobserved construct η according to a measurement model of the form $x_i = \lambda_i \eta + \epsilon_i$, where λ_i are factor loadings and ϵ_i represents idiosyncratic variation. A linear index I can then be viewed as an estimator of η . If the weights w_i do not align with the population-implied relationships between x_i and η , the index deviates from the optimal linear projection of η onto the space spanned by the observed variables. We can express the variance of the index as:

$$\text{Var}(I) = \mathbf{w}' \Sigma \mathbf{w},$$

where Σ is the covariance matrix of x . The variance of the estimation error $\epsilon = I - \eta$ therefore depends on the distance between the chosen weight vector $\mathbf{w}$ and the population-optimal weights implied by the covariance structure. Similarly, the covariance between the index and any individual indicator is:

$$\text{Cov}(I, x_k) = \sum_{i=1}^n w_i \sigma_{ik},$$

which shows that the interpretation of every index–indicator relationship is directly shaped by the imposed weights. Arbitrary weighting schemes thus produce indices that are suboptimal linear summaries of the observed data. They need not correspond to the principal directions of variation in Σ , nor to the optimal linear predictor of the latent factor.

The GCI addresses this by using PCA²⁰⁵ to select $\mathbf{w}$ as the eigenvectors of Σ , ensuring that the resulting index is the linear combination of observed variables that maximizes explained variance. The subindex indicators are reflective measures of an underlying capitalist institution: the latent construct causes the indicators rather than the reverse, so the dominant covariance structure in the data corresponds to the theoretical construct of interest, and PCA derives weights directly from that structure rather than from assignment. Where a subindex is theorized around more than one institutional dimension, that covariance structure is multidimensional by construction, which is why the procedure retains two components per subindex rather than forcing the indicator set onto a single one. We apply an adaptive procedure to address four further challenges: temporal variation in data availability, heterogeneous measurement reliability, missing data, and uncertainty quantification.

²⁰⁵This method is suggested for composite indices by the OECD in their handbook: OECD/JRC. 2008. Handbook on Constructing Composite Indicators: Methodology and User Guide. Paris: OECD Publishing.

Period Identification and Clustering

To account for structural differences in data availability over time, we do not use PCA over the full time series at once. Instead, we compute the percentage of non-missing entries for each variable-year combination, and years with similar availability patterns are grouped into periods using hierarchical clustering. This clustering is based on the similarity of missingness patterns across variables, ensuring that each period contained years with comparable data structures. We then segment the time series into between one and four periods, depending on each subindex's data availability structure, with periods ranging from one (Market-Supporting Policy, which uses a single joint model across all years) to four (Free Flow of Goods and Capital). This segmentation allows for more accurate estimation of principal components within each period and prevents distortions that could arise from inconsistent variable inclusion.

Reliability Adjustment

Before PCA estimation, we adjust the empirical correlation matrix $\mathbf{R}$ among variables in each period for known reliability differences. Let r_{ij} denote the correlation between variables i and j , and let $\rho_i \in [0, 1]$ denote the reliability of variable i . The adjusted correlation matrix $\mathbf{R}^*$ is constructed as:

$$r_{ij}^* = r_{ij} \cdot \sqrt{\rho_i \cdot \rho_j}$$

This method reduces the influence of less reliable indicators while preserving the correlation structure, ensuring that the resulting principal components are more robust to measurement error.

Principal Components Estimation and Scoring

For each temporal period, PCA is estimated on the reliability-adjusted correlation matrix $\mathbf{R}^*$. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ denote the matrix of mean-centered indicators and $\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}$ the sample covariance matrix. PCA solves:

$$\mathbf{S} \mathbf{w}_k = \lambda_k \mathbf{w}_k,$$

where λ_k is the variance explained by component k and $\mathbf{w}_k$ its loading vector. Scores are computed as:

$$z_{ik} = \mathbf{w}_k^\top \mathbf{x}_i, \quad \mathbf{Z} = \mathbf{X} \mathbf{W}.$$

We retain two components per period in every subindex and rotate them using oblimin to allow correlated institutional dimensions. The subindex score is the variance-weighted combination of both component scores:

$$s_i = \sum_k \left(\frac{\lambda_k}{\sum_k \lambda_k} \right) z_{ik}.$$

We report loadings using a 0.30 display threshold. Variables below this threshold remain in the underlying estimation and continue to affect scores. This follows common practice in PCA reporting, which treats 0.30 as the conventional minimum for a loading to be considered interpretable.²⁰⁶

Temporal Harmonization and Scaling

We ensure period-specific subindex scores are standardized within period before being harmonized to a common 0–100 scale via min-max normalization. Before estimation, each input variable is capped at its within-year 95th percentile, a one-sided winsorization that limits the influence of extreme high values while preserving within-period rank structure among the remaining observations. The resulting harmonized subindex scores are used as inputs to the second-order PCA that produces the composite GCI.

Missing Data and Imputation

Coverage is uneven at every level of the index. Source datasets begin and end in different years, several are discontinued partway through the panel, and some countries never report on a given domain at all. We address this separately at the indicator and subindex levels.

Within each subindex, indicator-level gaps are filled first by within-country linear interpolation, restricted to interior gaps of at most three years so that no value is carried beyond the span its neighbouring observations support. Values that remain missing are imputed by multiple imputation with predictive mean matching, which draws each imputed value from observed cases with similar predicted values rather than assigning a measure of central tendency, and so does not attenuate the correlations the PCA is estimated on. The imputation is repeated inside every bootstrap draw, so imputation uncertainty propagates into the subindex confidence intervals.

The second-order PCA requires a complete country-by-subindex matrix. We first exclude country-years observed on fewer than five of the eight subindices. This coverage count is computed before any imputation, so every scored country-year rests on at least five directly observed subindices and no score depends on more than three imputed ones. Of the 22,472 subindex cells in the scored panel, 923 (4.1%) require filling, affecting 526 of the 2,809 scored country-years; the remaining 81.3% of country-years are observed on all eight subindices. For those gaps we again interpolate within country first, which resolves 456 of the 923 cells. The 467 cells interpolation cannot reach (those where a country is observed only once on a subindex, or never) are imputed by predictive mean matching with $m = 20$ completed datasets and ten iterations each. Estimates from the twenty datasets are pooled by Rubin’s rules. Imputation substitutes for measurement and is not equivalent to it. Imputed subindex values are therefore withheld from the published data file, which instead reports, for each country-year, how many subindices were directly observed.

Bootstrap Uncertainty Quantification

Bootstrap procedures are used to quantify uncertainty in both subindex and final index scores. For each subindex, we generate bootstrap confidence intervals through resampling with replacement across 500 iterations. We resample observations with replacement from the analysis dataset, re-estimate the reliability-adjusted correlation matrix, apply PCA with oblimin rotation to the bootstrap sample, score the original observations using bootstrap factor loadings, and repeat for all

²⁰⁶See Joseph F. Hair Jr., William C. Black, Barry J. Babin, and Rolph E. Anderson, *Multivariate Data Analysis*, 8th ed. (Boston: Cengage Learning, 2018).

bootstrap iterations. We calculate confidence intervals using percentile methods from the bootstrap distribution:

$$\text{CI}_\alpha = \left[\text{Quantile}_{\alpha/2}(\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(B)}), \text{Quantile}_{1-\alpha/2}(\hat{\theta}^{(1)}, \dots, \hat{\theta}^{(B)}) \right]$$

where $\hat{\theta}^{(b)}$ represents the index estimate from bootstrap sample b . For 95% confidence intervals, $\alpha = 0.05$, yielding the 2.5th and 97.5th percentiles of the bootstrap distribution.

We winsorize outlier bootstraps at the 2.5% level to improve interval stability. When insufficient bootstrap samples are available (less than 30% success rate), we construct conservative confidence intervals using normal approximations based on the overall sample standard deviation.

The percentile procedure above applies to the subindices. At the composite level uncertainty arises from both variation and imputation. We draw 250 bootstrap replicates within each of the $M = 20$ imputed datasets, taking from each a point estimate $\hat{Q}_m$ and a within-imputation variance $\hat{U}_m$, and pool them by Rubin’s rules:

$$\bar{Q} = \frac{1}{M} \sum_{m=1}^M \hat{Q}_m, \quad \bar{U} = \frac{1}{M} \sum_{m=1}^M \hat{U}_m, \quad B = \frac{1}{M-1} \sum_{m=1}^M \left(\hat{Q}_m - \bar{Q} \right)^2, \\ T = \bar{U} + \left(1 + \frac{1}{M} \right) B, \quad \text{CI}_{95} = \bar{Q} \pm t_{0.975, \nu} \sqrt{T},$$

where B is the between-imputation variance, T the total variance, and ν the Barnard–Rubin degrees of freedom. Country-years requiring no imputation have $B = 0$ and reduce to a bootstrap-only interval, with those resting on imputed subindices receive correspondingly wider ones.

Subindex-Specific Methodologies

Each subindex follows the general framework described above but adapts it to its specific data structure and measurement challenges.

Strength of Property Rights and Private Ownership Subindex

The Property Rights subindex draws on 31 indicators spanning judicial independence, anti-corruption, intellectual property protection, contract enforcement, and state ownership. Sources include the V-Dem judicial and political indicators, World Bank Governance Indicators (Rule of Law, Government Effectiveness, Control of Corruption, Voice and Accountability, Political Stability), the Corruption Perceptions Index, the World Justice Project Rule of Law Index, the WEF Intellectual Property Protection survey, the Fragile States Index, the RSF Press Freedom Index, the IPRA Patent Index, the Wharton Political Constraints Index (POLCON), and the World Bank Resolving Insolvency and Enforcing Contracts scores.

The subindex is structured around two temporal periods: Period 1 (2009–2021) and Period 2 (2022–2025), reflecting the shift in data availability for several newer indicators after 2021. Period 1 uses 29 variables and Period 2 uses 31 variables. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations. The leading contributors to both periods are the WJP Rule of Law Index, World Bank Control of Corruption, Corruption Perceptions Index, World Bank Rule of Law, Fragile States Index, and World Bank Government Effectiveness.

Market-Supporting Policy Subindex

The Market-Supporting Policy subindex measures government policies, institutional quality, and physical infrastructure that enable market activity. It draws on 20 indicators from two domains: Market Initiatives (WEF indicators of regulatory efficiency, policy stability, government responsiveness, legal framework effectiveness, ICT development, road density, shipping connectivity, and air freight volume, and support for start-ups) and Operational and Institutional Quality (World Bank Regulatory Quality, V-Dem Public Sector Corruption Index, the Hamilton Indicator, and the Central Bank Independence Index).

The subindex covers a single temporal period (2009–2025). All 20 variables are used across the single period. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations.

Labor Market Openness Subindex

The Labor Market Openness subindex measures the flexibility, accessibility, and inclusivity of labor markets. It combines 13 indicators drawn from three thematic domains: Labor Market Flexibility (WEF Hiring and Firing Practices, WEF Labor–Employer Relations, Ease of Hiring Foreign Labor, Flexibility of Wage Determination, Pay and Productivity); Labor Protections (ILO Compliance with Labor Rights, Freedom from Forced Labor, Wage Inequality Between Men and Women, Inclusion of Women in Leadership); and Labor Availability (Foreign-Born Share of the Labor Force, Internal Labor Force Mobility, Fill Vacancies by Hiring Foreign Labor, WEF Ease of Finding Skilled Labor in Local Labor Markets).

The subindex is structured around two temporal periods: Period 1 (2009–2017) and Period 2 (2018–2025). Period 1 uses 12 variables and Period 2 uses 13. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations. The most influential variables across periods are the Inclusion of Women in Leadership, WEF Labor–Employer Relations, Pay and Productivity, Wage Inequality Between Men and Women, and WEF Hiring and Firing Practices.

Market Competition Subindex

The Market Competition subindex measures the competitive intensity of markets, the dynamism of creative destruction, and the regulatory and governance environment supporting competition. It draws on 31 indicators across three domains: Creative Destruction (WEF market dynamism, company adaptability, R&D collaboration, entrepreneurial risk, competitive advantage, and business density rates); Regulatory Analysis (World Bank Governance Indicators: Regulatory Quality, Control of Corruption, Political Violence, Government Effectiveness, Voice and Accountability, Rule of Law, and Resolving Insolvency); and Human Capital Competitiveness (Global Talent Competitiveness Index, firm collaboration, digital skills, and workforce mobility indicators).

The subindex is structured around two temporal periods: Period 1 (2009–2017) and Period 2 (2018–2025). Both periods use 31 variables. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations.

Capital Market Sophistication Subindex

The Capital Market Sophistication subindex evaluates the depth, accessibility, and resilience of financial markets. It draws on 29 indicators across seven domains: Access and Participation – Infrastructure (WEF local capital market access, IMF bank branches, IMF ATMs, Global Findex savings and borrowing rates); Access and Participation – Transactions (WEF access to loans, WEF financial services for business needs, syndicated loan issuance and maturity, domestic credit to the private sector, credit to government and SOEs, and mobile money and mobile/internet banking transaction volumes); Equity Market Depth & Scope (mutual fund assets, stock market turnover, stock price volatility, WEF regulation of security exchanges); Bond Market Depth and Scope (interest rate spread, corporate bond issuance and maturity); Venture Capital Availability (WEF VC availability); M&A Prevalence (cross-border M&A sales and purchases); and Liquidity and Resilience (IMF reserve adequacy ratio, NYU country risk premium from ratings and CDS, and liquid liabilities to GDP).

The subindex is structured around three temporal periods: Period 1 (2009–2015), Period 2 (2016–2020), and Period 3 (2021–2025). Period 1 uses 25 variables, Period 2 uses 25 variables, and Period 3 uses 17 variables. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations. The most consistently influential variables across all three periods are WEF regulation of security exchanges, domestic credit to the private sector, WEF local capital market access, WEF ease of access to loans, WEF venture capital availability, and NYU country risk premium (ratings-based).

Depth and Stability of Banking System Subindex

The Depth and Stability of Banking System subindex assesses the size, accessibility, and prudential soundness of the banking sector. It draws on 22 indicators across two domains: Banking System Depth (IMF Financial Integration indices of access, depth, and efficiency; World Bank five-asset index; private credit and deposit assets as a share of GDP; commercial bank branches per 100,000 adults; credit/debit card penetration; digital payments; mobile money transfers; WEF access to and innovation in financial services); and Banking System Stability (WEF soundness of banks, WEF financial resilience, World Bank Z-score, bank capital-to-assets ratio, bank regulatory capital to risk-weighted assets, nonperforming loans as a share of gross loans, bank liquidity coverage ratio, bank return on assets, and liquid assets to short-term liabilities).

The subindex is structured around three temporal periods driven by structural shifts in data availability: Period 1 (2009–2016), Period 2 (2017–2022), and Period 3 (2023–2025). Period 1 uses 18 variables, Period 2 uses 17 variables, and Period 3 uses 21 variables. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations. The most consistently influential variables are IMF financial integration depth and access, WB private credit, deposit assets per GDP, credit card penetration, debit card penetration, and commercial bank branch density.

New Business Formation and Growth Subindex

The New Business Formation and Growth subindex evaluates the ecosystemic conditions for entrepreneurship, firm creation, and business growth. It draws on up to 56 indicators across six domains: Human Capital (WEF indicators of workforce skills, education quality, internal mobility,

and staff training; UNDP average years of schooling; UNESCO tertiary enrollment and government education expenditure; V-Dem educational equality); Cultural Acceptance of Entrepreneurship (WEF attitudes toward entrepreneurial failure; GEM indicators of perceived opportunities, perceived capabilities, fear of failure, entrepreneurial attitudes, and status of entrepreneurs); Market Access (HHI market concentration, WEF intensity of local competition, market dominance, and state of cluster development, V-Dem state ownership, WB nominal GDP, logistics performance index, trade taxes, and WEF antitrust effectiveness); Access to Startup Capital (WEF access to loans and VC availability, IMF deposit accounts and borrowing from commercial banks, credit unions, and microfinance institutions, domestic credit to the private sector, WEF protection of minority shareholders, and SME financing); Government Supporting Policy (WEF regulatory compliance, legal framework efficiency, government long-term vision, hiring and firing flexibility, distortive effects of taxes and subsidies, ease of hiring foreign labor, non-tariff barrier prevalence, and growth of innovative companies; WB women business law index); and R&D Transfers (GERD as a percentage of GDP, WEF university–industry collaboration, WEF company R&D spending, GEM R&D transfer, and patent families by origin).

The subindex is structured around two temporal periods identified via hierarchical clustering of data availability: Period 1 (2009–2019) and Period 2 (2020–2025). This reflects the entry or restructuring of several WEF Competitiveness Report indicators beginning in 2020. Period 1 uses 52 variables and Period 2 uses 46 variables, of which 45 and 43 respectively load above the 0.30 display threshold. We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations.

Free Flow of Goods and Capital Subindex

The Free Flow of Goods and Capital subindex measures the openness of goods, capital, and human capital flows across borders. It draws on up to 11 indicators across three domains: Free Flow of Trade (the Bertelsmann Transformation Index trade openness subindex, the Economic Complexity Index, the Hamilton Indicator, trade as a percentage of GDP, Herfindahl–Hirschman Index of export concentration, and the Logistics Performance Index); Free Flow of Capital (the Chinn–Ito Financial Openness Index, FDI net flows as a percentage of GDP, and the OECD FDI Regulatory Restrictiveness Index); and Free Flow of Human Capital (net migration per 1,000 population and WEF Ease of Hiring Foreign Labor).

The subindex is structured around four temporal periods reflecting data availability shifts and the granular structure of the 11-variable dataset: Period 1 (2009–2012), Period 2 (2013–2016), Period 3 (2017–2020), and Period 4 (2021–2025). We extract two components via oblimin rotation in each period, with the final score formed as a variance-weighted combination of both. We also calculate bootstrap confidence intervals from 500 iterations. The most influential variables across all periods are the BTI trade openness subindex and the Economic Complexity Index, followed by the Chinn–Ito financial openness index, the Hamilton Indicator, export concentration, trade as a percentage of GDP, and FDI net flows.

Final Index Construction Using Second-Order PCA

We construct the composite GCI score by applying second-order PCA to the eight subindex scores. Each subindex score is produced by the adaptive PCA described above and scaled 0–100.

The eight subindex scores serve as inputs to a second-order PCA that extracts the common dimen-

sion underlying capitalist development. Let $\mathbf{z}_i = (z_i^{(1)}, z_i^{(2)}, \dots, z_i^{(8)})^\top$ be the vector of subindex scores for country-year i , where the eight components correspond to the subindices listed above.

We conduct a second-order PCA on the matrix of subindex scores $\mathbf{Z} \in \mathbb{R}^{N \times 8}$, where N is the number of country-year observations. Unlike the subindices, which retain two components each, the second-order model extracts a single principal component. Rotation is undefined for a one-component solution, so none is applied. That component explains 73.9% of variance across the eight subindices (73.2% before attenuation correction) and serves as the composite index score:

$$\text{GCI}_i = \mathbf{w}_1^\top \mathbf{z}_i$$

where $\mathbf{w}_1$ is the first eigenvector of $\mathbf{R}^*$ and $\mathbf{z}_i$ is the vector of subindex scores for observation i . Subindex loadings on this component range from 0.677 (Labor Market Openness) to 0.952 (Market Competition).

Raw composite scores are then transformed using a Box-Cox power transformation with $\lambda = 0.400$ to reduce right skew before min-max normalization to the 0–100 scale. The exponent is re-estimated by maximum likelihood on the final score distribution in each run rather than fixed in advance, and improves normality without distorting country rankings. Normalization uses the global minimum and maximum of the transformed panel, so the 0–100 endpoints are anchored on the lowest- and highest-scoring country-years rather than on within-year extremes. No winsorization is applied at the composite level.

The final index bootstrap procedure is more complex than those used for individual subindices because of its hierarchical structure. Within each of the $M = 20$ imputed datasets we resample observations with replacement from the combined subindex matrix, re-estimate the second-order PCA on the bootstrap sample, and score the original observations using bootstrap-derived weights, for 250 replicates per imputation and 5,000 draws in total. Rather than taking percentiles of the pooled draws, we summarize each imputation by its point estimate and within-imputation variance and combine them by Rubin’s rules as set out above, so that the reported interval carries both sampling and imputation uncertainty.

Our approach ensures that the final index reflects the dominant axis of variation across all subindices, with weights determined by the observed covariance structure. The resulting index is continuous and interpretable as a summary measure of capitalistic strength. The use of second-order PCA is justified on several grounds. The principal components are uncorrelated, ensuring that the index is not unduly influenced by redundant information across subindices. The first principal component captures the maximum possible variance in the subindex space, providing an efficient summary. The loadings v_{1d} are determined by the data, not by subjective judgment, and reflect the relative importance of each subindex in explaining overall variation. Because all subindices are standardized, the resulting index is not sensitive to differences in scale or units across datasets.

GCI scores are continuous 0–100 estimates for each country-year, where higher values indicate stronger alignment with the institutional features of market capitalism. Scores are accompanied by 95% bootstrap confidence intervals.

Peer Review Process

After the completion and composition of each subindex and the composite capitalism index scores, we conducted an extensive peer review process. Over 7 months between September 2025 and March

2026, we solicited and received review reports from 10 expert scholars covering all 8 subindices.²⁰⁷ We asked each reviewer to provide a comprehensive assessment of one designated subindex in which subject they had particular expertise, as well as the broader capitalism index. Comprehensive instructions as provided to the reviewers is available in Appendix B. In summary, we requested that reviewers evaluate their subindex’s theoretical framework, assess the validity of each domain, and consider the merit of each data used in each domain. For the composite capitalism score, we requested they evaluate the full index design and review our composition/weighting methodology. We then solicited a final evaluation of 1) recommend approval, 2) recommend approval with the addressing of minor concerns, 3) recommend major revision and resubmission, or 4) Reject publication.

In 2 cases, the reviewer recommended approval of the Index model without major revisions. The remaining 8 recommended significant changes, either to the theoretical model, dataset inclusion or exclusion, or quantitative methodology. In all 8 cases, we conducted extensive revisions based on the review reports, often in direct consultation with the reviewers themselves, and eventually produced an detailed response to the reviewer. The reviewer reports and our responses can be accessed at www.capitalismindex.com. In the end, nine of the 10 reviewers recommended full publication. The one expert who did not recommend publication did so due to a “lack of clear substantive contributions,” not theoretical or methodological errors. As such, and due to a full recommendation for publication of that subindex by a second scholar, we have proceeded with publication.

Results

The overall GCI covers 170 countries from 2009 to 2025, with annual coverage ranging from 158 to 169 countries. All scores are scaled 0–100.

Most Recent Results (2025)

The 2025 composite mean is 63.5 (SD = 17.5), with a median of 62.5 and an interquartile range of approximately 25 points. Table 4 lists the top-20 and bottom-20 countries.

The top tier consists of advanced liberal market economies and open financial centers. The bottom tier is concentrated in Sub-Saharan Africa, conflict-affected states, and countries with sustained authoritarian suppression of markets. Switzerland leads at 96.9; Venezuela scores 8.9.

Second-Order Latent Composite Structure and Composite Validity

The composite GCI is constructed via second-order PCA over the eight subindex scores. A single principal component (PC1) explains 73.9% of shared variance across the subindices (73.2% before attenuation correction; see Methodology). All eight subindices load positively on this component with loadings ranging from 0.677 (Labor Market Openness) to 0.952 (Market Competition), confirming that they measure a coherent underlying construct rather than orthogonal dimensions. The convergence of high first-component variance explanation and uniformly positive loadings confirms that the eight subindices behave as reflective indicators of a single underlying construct, consistent with the theoretical framework motivating the index design.

²⁰⁷See Appendix B for a list of our peer reviewers and the subindices they reviewed.

Rank	Country	Score	Rank	Country	Score
1	Switzerland	96.9	142	Burkina Faso	45.9
2	United States	95.0	143	Liberia	45.1
3	Hong Kong	94.5	144	Guinea-Bissau	44.2
4	Canada	93.8	145	Madagascar	43.2
5	Singapore	93.3	146	Bangladesh	42.2
6	Luxembourg	92.5	147	Nigeria	41.9
7	Denmark	91.9	148	Bolivia	41.4
8	Australia	91.6	149	Equatorial Guinea	40.2
9	Sweden	90.6	150	Mozambique	39.4
10	Norway	90.3	151	Iran	39.3
11	New Zealand	90.3	152	Haiti	33.1
12	United Kingdom	90.1	153	Myanmar	31.3
13	Taiwan	89.8	154	Yemen	28.8
14	Israel	88.7	155	Angola	27.3
15	Iceland	88.5	156	Dem. Rep. Congo	26.3
16	Finland	88.3	157	Sierra Leone	24.7
17	Japan	87.5	158	Mauritania	24.2
18	Netherlands	87.4	159	Chad	24.1
19	Belgium	86.6	160	Libya	21.8
20	United Arab Emirates	85.3	161	Venezuela	8.9

Table 4: GCI 2025 Rankings: Top-20 and Bottom-20 Countries

Table 5 reports both estimation stages for each subindex. The within-subindex columns describe how well a subindex coheres as a measure in its own right: the mean pairwise correlation among its own indicators ($\bar{r}$), and the share of their joint variance carried by that subindex’s first component. The composite columns describe its relation to the second-order factor: its PC1 loading (attenuation-corrected), the proportion of subindex-score variance explained by the composite factor (communality, h^2), and the reliability coefficient (α) derived from the bootstrap variance-propagation procedure described in the Compositional Methodology section. Reliability coefficients reflect the precision of each subindex estimate; values above 0.97 for all eight subindices indicate that measurement error is a negligible source of uncertainty in the composite.

Market Competition (0.952) and New Business Formation (0.950) are the two strongest-loading subindices, capturing dimensions of market dynamism most closely associated with the general capitalism construct. Labor Market Openness loads at 0.677, the lowest of the eight, with a communality of 0.458. The lower communality indicates it measures institutional variation partly distinct from the other seven pillars and contributes unique information to the composite rather than duplicating it.

Subindex	Within subindex			On composite			2025 Mean
	$\bar{r}$	PC1	PC2	Loading	h^2	α	
Market Competition	0.525	47.4%	13.7%	0.952	0.906	0.995	65.3
New Business Formation	0.301	36.4%	17.3%	0.950	0.902	0.999	44.5
Market-Supporting Policy	0.481	30.0%	20.5%	0.884	0.782	0.987	63.4
Property Rights	0.486	37.9%	23.5%	0.870	0.757	0.998	55.6
Capital Market Soph.	0.150	36.9%	15.4%	0.859	0.739	0.981	43.2
Banking Stability	0.101	26.7%	15.6%	0.830	0.688	0.980	60.6
Free Flow of Capital	0.153	31.9%	17.6%	0.824	0.678	0.977	53.0
Labor Market Openness	0.240	43.9%	15.3%	0.677	0.458	0.987	50.4
<i>Composite PC1 variance</i>				73.9% corrected; 73.2% uncorrected			

Note: Rows are ordered by communality. *Mean $\bar{r}$* is the average pairwise correlation among a subindex’s own indicators, computed on observed values before reliability adjustment. *PC1* and *PC2* are the proportions of within-subindex variance carried by each of the two retained components after oblimin rotation; their sum is the share of indicator variance the subindex retains, and their ratio is the weight each component receives in the score. *Loading* is the attenuation-corrected second-order PC1 loading, h^2 the share of the subindex score explained by the composite factor, and α the reliability coefficient from the bootstrap variance-propagation procedure. We report $\bar{r}$ rather than Cronbach’s α , which rises mechanically with indicator count: New Business Formation’s 56 datasets yield $\alpha = 0.960$ at $\bar{r} = 0.301$. Indicator counts per subindex are given in Table 1.

Table 5: Subindex Internal Coherence and Second-Order Composite Structure

Per-Subindex Internal Validity

Each subindex is estimated independently via adaptive PCA (see Methodology). The within-subindex columns of Table 5 report how far that estimation recovers a single signal in each case.

The four highest-communality subindices are also the four whose indicators agree most strongly, correlating at $\bar{r} = 0.301$ to 0.525, with their first component carrying between 30% and 47.4% of their joint variance. The three financial and trade subindices are substantially lower on indicator agreement ($\bar{r} = 0.101$ to 0.153), though Capital Market Sophistication’s first component (36.9%) is stronger than Market-Supporting Policy’s (30.0%). This is expected because the three financial and trade subindices are theorized around more than one institutional dimension, meaning PC1 and PC2 will be closer in size. Banking Stability joins depth to prudential soundness, and Free Flow joins goods and capital mobility. A deep banking system is not thereby a stable one, and the subindex is inherently built to measure this rather than average them. Low mean inter-item correlation is also expected from a construct created from several dimensions. All three are nonetheless estimated with high precision and load between 0.824 and 0.859 on the composite GCI, demonstrating shared institutional strength.

Every subindex is estimated the same way: two components per period, oblimin-rotated, with the score formed as the variance-weighted combination of both. A variable’s contribution to its subindex is therefore that same weighted combination of its two loadings, and the dominant indicators below are reported on that common basis. Because the weighting is signed, a variable that loads negatively enters with a negative contribution, indicating a series informative in the reverse direction rather than one that detracts from the construct.

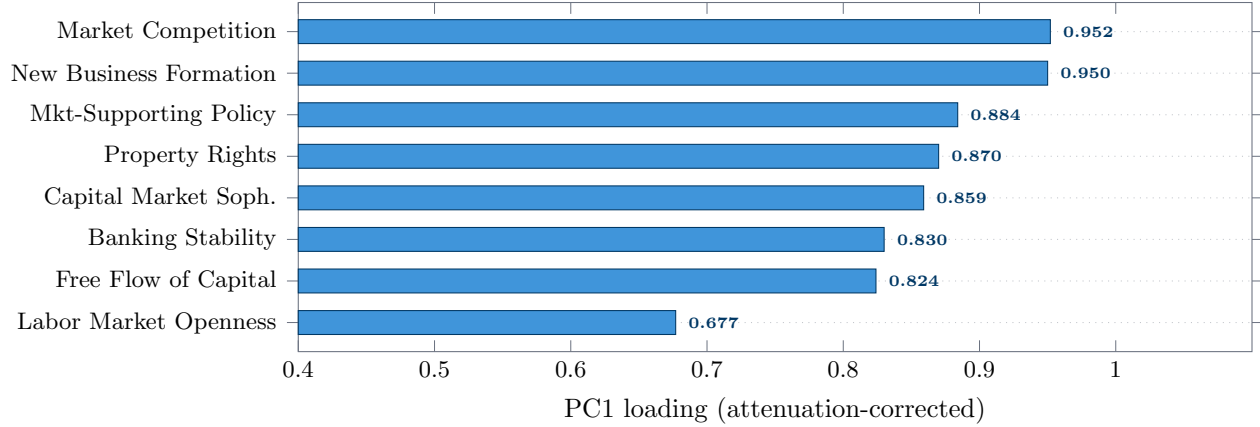


Figure 3: Second-order PC1 loadings by subindex, ordered by magnitude. All eight subindices load positively on the composite, from 0.677 (Labor Market Openness) to 0.952 (Market Competition). Values as reported in Table 5.

Cross-Sectional Distributions (2025)

Table 6 presents the cross-sectional distribution of each subindex and the composite GCI in 2025. Coverage varies by subindex (123–175 countries) and reflects data availability in each domain. The statistics are computed from the final harmonized scores after chain-linking, coverage-weighted shrinkage, and second-order PCA scaling.

Subindex	<i>N</i>	Mean	SD	Med	P10	P25	P75	P90	Min	Max
Market Competition	159	65.4	16.5	63.1	47.9	55.1	75.5	90.3	0.0	96.2
Mkt-Supporting Policy	158	63.4	15.1	63.2	45.9	54.4	73.7	85.2	14.3	93.5
Property Rights	175	53.4	22.6	53.2	24.4	37.2	68.2	85.6	8.3	97.2
Labor Market Openness	123	51.5	16.7	50.5	31.6	39.0	64.3	75.1	16.9	93.0
Free Flow of Capital	145	53.2	15.1	54.8	30.4	44.1	64.8	70.5	13.7	85.1
Banking Stability	166	60.2	17.1	61.9	39.3	49.4	71.0	81.8	2.7	96.0
New Business Formation	153	45.7	18.5	42.2	24.4	33.6	56.3	74.4	3.9	89.2
Capital Market Soph.	161	43.3	14.9	43.9	24.6	31.1	52.5	62.3	9.0	78.9
GCI Composite	161	63.5	17.5	62.5	43.2	52.3	77.4	87.5	8.9	96.9

Table 6: Subindex and Composite Distributions, 2025 (*n* varies by subindex coverage; *n* = 161 composite)

Market Competition (mean 65.4) and Market-Supporting Policy (mean 63.4) are the best-developed pillars globally. Property Rights has the widest standard deviation (22.6), making it the most differentiating dimension across countries. Capital Market Sophistication has the narrowest dispersion (SD = 14.9) and the lowest observed maximum (78.9) of any subindex, indicating that capital market development is constrained even in advanced economies. New Business Formation is right-skewed (mean 45.7, median 42.2), with a concentration of low-scoring countries and a thin upper tail.

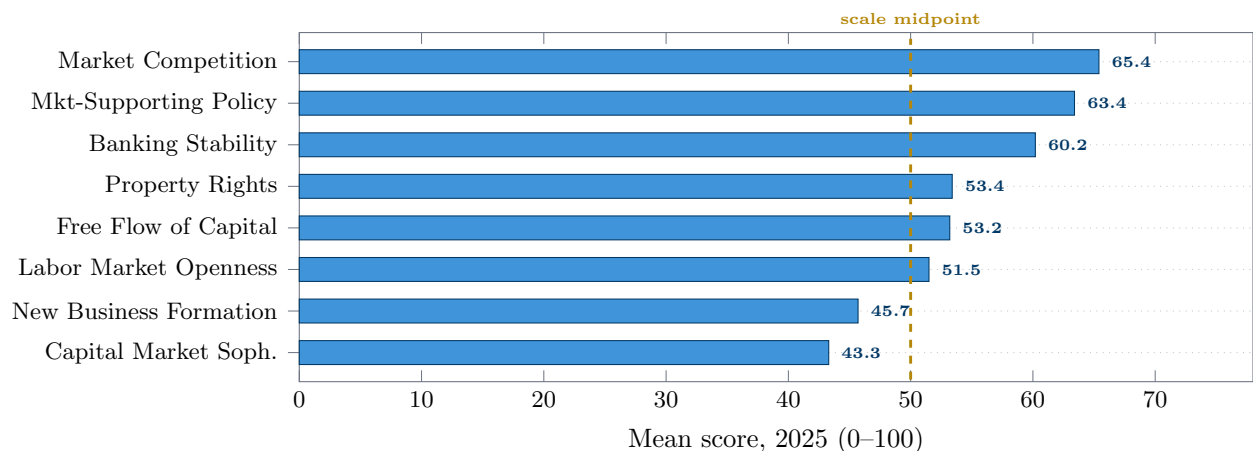


Figure 4: Mean subindex scores, 2025, ordered by magnitude; the dashed gold line marks the scale midpoint. Means are computed on each subindex’s own coverage sample ($n = 123\text{--}175$; see Table 6) and therefore differ slightly from the composite-sample means ($n = 161$) reported in Table 5.

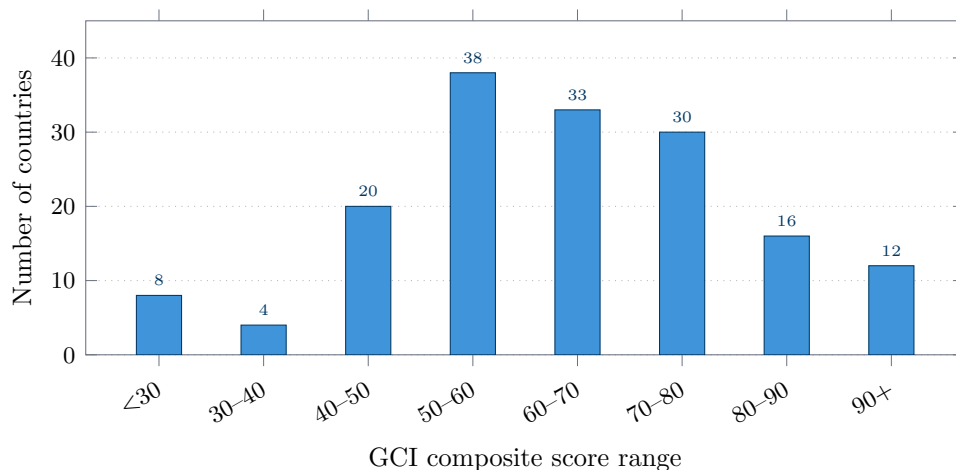


Figure 5: Distribution of GCI composite scores, 2025 ($n = 161$). Twelve countries score above 90 and eight below 30; the mass of the distribution lies between 40 and 80. The absence of a bimodal pattern indicates that the composite captures a continuum of market-system development rather than a binary split.

Longitudinal Trends, 2009–2025

Table 7 reports global mean scores for each subindex and the composite in each year from 2009 to 2025. All values are computed from the final harmonized scores in the full-index dataset.

Year	Bank	CapMkt	Free	Labor	MktComp	MktPol	NewBiz	PropRts	GCI
2009	50.2	52.2	58.2	54.1	65.2	62.8	41.8	51.9	64.1
2010	51.1	48.4	58.4	53.4	65.1	62.7	41.9	51.9	63.6
2011	51.1	47.2	57.6	50.5	65.8	62.4	42.5	52.0	63.0
2012	51.6	45.6	57.1	51.4	66.1	62.5	43.8	51.9	63.1
2013	51.7	45.1	57.3	53.5	65.9	62.7	43.6	52.1	63.3
2014	52.3	44.0	57.7	53.9	65.9	63.1	44.1	52.0	63.6
2015	52.6	44.4	57.5	54.1	65.8	63.4	44.6	52.5	63.8
2016	53.9	44.4	57.6	52.5	65.5	63.8	43.6	52.7	63.7
2017	53.9	45.0	57.4	52.1	65.5	63.0	46.1	52.9	63.8
2018	54.8	45.2	54.9	52.5	65.4	62.4	45.8	53.2	63.4
2019	54.9	45.0	53.6	52.1	65.7	63.0	46.2	53.2	63.4
2020	58.2	45.0	51.9	52.5	65.7	63.0	46.9	53.1	63.6
2021	58.8	44.3	50.4	53.3	66.7	63.7	47.8	53.2	63.9
2022	60.9	43.9	51.0	52.3	66.2	63.2	46.6	53.8	64.2
2023	60.5	43.0	51.0	51.3	66.4	63.4	47.1	52.9	63.5
2024	60.4	43.4	51.4	53.1	66.6	62.9	47.5	52.9	63.7
2025	60.2	43.3	53.2	51.5	65.4	63.4	45.7	53.4	63.5
Δ 2009–2025	+10.0	−8.9	−5.0	−2.6	+0.2	+0.6	+3.9	+1.5	−0.6

Bank = Banking Stability; *CapMkt* = Capital Market Sophistication; *Free* = Free Flow of Capital; *Labor* = Labor Market Openness; *MktComp* = Market Competition; *MktPol* = Market-Supporting Policy; *NewBiz* = New Business Formation; *PropRts* = Property Rights.

Table 7: Global Mean Subindex and Composite Scores, 2009–2025

The 17-year trends, plotted in Figure 6, fall into four patterns.

Decliners. Capital Market Sophistication fell 8.9 points (52.2 to 43.3). The drop occurs early as 8.2 points were lost between 2009 and 2014, consistent with post-GFC financial retrenchment and also with modest continued erosion after. Free Flow of Capital fell 5.0 points (58.2 to 53.2). The subindex’s sharpest decline, 3.0 points, occurred in 2018–2020, coinciding with documented increases in capital controls. Labor Market Openness declined 2.6 points (54.1 to 51.5), with most of the loss in 2010–2011 and resumed erosion from 2022.

Improvers. Banking Stability rose 10.0 points (50.2 to 60.2), the largest gain of any subindex, with steady increases from 2016 onward reflecting improvements in financial depth and stability indicators. New Business Formation rose 3.9 points (41.8 to 45.7) with consistent gains through 2021, followed by slight reversal. Property Rights improved 1.5 points (51.9 to 53.4), increasing gradually over the panel.

Stable pillars. Market Competition (+0.2) and Market-Supporting Policy (+0.6) each moved within a 2-point band over 17 years. Governance quality and competitive institutions are slow-moving; these results are consistent with that expectation.

Composite. The GCI fell 0.6 points (64.1 to 63.5) over 2009–2025. The large Banking Stability

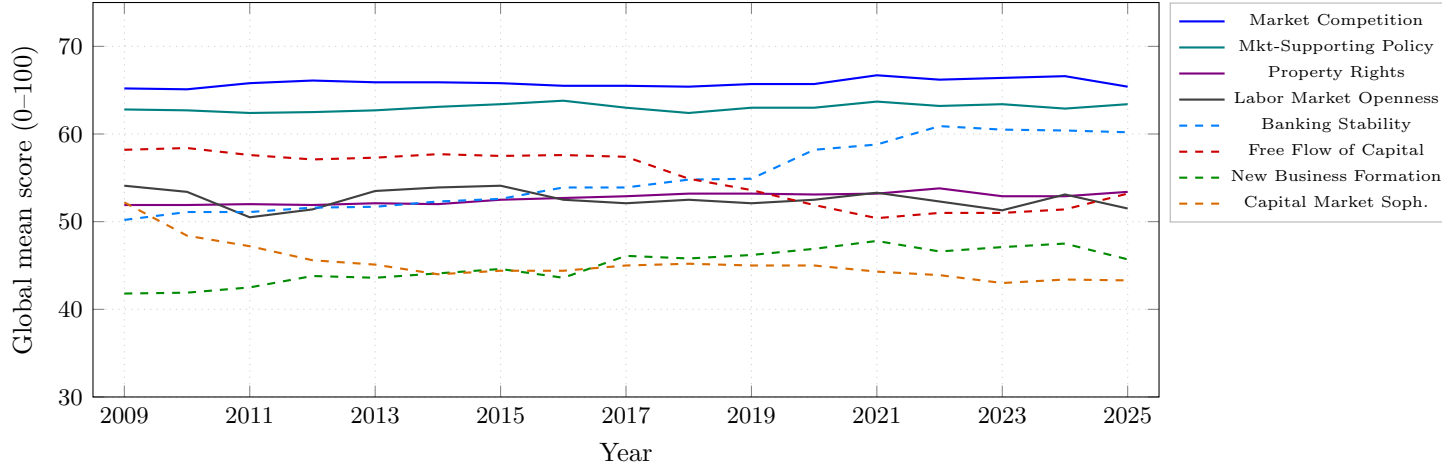


Figure 6: Global mean GCI subindex scores, 2009–2025 (all 17 annual observations)

improvement largely offsets the declines in Capital Market Sophistication and Free Flow of Capital, leaving the composite nearly stable. Market Competition and Market-Supporting Policy, the two strongest-loading subindices, are essentially flat, which further attenuates composite movement.

Robustness and Temporal Stability

Bootstrap confidence intervals. Each country-year GCI score is accompanied by a 95% confidence interval. Subindex intervals derive from 500 bootstrap resamples, with multiple imputation nested within each draw. Composite intervals combine 250 bootstrap replicates drawn within each of the 20 imputed datasets, pooled by Rubin’s rules, so that sampling variation and imputation uncertainty are both represented. In 2025, the mean 95% CI half-width across all 161 scored countries is 1.72 points (median: 1.20; P10: 0.74; P90: 3.81; maximum: 8.97). Across the full panel, intervals widen monotonically with the number of imputed subindices, from 1.34 points for fully observed country-years to 4.40 for those observed on five of eight.

Rank stability. The Spearman rank correlation between 2024 and 2025 GCI scores is $\rho = 0.995$ ($n = 161$). Over the full panel, the 2009-to-2025 Spearman correlation is $\rho = 0.906$, reflecting substantial rank persistence over 17 years alongside meaningful longitudinal mobility. Of the top-20 countries in 2024, 19 retained their position in 2025.

Coverage-robustness. Our main exclusion rule (country-years with fewer than 5 of 8 subindices are set to NA) removes country-years with insufficient subindex coverage. The scored dataset contains 2,809 observations representing 170 countries over 2009–2025.

Correlative Comparison with Economic Freedom Indices

The GCI measures market systems more broadly than existing economic freedom indices, grounding subindex construction in the theoretical distinctions between the institutional pillars of capitalism. However, some conceptual overlap with established economic freedom indices is expected given shared underlying constructs.

When the GCI is compared against The Heritage Foundation’s Index of Economic Freedom and the Fraser Institute’s Economic Freedom of the World Index on the 2024 cross-section ($n = 145$ countries covered by all three), the correlation of the GCI to Fraser is 0.807, GCI to Heritage

is 0.826, and the correlation between the two economic freedom indices themselves is higher at 0.873. The same ordering holds on the pooled 2009–2024 panel ($n = 2,380$ country-years), where the corresponding figures are 0.824, 0.841, and 0.866. This pattern is substantively informative, indicating that the GCI differentiates from both existing indices to a greater degree than they differentiate from each other.

This demonstrates and validates the claim that economic freedom and capitalism are related but not synonymous. Though the GCI is positively correlated with these two measurements, it captures significant additional variation. The conflation of economic freedom with broader capitalism can bias academic and policy research towards laissez-faire market policies and disguise the broader institutional frameworks needed to sustain free markets. Existing economic freedom indices are often treated as metrics of capitalism and used to signal a positive and linear relationship between free markets and societal outcomes. The GCI is designed to capture the totality of market systems and the supporting institutions that enable them, which is a clear prerequisite for rigorous empirical analysis of multidimensional capitalism.

Tax Policy and Market Systems

An important question for the GCI’s theoretical scope is whether it captures capitalism as a broad institutional system or merely reflects tax and fiscal policy preferences. The GCI does not include direct measures of tax rates or public expenditure levels in any subindex; the closest proxy is government final consumption expenditure (used in Property Rights as a measure of state ownership of the economy). This is by design: whether high taxes are compatible with strong capitalism is an empirical question, not a design assumption.

A forthcoming analysis regresses the GCI against the Tax Foundation’s International Tax Competitiveness Index and examines whether cross-national variation in income tax rates explains GCI score variation after controlling for other institutional factors. If the GCI is sensitive to tax levels, it would raise concerns about conflating capitalism with laissez-faire fiscal arrangements. If, conversely, tax-level variation explains little of the cross-national GCI variance, it would confirm the index’s capacity to distinguish capitalist institutional quality from fiscal policy choices, a key differentiator from existing economic freedom indices that weight tax measures heavily.

Discussion

The Results section established that the GCI is internally coherent, reliable across the panel, and a strong measure of capitalism. This section takes up four further external questions about capitalism. First, how does capitalism, as measured by the GCI, correlate with external variables of interest? Second, how does the GCI relate to V-Dem’s democracy indices? Third, what varieties of capitalism can be drawn out of the GCI’s eight-dimensional subindex space? Building on the varieties of capitalism literature, we identify four institutional configurations that recur across the data. Fourth, how do those varieties map onto measures of democracy? This categorical analysis reveals whether particular configurations of capitalist institutions align systematically with political regime types. We conclude with a discussion of the academic and pedagogical applications the GCI enables.

Correlative Comparison with External Variables of Interest

This section charts the relationship between the Global Capitalism Index and 5 widely cited data sets that measure inequality, human development, happiness, political corruption, and tax burden. These tables do not provide causal evidence, but they do document the strength of the relationship between two concepts using Pearson correlation analysis. Throughout this section, r represents the Pearson correlation coefficient (where -1 indicates a perfect negative relationship, in which one variable rises as the other falls, and $+1$ indicates a perfect positive relationship, in which both variables rise together), and n represents the number of country-year observations included in the calculation.

Inequality: GCI $\times$ Gini Coefficient ($r = -0.37$, $n = 1,254$)

The correlation between the composite GCI and the Gini coefficient is indicative of a moderate and negative relationship (-0.37). This shows that countries with stronger capitalist institutions tend to have lower income inequality.

This relationship holds consistent with an empirical literature showing that institutional quality, particularly rule of law, control of corruption, and secure property rights, reduces the ability of elites to extract rents and concentrate wealth. Chong and Calderón (2000) find that institutional quality is significantly negatively associated with income inequality.²⁰⁸ Chong and Gradstein (2007) extend the analysis and show that the relationship persists after accounting for the joint determination of the two variables.²⁰⁹ This correlation is best read as evidence that strong capitalist institutions across domains (consistent and enforceable rules, competitive markets, constrained rent-seeking) are associated with more egalitarian distributional outcomes.

Human Development: GCI $\times$ Human Development Index ($r = 0.79$, $n = 2,602$)

The correlation between the composite GCI and the Human Development Index is indicative of a very strong and positive relationship (0.79). Countries with more developed capitalist institutions score substantially higher on this composite measure of life expectancy, education, and income per capita. Since HDI incorporates income alongside health and education outcomes, this correlation captures both the direct impact and the parallel effects of institutional quality on public health capacity and human capital investment.

The strength of this association is expected and consistent with a large body of empirical work identifying institutional quality as a primary determinant of long-term development. Acemoglu, Johnson, and Robinson (2001) show that measures of institutional quality explain a large share of cross-country variation in income per capita.²¹⁰ Hall and Jones (1999) demonstrate that institutions and government policies that protect against private and public expropriation account for the majority of cross-country variance in output per worker.²¹¹

²⁰⁸Alberto Chong and César Calderón, "Institutional Quality and Income Distribution," *Economic Development and Cultural Change* 48, no. 4 (July 2000): 761–786.

²⁰⁹Alberto Chong and Mark Gradstein, "Inequality and Institutions," *Review of Economics and Statistics* 89, no. 3 (August 2007): 454–465.

²¹⁰Daron Acemoglu, Simon Johnson, and James A. Robinson, "The Colonial Origins of Comparative Development: An Empirical Investigation," *American Economic Review* 91, no. 5 (December 2001): 1369–1401.

²¹¹Robert E. Hall and Charles I. Jones, "Why Do Some Countries Produce So Much More Output per Worker than Others?" *Quarterly Journal of Economics* 114, no. 1 (February 1999): 83–116.

Happiness: GCI $\times$ The Happiness Index ($r = 0.71$, $n = 1,882$)

The correlation between the composite GCI and the Happiness Index is indicative of a very strong and positive relationship (0.71). Countries with more developed capitalist institutions have substantially higher average life evaluations from their populations.

This aligns with a large empirical literature identifying institutional quality as a primary determinant of subjective well-being. The *World Happiness Report* consistently finds that six factors—GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and freedom from corruption—together explain the large majority of variance in national life evaluations, and each is either produced by or directly measured within capitalist institutional systems.²¹² Ott (2010) reports a correlation of +0.75 between “technical quality” of governance (regulatory quality, rule of law, government effectiveness, control of corruption) and average happiness.²¹³ Frey and Stutzer (2002) show that institutional quality affects subjective well-being beyond its indirect effects through income.²¹⁴

Political Corruption: GCI $\times$ V-Dem Political Corruption Index ($r = -0.80$, $n = 2,737$)

The correlation between the composite GCI and V-Dem’s Political Corruption Index is indicative of a very strong and negative relationship (-0.80).

Political corruption diminishes competitive markets, the effective allocation of capital, and incentives for new business development as jockeying for rents commands greater shares of political and economic activity. While corruption may produce high profits for a small group of connected actors, perverse incentives and systemic disintegration diminish the effectiveness of capitalist institutions as a whole.

It is important to note that the GCI incorporates measures of corruption in two of its subindices. Transparency International’s CPI, the World Bank’s Control of Corruption Index, and multiple V-Dem judicial corruption indicators (Judicial Corrupt Decisions, Purges, Government Attacks on Judiciary, Court Packing) appear in the GCI’s Strength of Property Rights and Private Ownership subindex. Likewise, V-Dem’s Public Sector Corruption Index appears in the GCI’s Market-Supporting Policy subindex. Part of this correlation therefore reflects construct overlap rather than an entirely independent empirical association. It is best read as a consistency check confirming that a low-corruption environment is central to what the GCI measures.

Tax Burden: GCI $\times$ Tax Burden ($r = 0.31$, $n = 74$)

The correlation between the composite GCI and tax burden is indicative of a weak and positive relationship (0.31) across 74 country-year observations. The small sample size means that the true relationship could be anywhere from near-zero to moderate. However, both the direction and the small-to-moderate magnitude of this correlation suggest that low tax burdens are not, by themselves, foundational markers of thriving capitalist systems.

The GCI’s correlation is consistent with the idea that capitalism is an institutional system distinct

²¹² John F. Helliwell, Richard Layard, Jeffrey D. Sachs, Jan-Emmanuel De Neve, Lara B. Aknin, and Shun Wang, eds., *World Happiness Report 2024* (Oxford: Wellbeing Research Centre, University of Oxford, 2024).

²¹³ Jan C. Ott, “Good Governance and Happiness in Nations: Technical Quality Precedes Democracy and Quality Beats Size,” *Journal of Happiness Studies* 11, no. 3 (2010): 353–368.

²¹⁴ Bruno S. Frey and Alois Stutzer, “What Can Economists Learn from Happiness Research?” *Journal of Economic Literature* 40, no. 2 (June 2002): 402–435.

from any particular fiscal arrangement. The Nordic economies combine some of the world’s highest tax-to-GDP ratios with top-decile GCI scores, reflecting how strong property rights, competitive markets, and open trade can coexist with substantial welfare states. Lindert (2004) documents that high-tax, strong welfare states have historically achieved growth rates comparable to low-tax economies, suggesting fiscal policy is a distinct dimension rather than a defining feature of the capitalist system itself.²¹⁵

Correlative Comparison with Measures of Democracy

A secondary question is whether the GCI captures capitalism as distinct from political regime type. Capitalism and democracy are theoretically separable, seeing as liberal democratic institutions may support market systems, but market economies also function under more authoritarian regimes (as illustrated by Singapore, Qatar, and China’s positioning in GCI rankings). The presence of strong-loading governance variables in several subindices (e.g., V-Dem judicial independence and accountability in Property Rights; V-Dem public sector corruption in Market-Supporting Policy) raises the question of whether the GCI is simply measuring “democracy” by another name. The below correlations show moderate to strong and positive relationships, consistent with the expectation that democratic institutions support market functioning but are neither necessary nor sufficient for high GCI scores.

Electoral Democracy: GCI $\times$ V-Dem Electoral Democracy Index ($r = 0.57$, $n = 2,737$)

The correlation between the composite GCI and V-Dem’s Electoral Democracy Index is indicative of a moderately strong and positive relationship (0.57). Countries with more developed capitalist institutions tend to be more electorally democratic.

This aligns with the methodological premise that capitalism and democracy are separable institutional systems that benefit from common infrastructure (rule of law, constrained executive power, and checks on rent-seeking). The correlation being only moderately strong is consistent with the fact that numerous countries ranking high on the GCI also feature low scores on democracy metrics (Singapore, Qatar, UAE). Acemoglu and Robinson (2012) argue that inclusive political and economic institutions tend to co-evolve because each constrains the incentives that would otherwise erode the other.²¹⁶ North, Wallis, and Weingast (2009) frame the same phenomenon as a joint transition from “limited access orders,” in which political elites restrict competition in both politics and markets to preserve privilege, to “open access orders,” in which competitive elections and competitive markets emerge together.²¹⁷ The question of whether “low democracy” countries will be able to maintain high GCI scores over time remains to be seen.

Liberal Democracy: GCI $\times$ V-Dem Liberal Democracy Index ($r = 0.66$, $n = 2,737$)

The correlation between the composite GCI and V-Dem’s Liberal Democracy Index is indicative of a strong and positive relationship (0.66). Notably, this is higher than the correlation with V-Dem’s Electoral Democracy Index (0.57).

²¹⁵Peter H. Lindert, *Growing Public: Social Spending and Economic Growth Since the Eighteenth Century*, 2 vols. (Cambridge: Cambridge University Press, 2004).

²¹⁶Daron Acemoglu and James A. Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (New York: Crown Business, 2012).

²¹⁷Douglass C. North, John Joseph Wallis, and Barry R. Weingast, *Violence and Social Orders: A Conceptual Framework for Interpreting Recorded Human History* (Cambridge: Cambridge University Press, 2009).

The Liberal Democracy Index adds explicit measures of rule of law, judicial independence, and constraints on executive power to the Electoral Democracy Index. Limits on political power protect the property rights and contract enforcement that underlie functioning markets and help explain the difference in correlations between the two indices. Elections without liberal constraints (competitive but illiberal regimes) tend to preserve the discretionary power that undermines market confidence, while liberal constraints without full electoral competition can nonetheless sustain some level of functioning capitalist institutions. North and Weingast (1989) argue that credible constitutional constraints on the sovereign were necessary preconditions for the development of capital markets, as investors needed assurances against arbitrary expropriation.²¹⁸

Varieties of Capitalism

This section outlines a preliminary analysis identifying four distinct varieties of capitalism using a set of clustering techniques applied to eight subindices of the GCI across the 2025 cross-section. The following does not rank tiers of more or less capitalism, but rather crafts four qualitative shapes of capitalist systems. Each variety of capitalism exhibits a distinctive pattern of relative strengths and weaknesses across the eight GCI subindices, revealing how countries configure their capitalist systems in fundamentally different ways. The four varieties detailed in this section represent what each profile encapsulates and what organizational logic defines it rather than its position on a simplistic state versus market spectrum or a wealth ladder. We report this analysis to demonstrate what the subindex structure makes possible rather than as a finished typology.

This analysis maps the four varieties of capitalism based on the architecture of a country’s capitalist system, not the strength or intensity of the capitalism score. To fully understand both the shape and intensity of a country’s capitalist system, one must combine the typologies described below with the aggregate GCI score. A naïve clustering approach produces clusters that often restate the aggregate GCI ranking with richer countries in one cluster and poorer countries in another. Because the eight subindices are highly correlated with one another, the first axis of variation in the raw data is overall level and any clustering on it recovers wealth tiers. To solve this, we employed ipsative scoring. For each country, we subtracted that country’s own mean across the eight subindices. This centers every country at zero and retains only the relative profile detailing the factors at which each country is comparatively strong and weak, regardless of its overall GCI level. A country scoring 90 on everything and a country scoring 30 on everything both become flat lines at zero, with the only differentiation observed being deviations from their own average. We then z-standardized ipsative scores across countries (by each dimension’s standard deviation) to ensure equal weighting and performed clustering analysis. We estimate on the 2025 cross-section, covering the 113 countries observed on all eight subindices, and partition them with k -means, hierarchical, mixture-model and spectral solutions combined through a consensus ensemble.²¹⁹

We impose four clusters rather than deriving that number. Four is close to the resolution used across the comparative capitalism literature. Hall and Soskice distinguish two ideal types, Amable identifies five, and Whitley six.²²⁰ No fit statistic converges on a usable number of clusters.

²¹⁸Douglass C. North and Barry R. Weingast, “Constitutions and Commitment: The Evolution of Institutions Governing Public Choice in Seventeenth-Century England,” *Journal of Economic History* 49, no. 4 (December 1989): 803–832.

²¹⁹One country, Angola, is excluded. Its Market Competition score of exactly 0.0 produces the most extreme ipsative profile in the sample ($\|z\| = 7.13$) which every approach isolated as a single-member cluster.

²²⁰Peter A. Hall and David Soskice, eds., *Varieties of Capitalism* (Oxford: Oxford University Press, 2001); Bruno Amable, *The Diversity of Modern Capitalism* (Oxford: Oxford University Press, 2003); Richard Whitley, *Divergent Capitalisms* (Oxford: Oxford University Press, 1999).

The gap statistic, silhouette coefficient, and BIC all select a two-way split, while Davies–Bouldin points to nine. Four clusters allows us to begin investigations into the varieties while deferring to the existing theory. This analysis does not claim that capitalism comes in four kinds, but rather serves as a starting point for discussion. A subsequent paper will approach this subject with deeper statistical precision. These four clusters begin to separate countries in ways that are internally coherent and stable over time. After fitting the four varieties on the 2025 cross section, all 2,283 complete country-years from 2009 are assigned to those fixed centroids to measure persistence over time.

Variety	N	Mean GCI	GCI Range	Defining Character
Entrepreneurial	27	70.1	42.2–95.0	Firm formation, market-supporting policy
Corporate	21	62.6	24.1–87.5	Financial depth and capital market sophistication
Institutional	31	74.2	41.9–96.9	Property rights, competition and labor openness
Commercial	34	64.4	48.2–79.9	Trade openness and property protection
Total	113	68.1	24.1–96.9	

Note: 2025 cross-section. Membership explains 9.2% of the variance in composite GCI ($F(3, 109) = 3.68, p = 0.014$).

Table 8: The Four Varieties of Capitalism

Each variety is described below. Note that several dimensions deviate from the country mean in the same direction for nearly every country. Across all 113 economies, Market Competition averages +11.6 index points above the country mean, Capital Market Sophistication -12.8 points below, and New Business Formation -8.8 below. Raw figures should be read against those baselines rather than against zero.

Variety	Banking	Cap Mkts	Free Flow	Labor	Competition	Policy	New Biz	Prop Rights
Entrepreneurial	−0.24	−0.18	−0.35	+0.56	+0.46	+0.80	+1.07	−1.25
Corporate	+1.01	+1.33	+0.04	−0.86	−1.00	−0.39	−0.40	−0.24
Institutional	−0.87	−0.62	−0.70	+0.65	+0.56	+0.14	+0.23	+0.71
Commercial	+0.36	−0.12	+0.89	−0.50	−0.25	−0.52	−0.82	+0.50

Note: Column-centered ipsative z -scores. A positive value indicates a relative strength for that variety compared with the other three.

Table 9: Ipsative z -Score Centroids by Variety

Entrepreneurial Capitalism

27 countries | Mean GCI: 70.1 | GCI range: 42.2–95.0

Entrepreneurial Capitalism peaks on New Business Formation ($z = +1.07$) and Market-Supporting Policy ($z = +0.80$). Labor Market Openness ($z = +0.56$) and Market Competition ($z = +0.46$) provide two further positive pillars. The finance dimensions are mildly negative (Banking Depth $z = -0.24$, Capital Market Sophistication $z = -0.18$) and Free Flow of Goods and Capital sits at $z = -0.35$. Countries in this category demonstrate stronger relative weakness in Property Rights ($z = -1.25$).

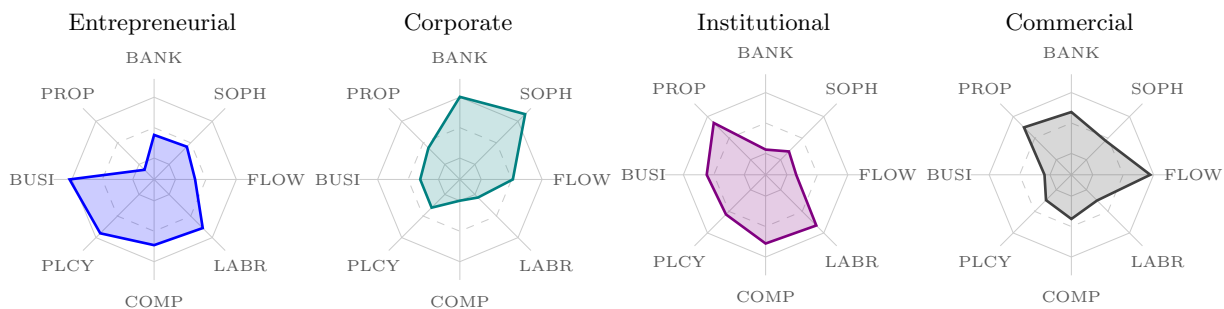


Figure 7: Profiles of the four varieties. Each shape is the variety’s centroid across the eight subindices. The dashed ring marks zero and the inner and outer rings -1 and $+1$.

The “Entrepreneurial” label captures what this variety does best: it creates the conditions for new enterprise. The institutional architecture focuses on building competitive marketplaces through active policy, with relatively available labor. On average, it sacrifices financial-property rights infrastructure: cross-border capital flows, deep capital markets, and robust property rights are all relative weaknesses.

The United States, Hong Kong, and Singapore sit at the top, but the same organizational logic appears in the Gulf states (the United Arab Emirates, Qatar, Oman, Saudi Arabia, Bahrain, Jordan), in emerging Asia (China, Indonesia, Vietnam, the Philippines, Cambodia, Laos, Bangladesh, Pakistan), in post-Soviet and Balkan reformers (Georgia, Ukraine, Albania), and in several of the world’s least developed economies (Rwanda, Mali, Zimbabwe, Cameroon).

These countries differ enormously in *level* of capitalism but they share an organizational logic configured around competitive marketplaces and supportive policy rather than around financial depth or property protection. Despite vast political and ideological differences, the United States and China both organize the capitalist elements of their economies around market competition supported by active government policy, with property rights and capital market infrastructure comparatively weak relative to their other strengths. The claim is not that these economies are alike, but rather that the *shape* of their institutional architecture is alike.

Corporate Capitalism

21 countries | Mean GCI: 62.6 | GCI range: 24.1–87.5

Corporate Capitalism presents the most extreme profile of any variety. Its strengths are Capital Market Sophistication ($z = +1.33$) and Banking Depth and Stability ($z = +1.01$). Free Flow of Goods and Capital is neutral ($z = +0.04$). Market Competition ($z = -1.00$), Labor Market Openness ($z = -0.86$), New Business Formation ($z = -0.40$), and Market-Supporting Policy ($z = -0.39$) are all weaknesses. Property Rights is close to neutral ($z = -0.24$). In raw terms, Banking Depth sits +16.7 points above each country’s own average while Labor Market Openness falls -15.0 points below and New Business Formation -11.7 below.

The “Corporate” label reflects a capitalist organizational logic centered on large, established firms and financial institutions rather than on entrepreneurial creation or competitive dynamism. The banking system is solid and effective, capital markets are relatively deep, and cross-border trade flows relatively freely. However, it is difficult to start new businesses, competition is muted, and labor markets remain stolid.

Japan and South Korea lead this group with two economies famously organized around large corporate conglomerates (keiretsu, chaebol). France, Malaysia, Poland, Thailand, Kuwait, Panama, and Armenia share the pattern. Brazil, Mexico, Guatemala, and Bolivia all appear here as well. These economies feature formal banking and capital market structures that serve incumbent firms rather than generating new ones. France is the most marginal member of any cluster in the analysis, sitting almost equidistant from the Corporate and Commercial centroids and best read as an intermediate case.

Institutional Capitalism

31 countries | Mean GCI: 74.2 | GCI range: 41.9–96.9

Institutional Capitalism features strong performance in the “real economy” and significant weakness in the financial sector. Property Rights ($z = +0.71$) and Labor Market Openness ($z = +0.65$) are the twin peaks, with Market Competition ($z = +0.56$) close behind and Market-Supporting Policy ($z = +0.14$) and New Business Formation ($z = +0.23$) mildly positive. Banking Depth ($z = -0.87$), Free Flow of Goods and Capital ($z = -0.70$), and Capital Market Sophistication ($z = -0.62$) are all weaknesses. In raw terms, Market Competition sits +13.9 points above each country’s own average and Property Rights +10.5 above, while Capital Market Sophistication falls –17.8 points below.

The “Institutional” label captures a capitalist model built on strong non-financial institutions: property rights, competitive markets, open labor markets, and the rule of law. The financial sector is relatively weak not because it is unimportant but because institutional energy flows toward building legal, regulatory, and competitive foundations rather than a financial superstructure.

Switzerland leads the cluster, followed by Canada, Denmark, Australia, Sweden, and New Zealand. The Nordics appear here in full, alongside the United Kingdom, Iceland, Finland, the Netherlands, Belgium, Ireland, Austria, Germany, Estonia, and Lithuania. Capitalism in these countries rests on institutions and norms, while banking systems and capital markets play a secondary role.

Notably, a substantial group of African economies clusters here (Botswana, Namibia, Cabo Verde, Benin, Zambia, Côte d’Ivoire, Kenya, Tanzania, Ghana, Senegal, Malawi, Liberia, and Nigeria), along with Uruguay and Serbia. These are economies that have invested in institutional development (property rights, competitive markets, effective policy frameworks) ahead of financial depth. Germany and Ghana belong to the same cluster but differ by 48 GCI composite points, illustrating how configuration and level are separable. Institutional Capitalism has the highest average composite score of the four varieties (74.2).

Commercial Capitalism

34 countries | Mean GCI: 64.4 | GCI range: 48.2–79.9

Commercial Capitalism is defined by the primacy of Free Flow of Goods and Capital ($z = +0.89$) and Property Rights ($z = +0.50$). Banking Depth ($z = +0.36$) is mildly positive. Every other subindex is negative, leading with New Business Formation ($z = -0.82$), Market-Supporting Policy ($z = -0.52$), Labor Market Openness ($z = -0.50$), Market Competition ($z = -0.25$), and Capital Market Sophistication ($z = -0.12$). In raw terms Banking Depth sits +10.1 points above each country’s own average, Property Rights +8.1 above, and Free Flow +6.4 above against a sample-wide –1.0. New Business Formation falls –14.7 points below and Labor Market Openness –11.4 below.

The “Commercial” label reflects an economy organized around the conditions traders and investors value (property protection, capital mobility, functioning banks) rather than around competition, labor mobility, or firm creation. This is capitalism configured to attract trade and protect capital, not necessarily to deploy it into new enterprises. Money, goods, and services transit the country and are protected, while the entrepreneurial and competitive mechanisms that would deploy them at scale are comparatively absent.

Spain, Portugal, Italy, and the Czech Republic represent the Continental European model with strong property rights and open capital flows alongside relatively rigid labor markets and less entrepreneurial dynamism. The Central and Eastern European contingent (Slovenia, Bulgaria, Latvia, Slovakia, Croatia, Hungary, Romania, North Macedonia, Montenegro, Bosnia and Herzegovina) represents post-transition economies that built trade- and investment-friendly legal frameworks faster than they built entrepreneurial ecosystems. Chile, Costa Rica, Mauritius, Jamaica, and Trinidad and Tobago bring a Latin American and island-economy variant of the same logic, with Colombia, Peru, Argentina, Ecuador, Honduras, El Salvador, Paraguay, and the Dominican Republic completing the group. Commercial Capitalism has the narrowest composite range of the four varieties (32 points). It is the most homogeneous variety in level and in shape, and the one whose members are most consistently middle-income countries.

Internal Coherence and Persistence

The four varieties differ significantly on all eight dimensions, with membership explaining between 27% and 60% of the variance in each dimension’s shape score (Table 10). Property Rights and New Business Formation discriminate most sharply, while Market-Supporting Policy does the least. Assignment confidence under the Gaussian mixture specification is high, with a mean maximum posterior probability of 0.981 and 109 of 113 countries assigned above 0.90. However, separation between clusters is modest (overall silhouette 0.169).

Dimension	$F(3, 109)$	p -value	η^2
Banking Depth & Stability	29.9	3.5×10^{-14}	0.452
Capital Market Sophistication	29.1	6.6×10^{-14}	0.445
Free Flow of Goods & Capital	24.3	4.0×10^{-12}	0.401
Labor Market Openness	24.8	2.7×10^{-12}	0.405
Market Competition	18.8	7.0×10^{-10}	0.341
Market-Supporting Policy	13.4	1.7×10^{-7}	0.269
New Business Formation	39.6	2.2×10^{-17}	0.521
Property Rights	54.1	1.7×10^{-21}	0.598

Note: η^2 is the share of variance in each dimension’s ipsative score explained by variety membership.

Table 10: One-Way ANOVA on Ipsative Dimensions

The clusters are persistent across the GCI panel. Holding the four clusters fixed and assigning every complete country-year in the 2009–2025 panel to the nearest, 80.6% fall in their own country’s modal variety, 47 of 149 countries never change variety across the seventeen years, and 92 remain in their modal variety in at least four-fifths of observed years. Persistence is uneven with 32 countries falling below 60% stability. These countries are largely middle-income economies undergoing rapid institutional change. Whether those cases represent genuine transitions or instability at cluster boundaries is a question for further analysis.

Measure	Value
Country-years in their country’s modal variety	80.6%
Countries never changing variety	47 of 149
Countries in modal variety $\geq 80\%$ of years	92 of 149
Countries in modal variety $< 60\%$ of years	32 of 149
Mean variety changes per country over 17 years	1.83

Note: Centroids are fixed from the 2025 cross-section; all 2,283 complete country-years in the panel are then assigned to the nearest centroid.

Table 11: Persistence of Variety Membership, 2009–2025

GCI Limitations

Theoretical

Any quantitative measurement instrument is only as good as the theoretical construct it operationalizes. The PCA methodology underlying the GCI measures how well our observable indicators cohere around the latent construct we specify, but it cannot adjudicate whether that construct is the correct one. Our framework is based heavily in western liberal theory, with Locke, Smith, Schumpeter, and others cited throughout the index. This is largely due to these theorists being the ones most directly tied to our institutional definition of capitalism, but it could be argued this is a narrow theoretical approach that needs expansion.

The distinction between institutional inputs and economic outputs, which we identify as one of the GCI’s central methodological contributions, is theoretically clean but operationally porous. Several indicators are practically ambiguous. We use YoY GDP growth as a proxy for creative destruction in the Market Competition subindex, but growth is arguably an outcome of institutional configurations. Business density rates, stock market turnover, trade-to-GDP ratios, and foreign-born share of the labor force could reasonably be interpreted either as institutional features or as outcomes of underlying institutional arrangements. The Hamilton Indicator draws on tariff rates, which are policy outputs of political bargaining rather than fixed institutional structures. We have attempted throughout to maintain the focus on inputs when possible.

The nation-state as a unit of analysis may mask meaningful subnational variation (firm, sector, region, metro area, and supranational blocs). Capitalist institutions can vary substantially within countries: policy differences across various US states, the divergence between China’s special economic zones and its provinces, and the broader urban-rural institutional divide across much of the globe. The GCI’s national-level scores should be read as characterizations of an aggregate, and users interested in subnational institutional variation will need to supplement the GCI with more granular data. We use national-level data because that is the most available data unit for global comparison, but it could still obscure deeper heterogeneity. Similarly, informal economies and informal institutions are largely invisible to the GCI’s measurement framework. Black market economies tend to be very capitalist, but our data covers virtually none of them. Property rights may be enforced through customary law and businesses may form and compete without formal registration. Therefore, the GCI’s reliance on formal institutional measures likely underestimates capitalist activity in economies with large informal sectors and may risk conflating formalization with capitalism itself.

Two further limitations concern the scope of what the GCI can meaningfully describe. First, the choice to identify capitalism with eight specific subindices is itself a contestable theoretical claim.

We have provided theoretical justification for each subindex in the Theory section, but the decision that these eight dimensions exhaustively constitute capitalism is debatable. Second, the temporal window constrains what the GCI can reveal about institutional formation and decay. The theory draws on centuries of scholarship about how capitalist institutions emerge, evolve, and break down in a very particular geoeconomic context. The period covered by the GCI contains the aftermath of the Global Financial Crisis, the COVID19 pandemic, and the geopolitical realignments of the post-pandemic period. Any inferences drawn from the GCI about institutional dynamics reflect this particular historical window and should be treated with appropriate caution as generalizations beyond it.

Methodological

There are several methodological limitations worth addressing. Missingness at the indicator level is more pervasive than the subindex figures suggest. Across the eight analysis panels, 57.9% of country-year-variable cells are observed, and only 104 of the 213 datasets span all seventeen index years: 63 enter the panel after 2009 and 72 cease before 2025. The gaps take three distinct forms: (i) Datasets are discontinued or restructured mid-panel, as when a block of twenty WEF Competitiveness Report items ends in 2019 and is replaced by reformulated questions in 2020; (ii) New sources enter partway through, with five datasets first appearing in 2024; and (iii) Individual countries fail to report in particular years even for indicators that are otherwise continuously available. Fill rates vary widely by subindex, from 81.7% for Property Rights to 42.7% for Capital Market Sophistication.

We address missing data constraints through two methods. Structural entry and exit are handled by the period segmentation described above. We perform hierarchical clustering on missingness patterns groups years with comparable data structures, so each variable contributes only within periods where it exists, and scores are chain-linked across boundaries. Sporadic country non-reporting is handled by MICE multiple imputation nested within each of the 500 bootstrap draws, so that imputation uncertainty propagates into the reported confidence intervals rather than being suppressed. Neither procedure is a substitute for complete data. Imputation cannot create information, but instead redistributes the observed correlation structure, which means imputed values are drawn toward what the correlated indicators already imply and therefore tend to reinforce the index’s internal coherence rather than test it. MICE further assumes that missingness is conditionally random given the observed data, an assumption that is difficult to defend here, since failure to report is plausibly a function of administrative and statistical capacity, which is itself among the institutional features the index is designed to measure. Period segmentation likewise buys within-period estimability at the cost of re-estimating weights across boundaries, so cross-period comparisons rest on the chain-linking procedure rather than on a single common model. Complete panels would be strictly preferable, and the reported intervals should be read with that in mind.

Another limitation is common-source variance. The WEF Executive Opinion Survey (EOS) supplies indicators to almost every subindex, and the WGI datasets appear in Property Rights, Market Competition, and Market-Supporting Policy. Correlated measurement error from shared instruments can inflate both the second-order loadings and the 73.9% composite variance share. We tested this by extracting a method factor from the 79 pooled EOS items, orthogonalizing it against the 134 non-survey items, and residualizing each subindex on the remainder. The variance share does not fall (74.5% to 74.9% on the 2,240 country-years where the factor is scoreable), and no loading moves by more than 0.007. Market Competition and New Business Formation, the two most exposed, shift by +0.001 and +0.004.

The bootstrap confidence intervals are a lower bound on total uncertainty. Our resampling procedure (500 iterations at the subindex level, and 250 within each of the 20 imputed datasets at the composite level) propagates sampling and imputation variance, but it does not account for uncertainty in period boundaries, component-count choice, reliability scores, or source measurement error. Period boundaries and the number of retained components are treated as fixed inputs rather than estimated quantities, and because resampling country-years leaves the correlation structure among shared data sources intact, systematic source error is invisible to the procedure by construction. The 2025 mean half-width of 1.72 points should therefore be read as the precision of the estimator conditional on the specification, not as complete uncertainty about a country's position.

Our year-relative normalization makes level comparisons frame-dependent. Min-max and z-scoring are deliberately computed year by year, which preserves genuine cross-national variation but means a country's score can move because the distribution around it shifted rather than because its own institutions changed. The one-sided cap applied to each input variable at its within-year 95th percentile compounds this at the upper tail, since the cap itself moves with the annual distribution. Longitudinal change figures, including the Δ 2009–2025 column in Table 7, should accordingly be read as movement relative to each year's distribution rather than against a fixed absolute standard. Where the substantive claim concerns relative position, the rank series is the safer instrument as rank statistics are unaffected by the frame.

Academic Applications

We expect the GCI to find its principal early applications in the domains surveyed in the literature review, but the index is constructed to support any cross-national empirical project where the relevant explanatory variable or control is the depth and configuration of capitalist institutions. Its most consequential design feature for empirical work is the separation of institutional inputs from economic outputs. When we approached each element of the index structure, we asked whether that input could be prominent only in capitalist economies. Metrics that could be prominent in any economic system (GDP per capita, educational attainment, social mobility, inequality, gender equity, health, innovation, etc.) are output variables against which the GCI can be analyzed for correlative or causal relationships. Each of the individual subindices and the composite index can be used as dependent variables without concern that the independent variable mechanically encodes the outcome.

Because PCA scores are continuous, standardized, and computed without strong pre-defined modeling assumptions, they are especially suited for empirical work. Panel regressions can estimate the effect of institutional configuration on outcome variables of interest, using either the composite index or individual subindices as regressors and exploiting within-country variation over the 2009–2025 period. Instrumental variables strategies can use the GCI or its subindices as instruments where institutional quality plausibly satisfies exclusion restrictions. Event studies and difference-in-differences designs can examine outcomes around major institutional reforms captured by year-over-year changes in specific subindices.

Pedagogical Applications

Capitalism is notoriously difficult to teach. It is conceptually contested, inherently interdisciplinary, dynamic across time, and politically loaded.²²¹ The GCI aims to sidestep these issues by providing

²²¹Robert F. Bruner, "Teaching Democracy and Capitalism: High Engagement and 'Doing Economics'," *The Journal of Economic Education* 56, no. 1 (2024): 76–86; see also John Dewey, *Democracy and Education* (New York:

tools that are fully compatible with “high engagement” or “doing economics” methods of organizing teaching around real datasets, decision-relevant problems, and student-led discovery rather than lecture-based exposition.²²²

By presenting capitalism as a system whose properties vary observably across countries and over time, the GCI represents a tool to convert abstract concepts into empirical and contrastable measurements. In so doing, the GCI gives students a concrete vocabulary for the components of capitalist systems (the 8 subindices and 33 domains), and extensive country observations per year to see cross-sectional variation rather than a single score.

Since the GCI features a transparent two-stage PCA, it also supports instruction at multiple levels of analytical commitment. Simpler work can use published subindex scores to test hypotheses about institutions and outcomes. Intermediate work can examine the indicators within a subindex and propose alternative weightings or combinations of subindices. Advanced work can replicate the index construction from raw inputs, propose new indicators, or extend the panel. Each level exercises the data analysis and statistical skills that are central to economic literacy.²²³

The GCI also encourages and develops judgment. Every methodological input or technique in the index represents a normative decision at some level, from the dimensions we identified as elements of capitalism to the datasets we use to measure or proxy concepts. Our theoretical and methodological transparency on these decisions allows teachers and students to question assumptions and propose solutions to errors. Classroom exercises organized around “should subindex X weigh more heavily?” or “which dimensions are constitutive of capitalism, and which are consequences of it?” exercise judgment on the values latent in measurement, identification of tradeoffs, and how to form defensible arguments under ambiguity.

To support adoption in classrooms, we are developing a comprehensive suite of teaching resources for secondary, undergraduate, and graduate levels that will be available for free on our demcap-classroom.com website (and linked on our public GCI website). These materials will lower the cost of adoption and position the GCI not only as a research instrument, but as the infrastructure for instruction about how capitalist systems work at every level.

Conclusion

The Global Capitalism Index measures the strength and change of eight institutional dimensions that constitute a capitalist economic system. The GCI covers 170 countries from 2009 through 2025 drawing on 213 datasets from international organizations, academic sources, and private-sector data providers. Each of the eight subindices is constructed through an adaptive principal component analysis that weights indicators by their reliability and shared variance. The composite index is constructed by a second PCA using the eight subindices and generates 95 percent bootstrap confidence intervals for every country-year score. The methodology deliberately separates

Macmillan, 1916), 191; Raj Chetty, “Using Big Data to Solve Economic and Social Problems” (course materials, Harvard University, 2019); Simon D. Halliday, “Data Literacy in Economic Development,” *The Journal of Economic Education* 50, no. 3 (2019): 284–298; and Tomas Dvorak, Simon D. Halliday, Michael O’Hara, and Aaron Swo-boda, “Efficient Empiricism: Streamlining Teaching, Research, and Learning in Empirical Courses,” *The Journal of Economic Education* 50, no. 3 (2019): 242–257.

²²²Robert F. Bruner, “Teaching Democracy and Capitalism: High Engagement and ‘Doing Economics,’” *The Journal of Economic Education* 56, no. 1 (2024).

²²³Judith D. Singer and John B. Willett, “Improving the Teaching of Applied Statistics: Putting the Data Back into Data Analysis,” *The American Statistician* 44, no. 3 (1990): 223–230.

institutional inputs from economic outputs, which allows researchers to use variables such as GDP, inequality, and human development as dependent variables in analyses without concern that the independent variable mechanically encodes the outcome. We expect the GCI to be used as a tool in empirical work on the determinants and consequences of capitalist institutions, as a diagnostic tool for policymakers and the public seeking to understand how specific institutional dimensions shape economic performance, and as a pedagogical resource for teaching about capitalism.

Appendix A: Structure of the Global Capitalism Index

The Global Capitalism Index is organized in three layers. Individual datasets aggregate into domains, domains aggregate into one of eight subindices, and the eight subindices aggregate into the composite GCI score. Aggregation at the subindex and composite levels is performed by adaptive principal component analysis rather than by assigned weights, so the contribution of any dataset or domain is determined by the observed covariance structure of the data in that period. The justifications for each of the eight subindex concepts are outlined in the Theory section of this paper. The following table maps each of those subindices to their constituent domains, with a brief description of each domain's role within the subindex.

Subindex	Domain	Description
Strength of Property Rights & Private Ownership	Intellectual Property Protection	The ability of firms and individuals to protect their intellectual property, measured through both the formal strength of patent and copyright regimes and the protection firms experience in practice. Without enforceable claims on the non-physical products of human and financial investment, the incentive structure that drives innovation does not form.
	Lack of Corruption	For property rights and rule of law to function effectively, the system needs to operate with universal and impartial application of the law, where larger or more powerful actors are unable to bend the system to their will. This domain captures the de facto dimension of property rights. Formal institutions, laws, and courts must exist for rights to be exercised, but they must also possess the capacity for enforcement free of practical corruption. Judicial corruption indicators are grouped here rather than under Rule of Law because they measure the integrity of enforcement institutions rather than the substance of legal protection.

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Subindex	Domain	Description
	State Stability	Without a stable and relatively dependable state, the system of property rights is fallible and susceptible to external pressure and risk. This would damage or eradicate the reliability of the system, rendering it unusable and undependable. This domain measures the capacity for stability and also includes a measure of political constraint against discretionary policy change.
	Political Expression	The public accountability of companies and the ability for in-depth criticism and discourse of business sentiment without retributive legal action, both upside and downside, is crucial to the vibrancy and multi-directionality of markets. This ability for investigators to produce and publish market and political sentiment free of political interference is crucial for an accurate reflection of property value, rights, and market functioning. This domain measures the information environment rather than electoral competition.
	Rule of Law	The substantive strength and predictability of the legal order through which individuals, firms, and the state claim or dispute rights of ownership. Rule of law is the mechanism by which property rights are enforced in practice. This domain captures constraints on government power, judicial accountability, compliance with court decisions, judicial review, transparency and predictability of law enforcement, and access to justice.
	Contract Enforcement	The ability of actors within a legal system to create and rely on contracts, measured through court efficiency, insolvency proceedings, and rule-based governance assessments. Contract enforcement is what allows property rights to be honored in practice rather than merely enshrined in law.

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Subindex	Domain	Description
	Private Ownership	The extent to which private ownership is both legally permissible and commonly practiced. This signals broader system protection of ownership access, claims, and protections. It also measures the degree of ownership which bridges the gap between de jure and de facto property rights. For two datasets in this domain, we employ a parabolic maximization approach. Please see Appendix C for more details.
Market-Supporting Policy	Market Initiatives	Active initiatives to secure and support the free operation of market systems. This domain does not measure state involvement in markets but state policies that protect and buttress market operation largely through survey measures of antitrust effectiveness, public sector support to start-ups, active labor training, government procurement of advanced technologies, and long-term policy vision. It also incorporates physical infrastructure provisions that link centers of physical, financial, and human capital to enable market creation and operation.
	Operational and Institutional Quality	The quality, flexibility, and independence of government institutions that sustain markets once created. This includes central bank independence, regulatory quality, and how conducive legal systems are to market operations. This domain measures the capacity of the state to perform ongoing institutional maintenance and recalibration. This domain also contains the Hamilton Indicator, which scores low mean tariff rates combined with high tariff dispersion and then discounts that score by a measure of particularistic versus public-goods policy orientation, to account for rent-seeking in policy formulation. This indicator is detailed further in the Theory section of the this paper.

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Subindex	Domain	Description
Labor Market Openness	Labor Market Flexibility	The ability of both firms and laborers to allocate human capital as they see fit, measured through ability to hire and fire, the character of labor–employer relations, openness to foreign labor, wage-setting mechanisms, and the linkage of pay to productivity.
	Labor Protections	Freedom from forced labor and from unequal work arrangements, measured through compliance with internationally recognized labor rights, freedom from coerced labor, and gender equity in pay and economic leadership. Arbitrary constraints such as slavery, race- or gender-based discrimination, and unequal legal treatment artificially undervalue human capital by preventing equal bidding in an open market. A labor system that rejects property in one’s own person, impinges on competition, and suppresses entrepreneurial activity is not structurally capitalist regardless of how profitable it is for its beneficiaries.

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Subindex	Domain	Description
	Labor Availability	The supply of new entrants to the labor market and the circulation of workers within it, measured through the foreign-born share of the labor force, internal mobility, the ease of filling vacancies including through foreign hiring, and the availability of skilled labor locally. Like firms competing in a consumer market, labor markets require an open supply of entrants with minimal barriers in order to sustain competition. The domain is intended to capture whether workers possess the option to reallocate their human capital, not the absolute level of turnover. Because globally comparable, annually updated measures of migration policy do not exist at the scale this index requires, the domain proxies the conditions under which labor supply can expand using variables that reflect both policy choice and structural economic conditions.
Market Competition	Creative Destruction	The dynamics of firm growth and attrition, market concentration, competitive intensity, and market-relevant innovation. Creative destruction is the process that incessantly revolutionizes economic structure from within and is a foundational mechanism of capitalism rather than a byproduct of it. This domain treats firm entry and exit as necessary but insufficient evidence of creative destruction. Changes in firms' relative market positions (measured using YoY GDP growth as a proximate estimate of economy-wide turnover) also matters in capturing capacity for creative destruction. R&D expenditure and university–industry collaboration measure whether there is sufficient sustained research investment that can generate the technological and organizational advances that market turnover depends on.

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Subindex	Domain	Description
Capital Market Sophistication	Regulatory Analysis	The institutions that support competitive activity and protect firms and consumers from market manipulation, measured through governance quality, insolvency resolution, and the adaptability of legal frameworks to new business models. Capitalism presupposes a baseline regulatory infrastructure that preserves the incentives for success while ensuring consequences for failure. The domain evaluates not the quantity of regulation but its quality and orientation.
	Human Capital Competitiveness	The individual, entrepreneurial, and consumer dynamics that support market competitiveness, measured through talent competitiveness, innovative capacity, collaboration within and between firms, workforce mobility, and the teaching of digital and technological skills.
	Access and Participation – Infrastructure	Availability and public participation in formal banking and financial networks, measured through bank branch and ATM density, formal saving and borrowing rates, and the perceived ability of firms to raise capital by issuing shares or bonds. Infrastructure is measured separately from transactions so that the two can be compared against each other. Economies with limited infrastructure typically show low transactional engagement, indicating early-stage development. Economies with extensive infrastructure but modest transaction volumes may signal structural bottlenecks, regulatory friction, or underutilization. Economies dense in both are mature, highly intermediated systems.

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Subindex	Domain	Description
	Access and Participation – Transactions	Frequency and breadth of real financial activity among households and firms, measured through access to loans, the adequacy of financial services to business needs, syndicated lending volume and maturity, domestic credit to the private sector, credit to government and state-owned enterprises, and mobile money and mobile/internet banking activity.
	Equity Market Depth & Scope	The size, diversity, and trading intensity of public equity markets, alongside measures of market functioning and governance: stock market turnover, mutual fund assets relative to GDP, price volatility, and survey assessment of exchange regulation.
	Bond Market Depth and Scope	The development and maturity structure of corporate debt markets, measured through issuance volume relative to GDP, average maturity, and the interest rate spread as an indicator of broader lending conditions.
	Venture Capital Availability	The prevalence and ease of obtaining funding for innovative, high-growth enterprises, measured through survey assessment of how easily start-ups can find equity funding.
	M&A Prevalence	The level of corporate consolidation and capital reallocation, measured through cross-border merger and acquisition sales and purchases. Active markets for corporate control indicate that capital can be redeployed from lower- to higher-return uses without the dissolution of the underlying assets.

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Subindex	Domain	Description
	Liquidity and Resilience	The systemic capacity to withstand shocks and sustain financial stability, measured through reserve adequacy, country risk premia, and liquid liabilities relative to GDP. Liquidity is a precondition for efficient price discovery, financial asset creation, and open market functioning. Markets that lack it are prone to amplified volatility and diminished investor confidence. Resilience entails both the short-run ability to absorb volatility and macroeconomic change and the longer-run capacity to return to normal operation following crisis. Together the two determine whether capital markets can sustain their intermediation function under adverse conditions.

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Subindex	Domain	Description
Depth & Stability of Banking System	Banking System Depth	The degree to which access to the banking system extends across the income and firm-size distribution, measured through financial institution access, depth, and efficiency indices, private credit and deposit bank assets relative to GDP, branch density, card and digital payment penetration, mobile money activity, and survey assessment of access to and innovation in financial services. Increased banking penetration in the broader economy enables the economic dynamism and risk-taking that capitalist systems require, and the institutionalization of banking creates an accessible technology through which risk can be pooled and voluntary exchange made more widely available. Banking infrastructure established mechanisms for assessing creditworthiness and solvency, which created trust in impersonal credit markets and extended access beyond individual financial relationships. The concentration of assets among the five largest banks is used as a proxy for depth on the reasoning that deeper, more developed systems tend to feature larger, better-capitalized leading institutions reflecting broader financial sector expansion.

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Subindex	Domain	Description
	Banking System Stability	The capacity of a banking system to absorb shocks and avoid crises, measured through capital adequacy, regulatory capital to risk-weighted assets, asset quality, liquidity coverage, return on assets, liquid assets to short-term liabilities, default probability, and survey assessment of bank soundness and financial system resilience. Because bank operations contain inherent points of fragility, robustness to shocks determines whether the banking system empowers or periodically destroys capitalist enterprise. When banks fail or credit freezes, uncertainty contracts the money supply, consumer demand, and the flow of capital.
New Business Formation & Growth	Human Capital	A nation's level of education and basic skills, talent development, attraction, and productivity, all of which supply necessary inputs to new business formation. The embedded knowledge, skill, and education of a talent base is itself productive capital, and endogenous growth theory holds that its accumulation through education, experience, and research drives technological progress and firm formation.
	Cultural Acceptance of Entrepreneurship	The systemic acceptance and encouragement of entrepreneurship, measured through attitudes toward entrepreneurial failure, perceived opportunities and capabilities, fear of failure, and the social status accorded to successful entrepreneurs. New business formation is a social phenomenon requiring cultural legitimation of risk-taking, profit motivation, and innovation.

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Subindex	Domain	Description
	Market Access	The practical ability of entrepreneurs to develop companies domestically and internationally, measured through export market concentration, competitive intensity, market dominance, cluster development, state ownership, logistics performance, taxes on international trade, antitrust effectiveness, and market scale. Once ventures are created, competition and growth require access to a widening network of market systems to scale.
	Access to Startup Capital	The ease with which entrepreneurs can access formal and informal credit systems, measured through deposit account and borrower penetration, outstanding deposits and loans from commercial banks, credit unions, and microfinance institutions, domestic credit to the private sector, and survey measures of loan access, venture capital availability, minority shareholder protection, and SME financing. Availability of startup capital provides both the early-stage financing that translates ideas into production and a signal of an economy's support for entrepreneurship.
	Government Supporting Policy	The extent to which government policy enables entrepreneurship through business regulation, market initiatives, competition policy, labor regulation, and taxation, measured through regulatory compliance burden, legal recourse against government action and in commercial disputes, distortive effects of taxes and subsidies, non-tariff barriers, hiring flexibility, long-term policy vision, and legal equality in business.

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Subindex	Domain	Description
	R&D Transfers	The extent to which research and development funding exists and translates in practice into new commercial opportunities, measured through gross expenditure on R&D, patent families by origin, university–industry collaboration, company R&D spending, and expert assessment of whether national research reaches small and medium enterprises. Cooperation among industry, research centers, and the state stimulates new ideas and creates avenues for their commercial realization.
Free Flow of Goods & Capital	Free Flow of Trade	The ability of goods and services to flow across borders is a necessary component of international trade and openness to broader global markets. This domain also measures levels of trade participation in practice. Though this may be biased towards countries with more developed trade regimes and endowments, the level of participation in international trade, relative to the size of the economy, does provide a useful indicator of willingness to participate in international markets. It is worth noting that trade-to-GDP ratios are systematically lower for large economies, which tend to utilize extensive internal markets more than smaller countries more dependent on cross-border exchange. However, tariff policy, non-tariff barriers, infrastructure investment, and institutional commitments each shape trade participation independently of country and market scale. Additionally, nations with minimized import/export costs and logistics performance demonstrate an easier environment for trade facilitation.

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Subindex	Domain	Description
	Free Flow of Capital	The ability of financial capital to move across borders, measured through capital account openness, total foreign direct investment flows, and regulatory restrictiveness toward foreign investment. Cross-border capital mobility facilitates risk diversification, broadens markets, and creates active uses for stored savings and earnings. The free deployment of capital across borders reduces market distortions, strengthens private ownership, and encourages capital mobilization. The domain measures both the regulatory environment for capital movement and observed flows.
	Free Flow of Human Capital	The ability of labor and skills to move across borders, measured through net migration and the restrictiveness of regulations governing the hiring of foreign workers. Allowing human capital to be allocated toward its most productive uses enables economies to access global talent pools, and the empirical literature on labor mobility finds that both internal and cross-border movement improves human capital allocation, productivity, and innovation.

Appendix B: Peer Review Instructions to Reviewers

The following instructions and materials were provided to each of the ten expert scholars who participated in the GCI peer-review process between September 2025 and March 2026. Reviewers were asked to evaluate one designated subindex in which they held particular expertise, as well as the overall structure and methodology of the composite Global Capitalism Index.

Reviewer	Affiliation	Subindex Reviewed
Robert Bruner	University of Virginia	Strength of Property Rights & Private Ownership
Sonal Pandya	University of Virginia	Market-Supporting Policy
Chris Hughes	University of Pennsylvania	Market-Supporting Policy
David Leblang	University of Virginia	Labor Market Openness
Laura Phillips-Sawyer	University of Georgia	Market Competition
Pedro Matos	University of Virginia	Capital Market Sophistication
Yiorgos Allayannis	University of Virginia	Depth & Stability of Banking System
Christine Mahoney	University of Virginia	New Business Formation & Growth
David Audretsch	Indiana University Bloomington	New Business Formation & Growth
Kerem Cosar	University of Virginia	Free Flow of Goods & Capital

Table B.1: Peer Reviewers of the Global Capitalism Index

For Assigned Subindex

1. Review and evaluate the composition for the assigned subindex, including:
 - a. Evaluate the theoretical framework
 - i. Does our model of the subindex make sense, are we missing any vital elements of the concept?
 - b. Determine the validity of the selected components
 - i. Are these components exhaustive of the concept, are there others we are missing?
 - c. Evaluate the relevance of the datasets that comprise those components, including the justifications for using certain datasets as proxies for concepts
 - i. Are there other datasets we should include?
 - ii. Are there concerns with any particular datasets?

For Overall Global Capitalism Index

1. Conduct a general review of the subindex and index structure of the Global Capitalism Index design, with a particular focus on the following questions:

- a. Are any of the 8 subindices of capitalism not appropriate to include?
 - b. Are any major categories missing?
- 2. Review technical/weighting methodology for entire Index, in particular:
 - a. Evaluate the principal components analysis methodology
 - i. Do you concur with our decision to use PCA instead of a subjective weighting scheme?
 - b. Does the description of the PCA process provide enough transparency?
 - c. What additional information would you like provided?

To do so, we provided each reviewer:

- 1. **Subindex Overview:** a review of the assigned subindex, including details on the theoretical framework, technical approach, results, and data sources
- 2. **Subindex Datasheet:** the results and data for the assigned subindex
- 3. **Global Capitalism Index Datasheet:** the aggregate results for the overall Global Capitalism Index
- 4. **Technical Appendix:** a detailed description of the principal components analysis
- 5. **Replication Files:** R markdown files, if you wish to recreate the principal components analysis for either the assigned subindex or the overall Global Capitalism Index
- 6. **Overview Slides:** slides from a recent presentation providing an overview of the Global Capitalism Index, the development process, and initial results

In sum, we asked each reviewer to provide a report on their findings, to include:

- 1. Major categories of capitalism or datasets we should alter, add, or remove from the Index
- 2. Major methodological critiques
- 3. Final evaluation of whether you would:
 - a. Recommend approval of Index model as it is
 - b. Recommend approval of Index model once minor concerns are addressed and recommended changes made
 - c. Recommend major revision and resubmission of Index model (including what major alterations would be necessary for the reviewer to reconsider)
 - d. Reject the Index

Appendix C: Parabolic Scaling Methodology

The country set used to calibrate the parabolic transformations was selected through a structured review of the comparative capitalism literature and an evaluation against the eight institutional dimensions that define the GCI. The starting point was Bruno Amable's (2003) typology, which identifies five varieties of capitalism (market-based, social democratic, Asian, Continental European, and Mediterranean) and assigns example countries to each system.²²⁴ We retain Amable's groupings as the backbone of the set: the United States, United Kingdom, Canada, and Australia for the market-based model; Denmark, Finland, and Sweden for the social democratic model; Japan and Korea for Asian capitalism; France, Germany, Belgium, Austria, Ireland, and Norway for Continental European capitalism; and Greece, Italy, Portugal, and Spain for the Mediterranean model.²²⁵ We then extended the set to address the framework's confinement to largely the Global North. Through focused regional reviews of Latin America, Sub-Saharan Africa, Eastern Europe, and Southeast Asia, we added Chile, Mauritius, and Estonia on the basis of their sustained alignment with the GCI's institutional dimensions. Specifically, these countries demonstrate longstanding market liberalization, low average tariffs, private sector growth, and institutional frameworks oriented toward fair competition.²²⁶ Singapore, South Africa, and Slovenia were considered but excluded because of persistent and substantial state participation in core market activity. Specifically, the prevalence of state holding companies in Singapore, racially targeted ownership transfer mandates in post-apartheid South Africa, and the almost 25 percent state ownership of the economy in Slovenia, all differ from the intention of this particular metric.²²⁷

The purpose of establishing this set is not to identify the “best” capitalist countries or to make a normative judgment about which countries are most capitalist. Rather, the set is intended to cover the empirical range of institutional configurations that the comparative capitalism literature, across multiple traditions, recognizes as capitalist. By averaging across this range to calibrate the parabolic peak for variables, the post-transformation data reflects what capitalist economies actually do in practice rather than what any single ideological tradition prescribes. This approach

²²⁴Bruno Amable, *The Diversity of Modern Capitalism* (Oxford: Oxford University Press, 2003).

²²⁵Amable's typology builds on foundational work in the varieties of capitalism literature, particularly on Peter A. Hall and David Soskice, eds., *Varieties of Capitalism: The Institutional Foundations of Comparative Advantage* (Oxford: Oxford University Press, 2001); J. Rogers Hollingsworth, Philippe C. Schmitter, and Wolfgang Streeck, eds., *Governing Capitalist Economies: Performance and Control of Economic Sectors* (New York: Oxford University Press, 1994); Colin Crouch and Wolfgang Streeck, eds., *Political Economy of Modern Capitalism: Mapping Convergence and Diversity* (London: Sage, 1997); and Richard Whitley, *Divergent Capitalisms: The Social Structuring and Change of Business Systems* (Oxford: Oxford University Press, 1999).

²²⁶On Chile's post-1970s financial and trade liberalization, see Andrés Velasco, “Liberalization, Crisis, Intervention: The Chilean Financial System, 1975–1985,” IMF Working Paper WP/88/66 (Washington, DC: International Monetary Fund, 1988). On the institutional foundations of Mauritius's growth model, see Jeffrey Frankel, “Mauritius: African Success Story” (working paper, Harvard Kennedy School, 2012); and Pritish Behuria, “The Political Economy of a Tax Haven: The Case of Mauritius,” *Review of International Political Economy* 30, no. 2 (2023): 772–800. On Estonia's position within the post-communist varieties of capitalism literature, see Pavol Baboš, “Varieties of Capitalism in Central and Eastern Europe: Measuring the Co-Ordination Index of a National Economy,” *SEER: Journal for Labour and Social Affairs in Eastern Europe* 13, no. 4 (2010): 439–458.

²²⁷On Singapore, see Katalin Völgyi, “A Successful Model of State Capitalism: Singapore,” in *Seeking the Best Master: State Ownership in the Varieties of Capitalism*, ed. Miklós Szanyi (Budapest: Central European University Press, 2019), 275–296. On the South African race-based market economy, see Grietjie Verhoef, “‘Settlers and Comrades’: The Variety of Capitalism in South Africa, 1910–2016,” *Business History* 63, no. 8 (2021): 1413–1446. On Slovenia's state ownership, see Dorothee Bohle and Béla Greskovits, “Neoliberalism, Embedded Neoliberalism, and Neocorporatism: Towards Transnational Capitalism in Central-Eastern Europe,” *West European Politics* 30, no. 3 (2007): 443–466; and Bertelsmann Stiftung, *BTI 2024 Country Report: Slovenia* (Gütersloh: Bertelsmann Stiftung, 2024).

preserves the institutional pluralism documented across the varieties of capitalism literature while still allowing the index to flag values that fall outside the range any recognized capitalist system has sustained. Critically, the calibration is applied only to variables for which there is strong theoretical reason to expect a nonlinear measurement relationship, wherein both very low and very high values are inconsistent with sustained market functioning. This process is only applied to two datasets in the current iteration of the GCI, but the process can be expanded to accommodate subsequent dataset additions when relevant.

Mechanically, we take this list of countries, gather their average scores for each year of a given data series, and compute the simple average of all those values so that each model receives equal weight. That aggregate average per year is then treated as the central value for the dataset during 0–10 scaling.